

Field Manual Electrical Assessments, SPOF Analysis & Recommendations for MV/LV Process Plants

Medium-Voltage / Low-Voltage Industrial Plant — Indicative ~2 MW Demand

Standard: IEC Voltage: 11 kV MV / 400-415 V LV Frequency: 50 Hz

Demand: ~2 MW / ~2.35 MVA

Indicative / Training Material. All values, equipment ratings, and configurations in this manual are *indicative only* and are provided for training and familiarisation purposes. They do not constitute a design, specification, or safety document. All numeric values must be confirmed by detailed engineering studies before use in any real installation. Follow all applicable national and IEC standards, site safety rules, and employer procedures at all times.

About This Manual

This field manual is a self-paced reference for a junior electrical engineer assigned to assess the main electrical equipment of a medium-voltage / low-voltage (MV/LV) industrial process plant with an indicative demand of approximately 2 MW. It covers equipment identification, core electrical calculations, single-point-of-failure (SPOF) analysis, and practical recommendations.

How to Use This Manual

Work through the parts in order. Each part builds on the previous: start by identifying the equipment and understanding its function (Part 1), apply the key calculations (Part 2), then conduct the SPOF analysis (Part 3) and formulate recommendations (Part 4).

Use the sidebar to navigate between sections. Checklists in each part track your progress — tick items as you complete them; your progress is saved automatically in the browser. Print the manual (use the **Print** button) for field use; the sidebar is hidden and each section starts on a new page.

i This manual is a single offline HTML file. It requires no internet connection and no installed software beyond a modern web browser. Checklist state is stored in the browser's local storage — use the same browser session to preserve progress.

Scope and Plant Configuration

The indicative plant configuration used throughout this manual is:

Supply	11 kV utility incomer, metered at MV level
MV switchgear	11 kV ring-main unit (RMU) with protection relay
Transformers	2 × 1600 kVA, Dyn11, 11 kV / 415 V (TX-1 and TX-2)
LV main switchboard	Split-bus, 415 V, with normally-open bus-tie (LV-MSB)
Motor control	Multiple MCCs fed from LV-MSB sections
Standby generation	Diesel generator (DG) with automatic transfer switch (ATS)
Total demand	~2 MW (indicative); apparent power ~2.35 MVA at pf 0.85

Refer to the master single-line diagram (SLD) throughout for equipment locations and tag references.

The Assignment

Where you are in the workflow: before anything else, understand what the campaign is and what you are being asked to deliver.

The SPOF Assessment Campaign

You are one of **two Electrical Engineers** on a four-month campaign to deliver an **Electrical Single Point of Failure (SPOF) Assessment** across **21 groundwater desalination plants** in the Central & Eastern sector — **19 in Riyadh Province and 2 in the Eastern Province** (no plants in Madinah or Qassim), 16 sites, ~1.66 M m³/day total. The core team is:

Role	Count	Primary responsibility
Project Manager (Electrical)	1	Programme, quality, HSE, client interface; reviews all deliverables before submission.
Electrical Engineer	2	Per-plant SPOF assessment, plant calculation packages, concept modifications, assessment forms. That is you.
Electrical Technician	Optional	Non-intrusive field surveys and asset inventory support only — not part of the core team.

What you are asked to do per plant

For each of the 21 plants you are expected to work autonomously through the following:

- **Desk study:** review the existing single-line diagrams, equipment schedules and prior studies.
- **Plant calculation package:** produce *load-flow* (verifying bus voltages and cable loadings under normal and N-1 conditions), *short-circuit* (prospective fault current at each bus), *protection coordination* (relay grading), and *earthing & lightning* assessments with power-system software. *Arc-flash* per IEEE 1584 (incident energy at each working position) and *RAM/RBD* — Reliability, Availability, Maintainability / Reliability Block Diagram availability modelling ([Part 3](#)) — are advantageous.
- **SPOF assessment:** inventory and classify electrical single points of failure; assess criticality using *FMECA* — Failure Mode, Effects and Criticality Analysis per IEC 60812 ([Part 3](#)) — which ranks each failure mode by Risk Priority Number ($RPN = \text{Severity} \times \text{Occurrence} \times \text{Detection}$, max 125 on the 1–5 scales used here).
- **SLD verification:** red-line the single-line diagrams against the as-found plant during site visits.
- **Concept modifications:** for each recommended change, prepare a marked-up SLD, equipment outline specification, scheme description, sequence of operation and high-level bill of materials. Concept level only — not detailed design, procurement or construction.
- **Assessment forms:** complete the four per-plant deliverable forms (see [Section — The Four Deliverable Forms](#)).

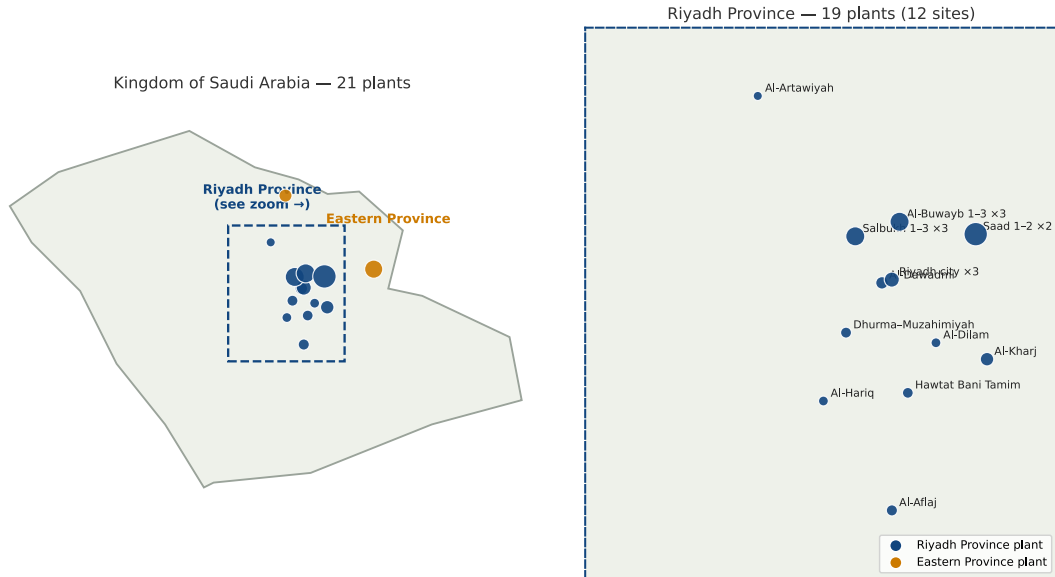
i Note — indicative figures: All ratings, capacities and figures in this manual are *indicative and tentative*. They illustrate method and typical practice. Confirm every value against the plant's actual load list, short-circuit study, utility connection agreement and governing specifications before acting on them.

Site logistics — Central & Eastern sector

The 21 plants sit at 16 sites — 19 in Riyadh Province (clustered around Riyadh, with outliers from Al-Aflaj in the south to Al-Artawiyah in the north) and 2 in the Eastern Province (Al-Hani and Hafar Al-Batin), ~1.66 M m³/day total. The map combines a national overview with a Riyadh-province zoom. City-level positions are indicative; marker size ~ capacity.

SWA Electrical SPOF / RAM — Site Logistics Map

21 groundwater plants · 19 Riyadh Province + 2 Eastern Province · 16 sites · ~1.66 M m³/day



Schematic / indicative, city-level positions (marker size ~ capacity). No plants in Madinah or Qassim. #Al-Hani (Eastern) location to verify.

Fig 0-1 — Site logistics map: 21 groundwater plants (19 Riyadh Province + 2 Eastern), ~1.66 M m³/day (indicative)

★ **Key takeaway:** You are an autonomous assessment engineer. Your job is to identify electrical single points of failure, quantify their risk, and recommend practicable mitigations — plant by plant, across 21 sites in four months.

The Four Deliverable Forms

Where you are in the workflow: understand the exact deliverables you must produce for every plant before you step on site.

For each of the 21 plants, the primary deliverables are four completed assessment forms plus the plant calculation package and any concept modification packages. The forms are described below. Blank templates are in [Appendix F — Blank Forms](#).

Form	What it contains	When completed	Who owns it
1. Document Inventory	List of all electrical documents obtained or requested: single-line diagrams (revision, date, verified), equipment schedules, protection setting records, prior studies, maintenance records, drawings. Records what was available, what was missing, and what was verified on site.	Started in desk study; updated on site visit; finalised after site.	Lead Engineer (you) — PM reviews.
2. FMECA Form	Failure Mode, Effects and Criticality Analysis (IEC 60812). One row per electrical item of significance: failure mode, local effect, system effect, existing controls, severity (S), occurrence (O), detectability (D), RPN or criticality number, and recommended action. Identifies and ranks the SPOFs.	Populated during analysis phase after desk study and site visit; updated iteratively.	Lead Engineer — PM quality-gates before submission.
3. Recommendations Register	Ranked list of recommended modifications or improvements. One row per recommendation: reference number, description, priority (high/medium/low), estimated scope, applicable standard, link to FMECA row, and status (open / in progress / closed). The basis for the concept modification packages.	Drafted after FMECA is complete; updated as concept modifications are developed.	Lead Engineer — PM approves priority ranking before issue.
4. Site-Visit Report	Factual record of the site visit: date, attendees, areas accessed, equipment observed, as-found conditions (nameplate data, breaker/switch status, visible condition concerns, any thermographic hot-spots if permitted), discrepancies against the SLD, and photographs index. Non-intrusive only — see the warning below.	Completed on the day of the site visit; reviewed by PM before issue.	Lead Engineer (with optional technician support for data capture).

✓ **Tip — form discipline:** Open all four forms as soon as a plant is assigned. Fill the Document Inventory and Site-Visit Report header during desk study so the template is ready on site. Do not wait until after the visit to create the form.

✓ **Tip — traceability:** Every row in the FMECA form should carry a unique reference that you can cite in the Recommendations Register. The PM uses this linkage when quality-checking your work.

i Form & row-ID legend. SF-01 = Document Inventory; SF-02 = FMECA Worksheet; SF-03 = Recommendations Register; SF-04 = Site-Visit Report. Within each form, rows carry the following prefixes: DOC-xx (SF-01 document rows) | FM-xx (SF-02 FMECA rows) | REC-xx (SF-03 recommendation rows) | OBS-xx (SF-04 observation rows) | PH-xx (SF-04 photolog rows) | ACT-xx (SF-04 action rows). These prefixes are consistent across the blank forms in [Appendix E](#) and the worked Jubail example in [Part 5](#).

★ **Key takeaway:** Four forms, completed in sequence — Document Inventory, FMECA, Recommendations Register, Site-Visit Report — are the non-negotiable deliverables for every plant. Start them early; update them iteratively.

Assessment Workflow at a Glance

Where you are in the workflow: map the full sequence so you can plan your time plant by plant.

Stage	What you do	Form(s) updated
1 — Pre-site desk study	Request and review all available documents: SLDs (confirm revision / date), equipment schedules, prior calculations, protection setting records. Identify missing documents to chase on site. Prepare calculation model shell. Mark up a working copy of the SLD with questions. Brief the optional technician if deployed.	Document Inventory (columns: document received, revision, status)
2 — Site visit (non-intrusive)	Verify SLD against as-found installation; red-line discrepancies. Capture nameplate data (ratings, make/model, serial numbers) for all main equipment. Record breaker/switch configuration (normally open / closed). Note visible condition, housekeeping and reliability concerns. Photograph (where permitted) and index photos. Collect any documents not previously received. Strictly observe host permit-to-work and QHSE rules.	Site-Visit Report (as-found data); Document Inventory (documents obtained on site)
3 — Analysis (office)	Update the calculation model with as-found data. Run load-flow, short-circuit, protection coordination and earthing/lightning studies. Identify and classify SPOFs from the verified SLD. Populate FMECA rows: failure mode, effects, controls, criticality (S × O × D). Rank items by criticality / risk.	FMECA form (all columns)
4 — Recommendations	For each high-criticality FMECA item, develop a concept modification: marked-up SLD, equipment outline specification, scheme description / sequence of operation, and high-level bill of materials. Re-run affected calculations on the modified arrangement. Populate the Recommendations Register with priority, scope and FMECA cross-reference.	Recommendations Register (all columns)
5 — Report & issue	PM reviews and quality-gates all four forms and the calculation package. Finalise the Site-Visit Report. Assemble the per-plant deliverable pack (forms + calculation package + concept modification packages + red-lined SLDs). Submit to PM for consolidation into the campaign report.	All four forms finalised; calculation package issued.

✓ **Tip — iterate freely:** The workflow above is a logical sequence, not a strict one-pass process. You will often loop back: a site finding may invalidate a desk-study assumption and require you to update the FMECA and re-run a calculation. Return to earlier stages whenever new information warrants it. The PM expects updated forms up until the submission milestone.

★ **Key takeaway:** Five stages — desk study, site visit, analysis, recommendations, report. The forms are living documents throughout; complete and finalise them in order.

How to Use This Manual

Where you are in the workflow: orient yourself to the manual before reading the technical content.

Manual structure — Parts 0 to 5 and Appendices

This manual is a single self-contained HTML file designed for field use (offline, printable). It is organised into six parts and a set of appendices:

Part	Title	What it gives you
Part 0	Orientation (this part)	Campaign context, deliverable forms, assessment workflow, conventions.
Part 1	Foundations	How to read a MV/LV SLD; every equipment tag and its function; power-flow walk-through; equipment schedule and applicable IEC standards.
Part 2	Calculations	All core electrical calculations — load, transformer sizing, full-load current, short-circuit, voltage drop, PFC, cable selection, protection coordination, arc-flash, earthing — with worked examples and a formula card.
Part 3	SPOF Analysis	SPOF identification method, a symbol library of failure points, FMECA and RBD techniques, reliability metrics, and a printable site checklist.
Part 4	Control Philosophy & Modifications	DCS/SCADA/PMS integration, best-practice modification templates, grid-instability mitigations, modification concept packages.
Part 5	Field Tasks	Step-by-step site procedures: desk study, site-visit checklist, drawing verification, analysis workflow, report assembly, and the first-assignment walk-through.
Appendices	A – G	Glossary, standards list, formula card, full tag table, blank forms, hardening guidance, MTBF reference data.

How to use the manual on a real assignment

- First time through:** read Parts 0 and 1 in full to internalise the workflow and the tag scheme. Skim Part 2 to confirm you are comfortable with the calculation methods.
- Starting a new plant:** open [Part 5 — Desk Study Procedure](#) and follow the checklist. Open the four blank forms from [Appendix F](#) and fill the header information.
- On site:** use the [Visual Inspection Checklist](#) and the [Drawing Verification Checklist](#) in Part 5. The SPOF symbol library in [Part 3](#) helps you spot failure points in real switchrooms.
- During analysis:** use Part 2 for calculations and Part 3 for the FMECA method. Cross-reference to the [Formula Card](#) and [MTBF Reference Data](#) as needed.
- New to this type of assignment?** Start with the [first-assignment walk-through](#) in Part 5 — it takes you step-by-step through one complete plant cycle.

✓

Tip — sidebar navigation: Use the sidebar (left panel) to jump between sections. Checkboxes in checklist sections persist across sessions in the same browser — use them to track progress per plant.

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On-screen vs field use. Checkbox tick-state is stored in the browser's `localStorage` — it persists only in the same browser on the same device, and only while on-screen at a desk. For **field use**, **print the relevant checklist section** (use the Print button or Ctrl+P) and tick on paper. Transfer your paper marks back to the on-screen form in the office.

★

Key takeaway: Parts 0 and 1 are the foundation — read them first. Parts 2–4 are your reference library. Part 5 is your task guide on every assignment. Use the first-assignment walk-through in Part 5 if you are new to this type of work.

Conventions Used in This Manual

Where you are in the workflow: agree the ground rules before diving into the technical content.

IEC basis, voltages and frequency

This manual follows **IEC conventions** throughout:

Parameter	Value used in this manual	Note
Standards basis	IEC (primary); IEEE referenced where relevant (e.g. IEEE 1584, IEEE 493)	ANSI device numbers (52, 87T, etc.) are used in SLD breaker notation as industry practice.
MV supply voltage	11 kV (indicative)	Actual utility voltage must be confirmed per plant; could be 6.6 kV, 13.8 kV or other.
LV distribution voltage	400 / 415 V, 3-phase	IEC utilisation voltage. Confirm per plant.
Frequency	50 Hz	Confirm per plant.
MV earthing	Resistance-earthed (NER)	Typical for industrial MV; confirm per plant.
LV earthing	TN-S (solid)	Confirm per plant.

Equipment tag scheme

Every piece of equipment is identified by a **tag**. Use only the tags in the table below; do not invent new tags without updating the project reference document. The full tag table with functions is in [Part 1 — Equipment Function Table](#) and [Appendix D — Tag Table](#).

Tag	Equipment	Voltage level
MV - MET	Utility revenue metering cubicle	11 kV (MV)
MV - SWGR	MV metal-clad switchgear	11 kV (MV)
MV - NER	Neutral Earthing Resistor	11 kV (MV)
TX-1 / TX-2	MV/LV power transformers (2 × 1600 kVA, indicative)	11 kV / 400 V boundary
LV - MSB	Main LV switchboard / PCC (split bus A/B)	400 V (LV)
BT	LV bus-tie circuit breaker (normally open)	400 V (LV)
MCC-1 ... MCC-n	Motor Control Centres	400 V (LV)
VFD	Variable Frequency Drive(s)	400 V (LV)
PFC	Power factor correction (detuned capacitor bank)	400 V (LV)
DG	Diesel generator set	400 V (LV)
ATS	Automatic Transfer Switch	400 V (LV)
UPS	Uninterruptible Power Supply	400 V AC / battery
DCDB	DC battery and charger system (110 V DC)	DC (MV switchgear supply)
DB / EDB	Distribution board / Essential distribution board	400 V (LV)

The "indicative figures" rule

Every numerical value in this manual — voltages, currents, kVA ratings, cable sizes, fault levels, reliability figures — is **indicative and tentative**. The values are sized to a representative ~2 MW industrial plant to illustrate calculation method and typical equipment selection. They are **not** a substitute for:

- the plant's actual load list and demand study,
- a site-specific short-circuit and load-flow study,
- the utility connection agreement and network data,
- the governing project specification and local regulations.

Always confirm figures against actual plant data before acting on them.

**Warning — NON-INTRUSIVE surveys only.** All site surveys under this campaign are **non-intrusive**. You must not open energised panels, operate switches or breakers, perform live electrical tests, or take measurements that require direct contact with live conductors. Visual inspection, nameplate reading, photograph capture and accessible meter readings are permitted. Strictly observe the host permit-to-work procedure, lock-out/tag-out rules and QHSE requirements (ISO 9001 / 14001 / 45001) at every plant. The NDA and applicable national cybersecurity data-handling requirements apply to all data collected on site.

★ **Key takeaway:** IEC basis, 11 kV / 400–415 V / 50 Hz, the tag scheme above, all figures indicative — and every site survey is non-intrusive. These four rules underpin everything in this manual. Part 1 now walks you through the standard MV/LV plant architecture you will encounter on most of these 21 sites.

The MV/LV Plant at a Glance

Where you are in the workflow: before you can assess a plant, you must be able to read its single-line diagram and name every item on it. Part 1 builds that foundation.

Design basis of the reference plant

Throughout this manual, calculations and examples are illustrated using a representative ~2 MW industrial process plant (water / desalination service, typical of the sites you will assess). The design basis is summarised below; all figures are *indicative and tentative*.

Parameter	Indicative value	Note
Total plant maximum demand	~2.0 MW	Confirm against actual load list.
Overall power factor (before correction)	0.85 lagging	Indicative; varies with motor mix and VFD loading.
Apparent power (uncorrected)	~2.35 MVA	$S = P / pf = 2000 / 0.85$
Installed transformer capacity (with margin)	2 × 1600 kVA	+20–25 % design/diversity margin; next standard size above 2 × ~1175 kVA.
Utility incoming voltage (MV)	11 kV, 3-phase, 50 Hz	Actual utility voltage confirmed per plant; could differ.
Distribution / utilisation voltage (LV)	400 / 415 V, 3-phase, 50 Hz	IEC standard; confirm per plant.
MV system earthing	Resistance-earthed via NER	Limits earth-fault current to ~300–400 A (indicative).
LV system earthing	TN-S (solid)	Confirm per plant.
Redundancy philosophy	N-1 partial — 2 × transformers + N.O. bus-tie; single DG for essential loads	One transformer can carry the essential load if the other is lost.

WORKED EXAMPLE — TRANSFORMER SIZING FROM DEMAND

Given ~2 MW demand at 0.85 power factor, size the transformer bank:

$$S = P / pf = 2000 \text{ kW} / 0.85 \approx 2353 \text{ kVA}$$

Apply a design and diversity margin of +20–25 %:

$$S_{\text{design}} = 2353 \times 1.25 \approx 2941 \text{ kVA}$$

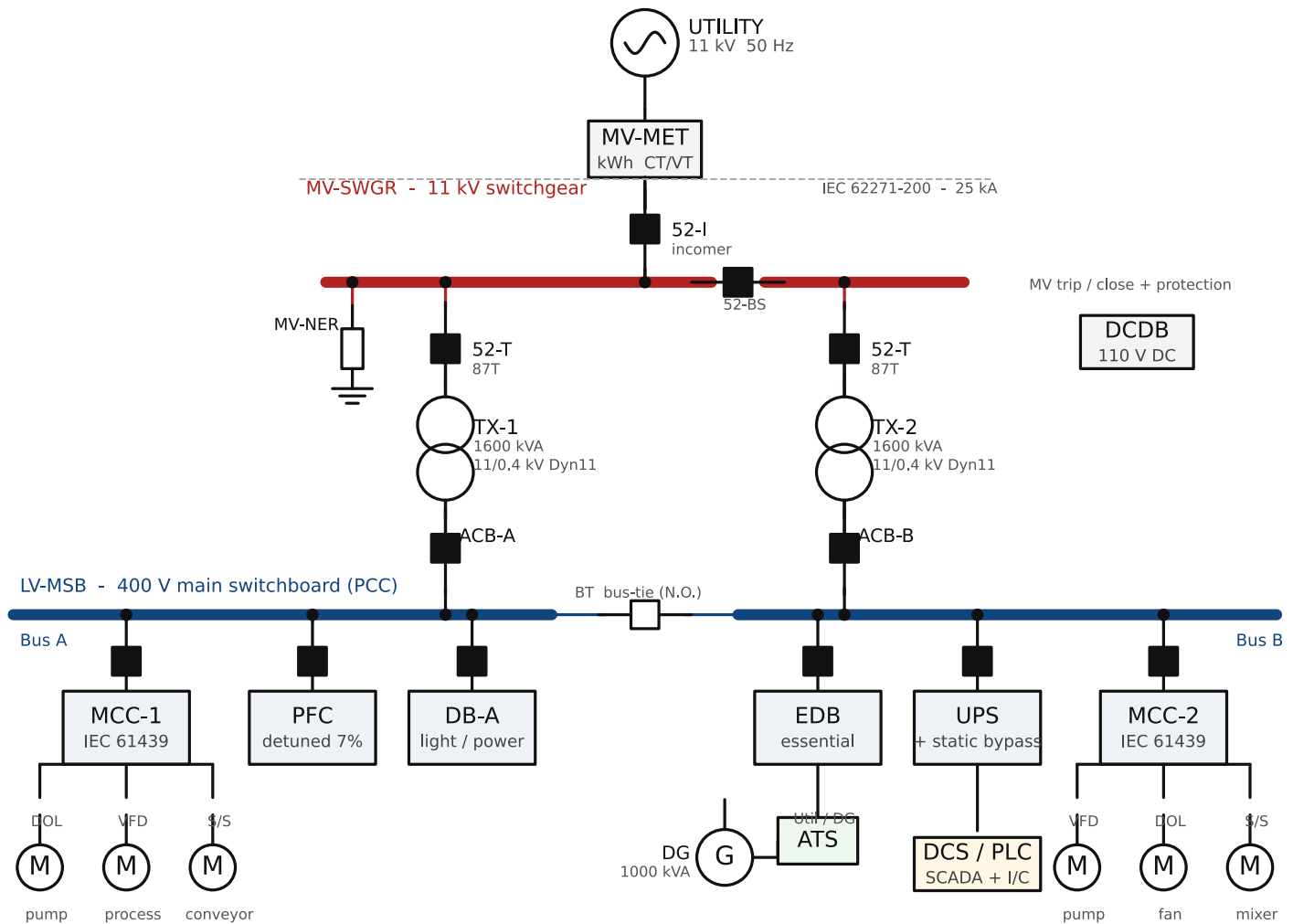
Split across two transformers (N-1 split-bus), select next standard size:

$$S_{\text{each}} = 2941 / 2 \approx 1471 \text{ kVA} \rightarrow \text{select } 2 \times 1600 \text{ kVA}$$

Result: 2 × 1600 kVA, Dyn11, 11 kV / 0.4 kV (tags TX-1 and TX-2). Values are indicative.

Power architecture — the master SLD

The master single-line diagram below shows the complete reference power architecture. Study it before proceeding — every section of Part 1 refers to items on this drawing.



MASTER SINGLE-LINE DIAGRAM - ~2 MW MV/LV PROCESS PLANT
 11 kV / 400 V - 50 Hz - 2 x 1600 kVA (Dyn11, Z~6%) - IEC 60617 - indicative

Fig 1-1 — Master SLD, indicative ~2 MW MV/LV process plant (IEC 60617 symbols; all ratings indicative)

The architecture has three tiers:

- MV (11 kV) tier:** Utility feed → metering (MV-MET) → MV switchgear (MV-SWGR) with protection IEDs (MV-PROT) and neutral earthing resistor (MV-NER).
- Transformation tier:** Two independent transformer feeders supply TX-1 and TX-2, each stepping 11 kV down to 400 V.
- LV (400 V) tier:** Main switchboard (LV-MSB) with a split bus (A and B) joined by a normally-open bus-tie (BT); Motor Control Centres (MCC-1, MCC-2), PFC bank, distribution boards, essential board (EDB), diesel generator (DG), ATS, UPS and DC battery system (DCDB).

★ **Key takeaway:** The ~2 MW reference plant is the template for everything in this manual. Two 1600 kVA transformers on a split LV bus with a normally-open bus-tie give partial N-1 resilience — understanding why this architecture is used, and where it is still vulnerable, is the core of the SPOF assessment.

How to Read an SLD

Where you are in the workflow: before the site visit, you must be comfortable reading any MV/LV SLD you are handed, not just the reference drawing.

Power-flow direction and voltage levels

Read an SLD **top to bottom** — energy flows from the utility source at the top to the loads (motors, boards) at the bottom. On the reference drawing there are two clearly separated horizontal zones:

- MV zone (upper band):** 11 kV — from the utility intake down to and including the transformer primaries. MV busbars are drawn in **firebrick red** on colour drawings.
- LV zone (lower band):** 400 V — from the transformer secondaries to all distribution equipment and loads. LV busbars are drawn in **navy blue**.

The two **transformers** (TX-1 , TX-2) are the physical and electrical boundary between the MV and LV zones.

IEC 60617 symbols used on the reference SLD

*i***Note — IEC 60617:** This is the international standard defining graphical symbols for diagrams. The symbols used on the reference SLD are drawn in accordance with IEC 60617 series. If you encounter a drawing using ANSI/IEEE symbols (common in North America), the layout is the same but some symbols differ — the most common example is the circuit-breaker symbol (IEC: a solid square; ANSI: two diverging lines). Always check the drawing legend before interpreting unfamiliar symbols.

Symbol / notation	Meaning	Where seen on reference SLD
Heavy horizontal line ()	Busbar — the common copper to which feeders and incomers connect.	11 kV bus in MV-SWGR; LV Bus A and Bus B in LV-MSB.
Solid square on conductor	Circuit breaker (closed / in service). IEC 60617-07.	All MV and LV breakers.
Hollow / dashed square	Circuit breaker normally open (N.O.).	Bus-tie  ; ATS open position.
 [52-I] ,  [52-T1] , etc.	MV circuit breaker — ANSI device number 52. Suffix identifies the function (I = incomer, BS = bus-section, T1/T2 = transformer feeder).	MV-SWGR breakers.
 [ACB]	Air Circuit Breaker — large LV breaker used for main incomers and bus-tie.	LV-MSB incomers from TX-1 and TX-2; bus-tie BT.
 [MCCB]	Moulded-Case Circuit Breaker — smaller LV breaker used for outgoing feeders.	LV-MSB outgoing feeders to MCC-1, MCC-2, PFC, DB, EDB.
Two interlinked circles with a line between () with  8)	Two-winding transformer (IEC 60617). The winding configuration label (e.g. Dyn11) shows delta primary (D), star secondary (y, solidly earthed), and 30° phase shift (clock position 11).	TX-1 and TX-2.
Resistor symbol to earth at neutral point ( [N])	Neutral Earthing Resistor — MV neutral earthed through a resistor.	MV-NER at the 11 kV neutral.
Three diminishing bars to earth	Earth / ground connection.	LV transformer secondary neutral; earthing points on equipment.
Circle with  CT or  VT	Current Transformer / Voltage Transformer — instrument transformers feeding meters and relays.	MV-MET (revenue CTs/VTs); MV-PROT (protection CTs).
Circle with  M	Motor load.	Process motors at the bottom of each MCC feeder.
Two parallel plates	Capacitor bank (PFC).	PFC on LV Bus A feeder.
Labelled rectangle (block)	Equipment assembly — used for MCC, VFD, UPS, ATS, DB, DCDB where a single-symbol representation is appropriate.	MCC-1, MCC-2, VFD, UPS, ATS, DCDB, DB, EDB.
 DOL /  S/S /  VFD label on motor feeder	Starter type: Direct-on-Line / Soft-Starter / Variable Frequency Drive.	Within MCC blocks.

Busbar convention and the split bus

The LV main switchboard (LV-MSB) is drawn as a **split bus**: **Bus A** (left, fed by TX-1) and **Bus B** (right, fed by TX-2) are two physically separate copper bars in the one board. They are joined only through the normally-open bus-tie (). In normal operation each transformer supplies its own half of the plant independently. The split-bus arrangement:

- limits the fault current that a single transformer must interrupt,
- contains the blast zone of any bus fault to one section, and
- allows the bus-tie to close automatically on loss of one incomer, maintaining supply to both buses from the surviving transformer.

Red-lining on site

On a site visit, you compare the SLD against the as-found installation and mark discrepancies in red on a working print or a PDF annotation tool. Common discrepancies to look for:

- Breaker labelling or device number differs from drawing.
- An extra or missing feeder — a circuit on site not shown on the drawing.
- Bus-tie in a different normal position (open or closed) from the drawing.
- Equipment substituted (e.g. RMU instead of metal-clad switchgear).
- Protection relay type or settings different from what is listed.
- Cables or transformers replaced with a different rating.

★ **Key takeaway: Read SLDs top-to-bottom (source to load), know the IEC 60617 symbols, and understand the split-bus concept. On site, every discrepancy between drawing and reality is a red-line — and potentially a SPOF waiting to be found.**

PCC vs MCC

Where you are in the workflow: this distinction is fundamental — confusing the two is one of the most common errors on assessment forms.

Both the PCC and the MCC are LV assemblies built to IEC 61439-1/-2, and they can look similar in a switchroom. They do very different jobs:

Feature	PCC (Power Control Centre)	MCC (Motor Control Centre)
Tag in reference plant	LV-MSB	MCC-1 , MCC-2
Primary function	Receives transformer output(s); distributes <i>bulk</i> LV power to downstream boards and feeders. The LV power hub.	Houses individual motor <i>starters</i> — DOL, star-delta, soft-starter, VFD — that control and protect specific process motors.
Incomers	Air Circuit Breakers (ACBs) from transformers; motorised bus-tie ACB.	Single MCCB incomer from the PCC (LV-MSB) outgoing feeder.
Outgoing feeders	MCCBs feeding MCCs, distribution boards, PFC bank, large individual loads.	Per-motor buckets: MCCB or fuse-contactor, contactors, overload relay, local/remote control, ammeter, run/trip indication.
Bus rating (indicative)	~2500–3200 A; 50–65 kA fault withstand.	~1000–2000 A; fault withstand matched to upstream board.
Form of segregation (IEC 61439)	Form 4b — busbars separated from functional units; terminals separated from other functional units.	Form 3b/4b — functional units (starters) separated from each other.
Where on the SLD	The central LV switchboard spanning Bus A and Bus B — immediately below the transformers.	On outgoing MCCB feeders from the LV-MSB Bus A/B; one MCC per bus section in the reference plant.
One-line description	"The LV power hub — it feeds the MCCs."	"Where the motors are started — it feeds the loads."

Note — vocabulary variation: In different organisations, the PCC is also called the Main LV Switchboard (MLVSB), Main Distribution Board (MDB), or simply the Main Board. The MCC may be called a Motor Control Panel (MCP) or a starter board. The *function* — bulk distribution vs. individual motor control — is the reliable distinguishing criterion, regardless of what the nameplate says.

Tip — SPOF implication: Because the PCC is the single point through which all transformer power passes before reaching the MCCs, a fault or loss of the PCC is a plant-level SPOF. The split bus with bus-tie is specifically designed to limit this vulnerability. When assessing a plant, check whether the PCC is truly split-bus or is effectively a single-bus board with cosmetic segregation.

★ Key takeaway: PCC (LV-MSB) = bulk distribution hub, fed by transformers, feeds MCCs. MCC = motor starter assembly, fed by PCC, feeds individual motors. The PCC feeds the MCCs; the MCCs feed the loads.

Equipment Function Table

Where you are in the workflow: this is the core reference table for the assessment — you will return to it every time you encounter a tag on site or in a document.

The table below covers all 17 tags in the reference plant (including **MV-PROT**). For each tag it states the equipment name, its function (what it does and why it is there), the indicative rating, and where it appears on the master SLD. All ratings are indicative; confirm against actual plant nameplates.

Tag	Equipment	Function / purpose	Indicative rating	Where on the SLD
MV-MET	Utility revenue metering cubicle	Measures energy and maximum demand at the 11 kV intake via utility-grade CTs and VTs feeding a tariff kWh/kVAh meter. The commercial boundary between utility and plant — the utility reads this meter for billing.	Utility CT/VT and tariff meter; class 0.2S or better.	MV section, first item on the 11 kV supply side (top of drawing).
MV-SWGR	11 kV metal-clad switchgear	Receives and distributes the 11 kV supply. Houses the MV incomer breaker [52-I], bus-section breaker [52-BS], and two transformer-feeder breakers [52-T1] / [52-T2], so that MV circuits can be switched and fault current cleared quickly. The 11 kV busbar inside this board is the MV power hub.	11 kV, 630–1250 A bus, 25 kA / 1 s; vacuum CBs; IEC 62271-200.	MV section, central board below metering.
MV-PROT	MV protection and control IEDs	Detects MV faults and trips the correct breaker to isolate them quickly, minimising damage and restoring supply elsewhere. Functions: overcurrent (50/51), earth-fault (50N/51N), transformer differential (87T), under/over-voltage (27/59). Relays are supplied from the DCDB so they remain operative even on AC failure.	Numerical IEDs; one per breaker or combined; IEC 60255 series.	MV section, relay panels associated with MV-SWGR breakers.
MV-NER	Neutral Earthing Resistor	Earths the 11 kV transformer star neutral through a resistor that limits the earth-fault current to a safe, detectable level (~300–400 A indicative). Without it, an undetected earth fault could cause equipment damage and personnel hazard. A neutral CT allows the protection relay to detect and measure the fault current.	Resistance sized for ~300–400 A earth-fault level (indicative); with neutral CT.	MV section, at the [N] neutral point — connected to the neutral of the 11 kV supply transformer or to the MV bus neutral where provided.
TX-1	MV/LV power transformer No. 1	Steps 11 kV down to 400 V to supply Bus A of the LV-MSB. One of the two parallel sources that give the plant its partial N-1 resilience. If TX-2 is lost, the bus-tie closes so TX-1 can carry both buses.	1600 kVA, 11 kV / 0.4 kV, Dyn11, Z ≈ 6 %, IEC 60076; cast-resin or oil-immersed.	Between MV feeder [52-T1] and LV incomer [ACB-A] on Bus A.
TX-2	MV/LV power transformer No. 2	Steps 11 kV down to 400 V to supply Bus B of the LV-MSB. Mirrors TX-1 on the Bus B side. If TX-1 is lost, the bus-tie closes so TX-2 can carry both buses.	1600 kVA, 11 kV / 0.4 kV, Dyn11, Z ≈ 6 %, IEC 60076.	Between MV feeder [52-T2] and LV incomer [ACB-B] on Bus B.
LV-MSB	Main LV switchboard / PCC (Power Control Centre)	The 400 V power hub: receives the two transformer outputs on a split bus (A and B) and distributes bulk LV power via MCCB feeders to the MCCs, PFC bank, distribution boards and large individual feeders. ACB incomers from both transformers; motorised normally-open bus-tie (BT) between the two bus sections.	400 V, split bus A/B, ~2500–3200 A bus, 50–65 kA; ACB incomers + motorised bus-tie; Form 4b; IEC 61439-1/-2.	LV section — the main board spanning both Bus A and Bus B, directly below the transformers.
BT	LV bus-tie circuit breaker (normally open)	Normally open during operation so the two bus sections are independent. Automatically closes on loss of either LV incomer (trip of ACB-A or ACB-B), allowing the surviving transformer to supply both buses. This is the single most important N-1 feature in the LV distribution — its correct operation (auto-close logic armed, correct setting) must always be verified.	ACB, motorised/automatic; matching LV-MSB rating.	LV-MSB — the link between Bus A and Bus B (drawn with hollow/N.O. symbol).
MCC-1	Motor Control Centre No. 1 (Bus A)	Houses the motor starters (DOL, star-delta, soft-starter and/or VFD buckets) that control and protect the Bus-A process motors. Each bucket contains: MCCB or fuse-contactor, contactor(s), thermal/electronic overload relay or motor protection relay, local/remote control switch, run/trip indication and ammeter.	400 V; bus rating matched to group motor demand; IEC 61439-1/-2; Form 3b/4b.	LV-MSB Bus A, on an outgoing MCCB feeder.
MCC-2	Motor Control Centre No. 2 (Bus B)	Mirrors MCC-1 on Bus B — houses the starters for Bus-B process motors. Distributing motors across two bus sections means the plant retains some motor availability even if one transformer section is lost.	400 V; bus rating matched to group motor demand; IEC 61439-1/-2.	LV-MSB Bus B, on an outgoing MCCB feeder.
VFD	Variable Frequency Drive(s)	Provides continuously variable speed control of pumps, fans, compressors and mixers by varying output voltage and frequency. Reduces energy consumption and mechanical starting stress compared with DOL starting. May be installed as individual MCC buckets or as standalone panels. Input chokes and EMC filters required to limit harmonic injection into the LV bus.	Rated per driven motor; input choke and EMC filter; IEC 61800 series.	Within MCC-1 / MCC-2 buckets, or as standalone panels; feeding speed-controlled motors.
PFC	Power factor correction (detuned capacitor bank)	Supplies reactive power locally to correct the plant power factor from ~0.85 to 0.95+ lagging, reducing demand charges and freeing transformer and cable capacity. Automatic switching (contactor-controlled	Automatic; detuned (7 % reactor); sized to correct ~0.85 → 0.95+; IEC 60831.	LV-MSB Bus A, on a dedicated MCCB feeder.

Tag	Equipment	Function / purpose	Indicative rating	Where on the SLD
		steps) matches reactive output to load variation. Detuned (7 % tuning reactor in series) to prevent resonance with VFD-generated harmonics.		
DB	Distribution board(s)	Distributes LV power to general (non-essential) loads — lighting, small power, HVAC, sockets — via MCBs and RCBOs. Fed from the LV-MSB via MCCB feeders. These loads are not critical: they are shed on a utility outage before or when the generator takes over the essential bus.	400/230 V; MCB/RCBO outgoing ways; IEC 60439 / IEC 61439.	LV-MSB Bus A (and/or Bus B), on MCCB outgoing feeders.
EDB	Essential / emergency distribution board	Feeds loads that must remain energised through a utility outage — escape lighting, fire systems, critical control auxiliaries, communication systems. Supplied from the LV-MSB Bus B via an MCCB feeder in normal operation, and backed by the ATS so that on utility loss it is transferred to the generator within the ATS changeover time.	400/230 V; MCB/RCBO outgoing; IEC 61439.	LV section — fed from Bus B (normal) and backed via ATS from DG.
DG	Diesel generator set	Standby generation sized to carry the essential load on utility failure — not necessarily the full 2 MW plant demand. Starts automatically (auto-start logic) on detection of utility loss; the ATS transfers the essential bus once the generator has reached rated voltage and frequency. Equipped with AVR, fuel day-tank, acoustic canopy and exhaust system.	~800–1250 kVA indicative (size confirmed by essential-load list); 400 V, 50 Hz, AVR.	Standby power path (lower band of drawing).
ATS	Automatic Transfer Switch	Automatically transfers the essential bus between the utility feed (normal source) and the generator (standby source) on utility loss, and retransfers when utility supply is restored and stable. Mechanically interlocked to prevent both sources feeding the bus simultaneously — an important safety feature. Changeover time is the key performance parameter (typically 10–30 s for diesel start + transfer).	Rated for essential bus load; mechanically interlocked; IEC 60947-6-1.	Standby path — between DG output and EDB incoming supply.
UPS	Uninterruptible Power Supply	Provides clean, uninterrupted AC power to the DCS, PLC, SCADA, instrumentation and critical communications with no break — not even the ~20 s it takes the diesel generator to start and stabilise. Operates from internal batteries during the outage bridge; static bypass allows maintenance without interrupting the load. Battery autonomy is typically 15–30 min (confirm per spec).	Rated per DCS/instrument load; static bypass; battery 15–30 min (indicative); IEC 62040.	LV-MSB Bus B feeder, dedicated UPS output to DCS/control system.
DCDB	DC battery and charger system (110 V DC)	Provides secure 110 V DC power to trip and close the MV switchgear breakers, and to power the protection relay IEDs and their communications. This DC supply must remain available even when the entire AC system (both utility and generator) is dead — battery float charging maintains the battery at full charge continuously, and the battery alone sustains protection and switching capability during a complete AC blackout.	110 V DC (typical); battery sized per IEC 62485-2; float/boost charger; IEC 60896.	DC control/protection supply band — feeds MV-SWGR trip/close circuits and MV-PROT IEDs.

i Note — tag MV-PROT : The protection IEDs are sometimes shown as part of the MV-SWGR cubicle rather than as a separate tag, depending on the drawing convention. On site, check whether the relay panel has its own tag plate. For FMECA purposes, MV-PROT is treated as a distinct item because its failure mode (relay malfunction or loss of DC supply) is independent of the switchgear mechanism failure.

★ Key takeaway: Seventeen tags, two voltage levels, three functional layers — generation of the supply (MV), distribution (LV-MSB), and control/protection (DCDB/UPS/DG/ATS). Memorise the function of each tag in one sentence; you will need that knowledge to populate the FMECA without hesitation.

Power-Flow Walk-Through

Where you are in the workflow: tracing the three power paths is what you do mentally at every site visit — it reveals where the architecture is robust and where it is not.

Keep the master SLD (Fig 1-1) open alongside this section. There are three distinct paths through the plant — trace each one on the drawing as you read.

Path 1 — Main (normal) path: utility to process motor

1. **Utility 11 kV feed** arrives at the site and passes through the revenue metering cubicle (**MV-MET**) where energy is metered for tariff billing.
2. The supply enters the **MV switchgear** (**MV-SWGR**) via incomer breaker [52-I]. The protection IEDs (**MV-PROT**) watch this incomer and the transformer feeders for faults. The 11 kV neutral is earthed through the resistor (**MV-NER**).
3. Two transformer feeder breakers [52-T1] and [52-T2] supply the two transformers (**TX-1** and **TX-2**), which step 11 kV down to 400 V.
4. Each transformer secondary feeds its LV incomer ACB: TX-1 → ACB-A → **Bus A**, TX-2 → ACB-B → **Bus B**. The bus-tie (**BT**) between them is normally open, so the two bus sections are independent.
5. Outgoing MCCB feeders from Bus A and Bus B supply the Motor Control Centres (**MCC-1** , **MCC-2**), the PFC bank (**PFC**), distribution boards (**DB**) and the essential board (**EDB**).
6. Inside an MCC, the appropriate starter (DOL, soft-starter or a **VFD**) energises the **process motor** — pump, fan, conveyor or mixer.

✓ **Tip — tracing on a real SLD:** On a site SLD that may be larger or more complex, follow the same top-to-bottom discipline and identify: (a) the metering point, (b) the MV incomer and switchgear, (c) the transformer(s), (d) the LV main board and its bus arrangement, (e) the MCC feeders, and (f) the standby and DC paths. This sequence works on any MV/LV plant drawing.

Path 2 — Standby path: on utility loss

When the utility supply is lost (or falls outside acceptable voltage/frequency limits):

1. The ATS detects utility loss and issues an **auto-start signal** to the diesel generator (**DG**).
2. The DG starts, ramps to rated speed, and the AVR brings the voltage to 400 V ± tolerance.
3. The **ATS transfers** the essential bus from the utility feed to the DG output — the transfer is mechanically interlocked to ensure only one source feeds the EDB at a time.
4. The **essential distribution board** (**EDB**) remains energised — escape lighting, fire systems and critical process auxiliaries stay live.
5. Within the LV-MSB, if one incomer ACB trips (loss of one transformer), the **bus-tie** (**BT**) auto-closes so the surviving transformer can supply both Bus A and Bus B (at reduced capacity — load shedding may be needed).

i **Note — ATS changeover time:** A diesel generator takes typically 10–30 s to start, synchronise and accept load (indicative). During this window, only the UPS-backed loads remain energised. The EDB is de-energised for this interval unless the ATS uses a static (electronic) transfer or the EDB has its own small UPS. Verify the actual changeover time on site and confirm it is acceptable for the essential loads.

Path 3 — DC / control path: always alive

The control and protection path is designed to be available at all times — even during a complete AC blackout:

1. The **DCDB** (DC battery and charger) is float-charged from the LV supply and holds its battery at full charge continuously. On loss of AC, the battery sustains the DC bus.
2. The **110 V DC bus** from the DCDB feeds the trip and close coils of the MV switchgear breakers and the MV protection relay IEDs. The protection system remains operative — it can still detect faults and trip breakers — even with no AC available.
3. The **UPS** is float-charged from the LV AC bus (typically Bus B). On loss of AC it switches seamlessly to its own battery (static changeover, no break). It supplies the DCS, PLC, SCADA and instrumentation at clean regulated AC.
4. Together, the DCDB and UPS ensure that both *protection* (MV DC) and *control* (AC to DCS/instruments) remain available for the full battery autonomy period, giving the operator visibility and the ability to operate the MV switchgear during a blackout.

✓ **Tip — SPOF on the DC path:** A common finding is that both the UPS and the DCDB charger are supplied from the same LV bus section or even the same feeder, removing their independence. On site, verify that the UPS and DCDB have segregated sources. Also check the battery condition and last test date — an aged or untested battery is a hidden SPOF that does not show up on the SLD.

★ **Key takeaway: Three paths — normal (utility to motors), standby (DG via ATS to essential loads), and DC/control (DCDB to MV switchgear; UPS to DCS). Trace all three on every SLD you review; the points where paths converge or where independence is lost are your SPOF candidates.**

Equipment Schedule and Standards

Where you are in the workflow: the equipment schedule is the quantitative companion to the function table — use it during the desk study to frame your calculations and check nameplate data on site.

Indicative equipment schedule

The schedule below summarises the principal electrical equipment for the reference ~2 MW plant. All ratings are **indicative and tentative**; confirm against the actual plant documents and nameplate data.

Tag	Equipment	Indicative rating / specification	IEC standard
MV-MET	Utility metering cubicle	11 kV; tariff-grade CTs/VTs (class 0.2S); kWh + kVAh meter; per utility specification.	IEC 62053; utility specification.
MV-SWGR	MV metal-clad switchgear	11 kV; 630–1250 A bus (indicative); 25 kA / 1 s fault withstand; vacuum CBs; 4 panels (incomer, bus-section, 2 × transformer feeder).	IEC 62271-200; IEC 62271-100.
MV-NER	Neutral Earthing Resistor	Resistance to limit earth-fault current to ~300–400 A (indicative); with neutral CT; outdoor or indoor housing.	IEC 60099-6 (guidance); manufacturer's specification.
TX-1 / TX-2	MV/LV power transformers	2 × 1600 kVA (indicative); 11 kV / 0.4 kV; Dyn11; Z ≈ 6 %; cast-resin (dry-type) or oil-immersed (ONAN); off-load taps ±2 × 2.5 %.	IEC 60076 series.
LV-MSB	Main LV switchboard / PCC	400 V; split bus A/B; ~2500–3200 A main bus (indicative); 50–65 kA fault withstand; ACB incomers + motorised N.O. bus-tie ACB; MCCB outgoing feeders; Form 4b segregation.	IEC 61439-1; IEC 61439-2.
MCC-1 / MCC-2	Motor Control Centres	400 V; horizontal bus 1000–2000 A (indicative); 50 kA fault withstand; withdrawable or fixed starter buckets: DOL, star-delta, soft-starter, VFD; MCCB or fuse-contactor, contactor, electronic overload/MPR; Form 3b/4b.	IEC 61439-1; IEC 61439-2.
VFD	Variable Frequency Drives	Rated per motor; 400 V input; input choke + EMC filter; integrated or remote keypad; RS485/PROFIBUS/PROFINET fieldbus interface.	IEC 61800-3; IEC 61800-5.
PFC	Power factor correction bank	Automatic; detuned (7 % tuning reactor); sized to correct ~0.85 → 0.95+ (indicative); contactor-switched capacitor steps.	IEC 60831; IEC 60831-1/-2.
DB	Distribution boards	400/230 V; MCB and RCBO outgoing ways; surface or flush-mounted; IP rating per environment.	IEC 61439-3.
EDB	Essential distribution board	400/230 V; fed via ATS; rated for essential load; MCB/RCBO outgoing.	IEC 61439-3.
DG	Diesel generator set	~800–1250 kVA (indicative — sized by essential load list); 400 V, 3-phase, 50 Hz; AVR; fuel day-tank; acoustic canopy; auto-start panel.	IEC 60034 (generator); ISO 8528 (genset performance).
ATS	Automatic Transfer Switch	Rated for essential bus full load; mechanically interlocked; open-transition or closed-transition (confirm per plant); changeover time per DG start time (indicative 10–30 s).	IEC 60947-6-1.
UPS	Uninterruptible Power Supply	Online double-conversion; rated per DCS/instrument load; static bypass; battery autonomy 15–30 min (indicative); 400 V or 230 V single-phase output as required.	IEC 62040-1; IEC 62040-3.
DCDB	DC battery and charger system	110 V DC (typical for MV switchgear); float/boost charger; VRLA or vented lead-acid battery; battery sized for ~8 h autonomy (confirm per spec) plus interruption allowance.	IEC 60896-11/-21; IEC 62485-2.

Applicable IEC standards summary

Equipment category	Primary standard(s)
MV switchgear (metal-clad)	IEC 62271-200; IEC 62271-100 (vacuum CBs)
Power transformers	IEC 60076 series
LV switchgear and MCC assemblies	IEC 61439-1; IEC 61439-2
LV circuit breakers (ACB, MCCB, MCB)	IEC 60947-2; IEC 60947-3
Motors	IEC 60034 series
Variable speed drives / VFDs	IEC 61800-3; IEC 61800-5
Power factor correction capacitors	IEC 60831
Uninterruptible power supplies	IEC 62040-1; IEC 62040-3
Automatic transfer switches	IEC 60947-6-1
DC battery systems	IEC 60896; IEC 62485
Earthing and bonding (general installation)	IEC 60364 series
Lightning protection systems	IEC 62305 series

Equipment category	Primary standard(s)
Protection relays / IEDs	IEC 60255 series
FMECA methodology	IEC 60812
Reliability / availability (RAM)	ISO 14224; ISO 20815
Arc-flash incident energy	IEEE 1584 (US basis, widely adopted)

i Note — standards confirmation: The standards listed are indicative. The governing standards for each plant are determined by the client specification and the applicable national regulations. Check the project specification and confirm with the client which revision of each standard applies.

★ Key takeaway: The equipment schedule gives you the quantitative framework for the desk study — use it to check whether the nameplate data you collect on site matches the design intent. Any significant discrepancy (wrong rating, different vector group, different impedance) is a calculation input that may change your load-flow or short-circuit result and should be noted in the Document Inventory and potentially the FMECA. Part 2 now shows you how to run the calculations.

Part 1 — Self-Check

Where you are in the workflow: test yourself before moving to the calculations in Part 2. If you cannot answer these confidently, re-read the relevant section of Part 1.

Attempt all six questions before revealing the answers. Use only the master SLD (Fig 1-1) and your memory — no notes.

Question 1

Which two tags mark the boundary between the 11 kV and the 400 V parts of the plant? State what each one does.

▼ Show answer

TX-1 and TX-2 — the two MV/LV power transformers (1600 kVA each, indicative; 11 kV / 0.4 kV, Dyn11). They step the 11 kV utility supply down to 400 V for the LV distribution system. TX-1 feeds Bus A and TX-2 feeds Bus B of the LV-MSB. Everything above them is MV (11 kV); everything below them is LV (400 V).

Question 2

What is the difference between LV-MSB and MCC-1? Which one feeds which?

▼ Show answer

LV-MSB is the PCC (Power Control Centre) / Main LV Switchboard — it receives the transformer outputs and distributes bulk 400 V power. Think "the LV power hub."

MCC-1 is a Motor Control Centre — an assembly of motor starters (DOL, soft-starter, VFD) that control and protect individual process motors. Think "where the motors are started."

The LV-MSB feeds MCC-1 (via an outgoing MCCB feeder from Bus A). MCC-1 feeds the individual motors.

Question 3

The bus-tie BT is normally open. What event causes it to close automatically, and why is that desirable?

▼ Show answer

The bus-tie closes automatically on the trip or loss of either LV incomer ACB (i.e. loss of one transformer or its MV feeder). The auto-close logic detects the incomer trip and closes BT so the surviving transformer can carry both Bus A and Bus B loads.

This is desirable because it prevents loss of supply to half the plant whenever one transformer is lost — it is the primary N-1 resilience feature of the LV distribution. Without it, loss of TX-1 would immediately shed all Bus A loads (MCC-1, PFC, DB-A and any Bus A feeders).

Question 4

Name the three tags on the standby/essential power path and state the order in which energy flows through them on a utility outage.

▼ Show answer

DG → ATS → EDB.

On a utility outage: the DG starts (auto-start signal from the ATS), ramps to rated voltage and frequency; the ATS transfers the essential bus from the utility source to the DG output; the EDB (essential distribution board) remains energised, supplying escape lighting, fire systems and critical process auxiliaries.

Question 5

Why does the plant need a separate DCDB when it already has a UPS? What does each supply?

▼ Show answer

They serve two different and independently critical supply circuits:

- UPS supplies AC power (400 V or 230 V) to the DCS, PLC, SCADA and instrumentation systems. These are AC-powered electronics that need a clean, regulated, uninterrupted supply.
- DCDB supplies 110 VDC to the trip and close coils of the MV switchgear breakers and to the MV protection relay IEDs. These must operate even when the AC system is completely dead — a DC battery is the only supply that can guarantee this.

If you replaced the DCDB with a UPS, the MV switchgear would lose its ability to trip or close once the UPS itself lost its input feed. The DCDB's battery is the last line of protection operability during a full AC blackout.

Question 6

What is the purpose of MV-NER, and what would happen if it were removed (neutral left floating)?

▼ Show answer

MV-NER (the Neutral Earthing Resistor) earths the 11 kV system neutral through a resistor to **limit earth-fault current** to approximately 300–400 A (indicative). This value is high enough for earth-fault protection relays to detect the fault reliably, but low enough to limit damage to equipment and reduce the step/touch voltage hazard at the fault location.

If the NER were removed and the neutral left floating (isolated neutral system):

- An earth fault would not produce a reliable fault current — the protection relay could not detect it dependably, meaning the fault might persist and worsen.
- The healthy phases would rise to line-to-line voltage above earth, stressing cable insulation and equipment rated for line-to-earth voltage.
- Transient overvoltages (ferro-resonance) could develop on the unearthed network, potentially damaging voltage transformers and other equipment.

★ **Key takeaway: If you answered all six correctly without hesitation, you are ready for Part 2 — Calculations. If any question exposed a gap, go back to the relevant section of Part 1 now. The calculations in Part 2 assume you know what every tag does and how the three power paths connect.**

Part 2 — Calculations

Where you are in the workflow: you have identified the equipment and read the SLD (*Part 1*); now you run the numbers that tell you whether the existing equipment is correctly rated before you write a single recommendation.

★ **Assessment calculations are verification, not green-field design. Your job in the field is to ask: "Is this transformer/breaker/cable/genset still adequate for what the plant is actually doing today?" — and to back the answer with arithmetic, not opinion.**

Why Calculations Matter in an Assessment

Every rating you see on the SLD — a transformer kVA, a breaker breaking capacity, a busbar bracing figure, a cable cross-section, a capacitor bank, a genset — is the output of a calculation. That calculation is the **audit trail** that justifies the rating. In an assessment you work in the opposite direction: you start from the nameplate, reverse-engineer the assumption that was made, and check whether reality still matches it.

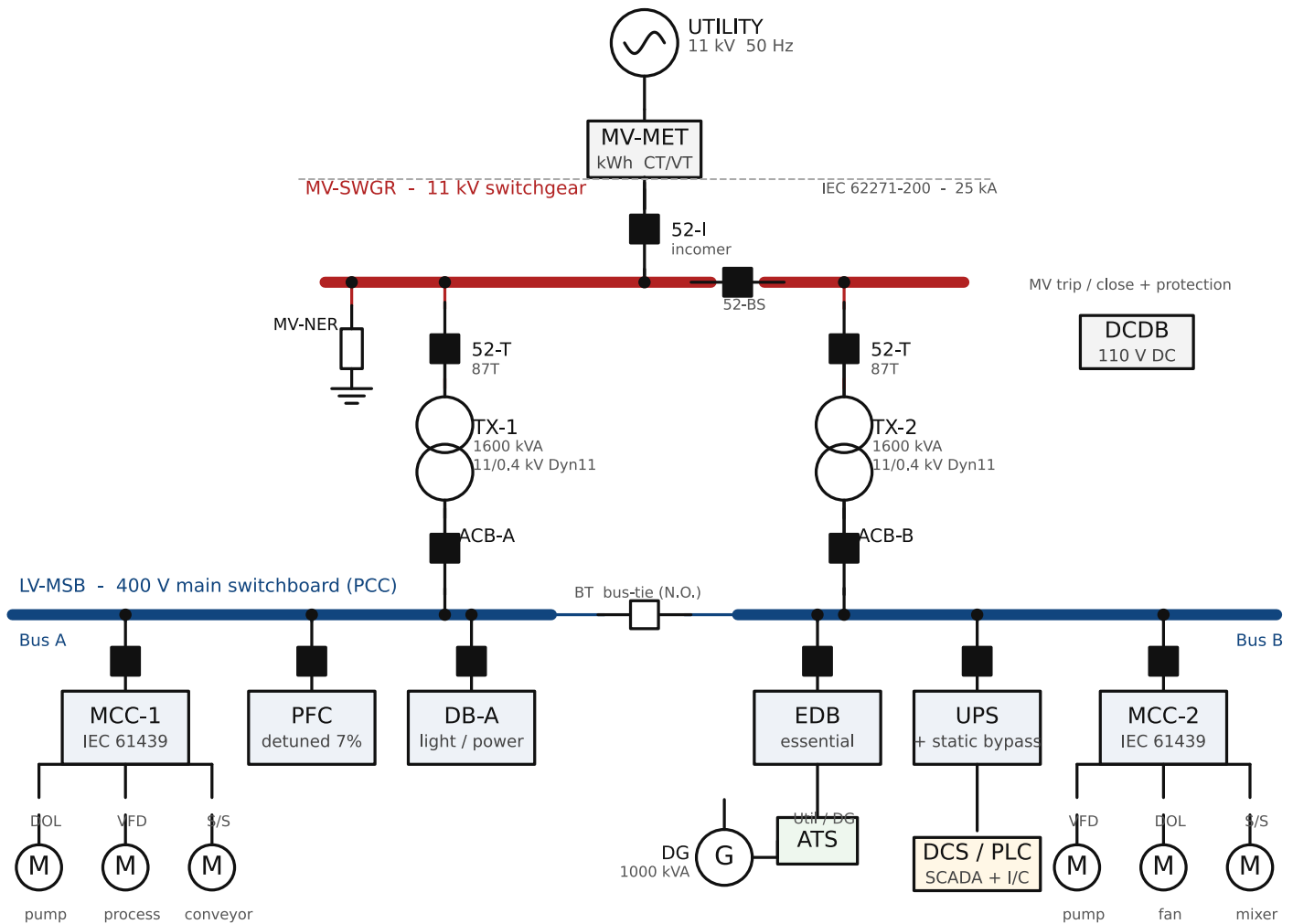
The two failure modes you are looking for:

- **Under-rating** — cables and busbars that overheat (fire risk), switchgear that cannot interrupt a fault (catastrophic failure), protection that mis-coordinates or nuisance-trips, motors that stall, process voltage too low to run.
- **Over-rating** — wasted capital tied up in oversized transformers, switchgear frames, cable and gensets the plant will never use; this matters most when the client is evaluating whether to add load or replace equipment.

The indicative design basis used throughout Part 2:

Parameter	Indicative value	Source
Plant maximum demand	~2.0 MW	Basis of design
Power factor (uncorrected)	0.85	Basis of design
Apparent power	~2.35 MVA	2.0 / 0.85
Transformer configuration	2 × 1600 kVA, Dyn11, Z ≈ 6 %	Basis of design
LV voltage	400 V, 3-phase, 50 Hz	IEC
MV voltage	11 kV	Utility connection
LV FLC per transformer	≈ 2 310 A	§c-flc
LV prospective I _{sc} (single tx)	≈ 38.5 kA	§c-shortcircuit

Embed the master SLD here so every sub-section below can refer back to the equipment being sized.



MASTER SINGLE-LINE DIAGRAM - ~2 MW MV/LV PROCESS PLANT

11 kV / 400 V - 50 Hz - 2 x 1600 kVA (Dyn11, Z~6%) - IEC 60617 - indicative

Fig 2-0 — Master SLD for the indicative ~2 MW MV/LV process plant (IEC 60617 symbols). All ratings indicative; confirm against as-built drawings and nameplates on site.

i Constants used throughout Part 2: $\sqrt{3} = 1.732$ | LV line voltage V = 400 V | MV line voltage = 11 kV | 50 Hz. Every numeric result is *indicative and tentative* — it illustrates method and typical practice, not a substitute for project-specific load lists, short-circuit/load-flow studies, and the governing standards.

★ If you master only four, master these: On most SPOF assessments, four calculations carry the most evidential weight: (1) Transformer sizing / N-1 check (§ 2) — confirms the fundamental redundancy claim; (2) Full-load current (FLC) (§ 3) — the basis for every breaker, cable, and relay sizing check; (3) Short-circuit level (§ 4) — verifies that switchgear can interrupt faults safely; (4) Protection coordination (§ 9) — confirms that faults isolate the minimum zone. The remaining sections are Reference — important for completeness but less frequently the direct driver of a P1 FMECA finding.

1 — Load & Demand Estimate Core

Goal: confirm the ~2 MW plant demand and convert it to apparent power (MVA) — the anchor point for every sizing calculation that follows.

Method

Assumptions (indicative): sum of connected loads (nameplate) ≈ 2.7 MW; overall demand/diversity factor ≈ 0.75; overall power factor before correction = 0.85.

WORKED EXAMPLE — DEMAND AND APPARENT POWER

Step 1 — Active demand:

$$P_{\text{demand}} = \Sigma P_{\text{connected}} \times \text{demand factor}$$
$$P_{\text{demand}} = 2.7 \text{ MW} \times 0.75 = \mathbf{2.025 \text{ MW}} \approx \mathbf{2.0 \text{ MW}}$$

Step 2 — Apparent power:

$$S = P / \text{pf}$$
$$S = 2.0 \text{ MW} / 0.85 = \mathbf{2.35 \text{ MVA}}$$

Step 3 — Installed transformer base (with diversity/spare margin):

$$S_{\text{installed}} = 2.35 \times 1.20 \text{ to } 1.25 \approx 2.8\text{--}3.0 \text{ MVA}$$

This drives the 2 × 1600 kVA selection (3.2 MVA installed).

Result: maximum demand ≈ **2.0 MW / 2.35 MVA** at 0.85 pf. All values indicative.

✓ **Field tip — What to capture:** obtain the latest load list or motor schedule from the client. If none exists, read each MCC ammeter at peak shift and multiply by $\sqrt{3} \times V \times \text{pf}$ to back-calculate kW per feeder. Sum the branch demands to build your own diversity estimate.

Assessment interpretation

If as-found...	Likely meaning	Action
Demand > transformer capacity	Plant has grown since original design	Flag overload risk; recommend load-flow study
Demand < 50 % capacity for years	Oversized original design or load reduction	Check transformer no-load losses; note in report
No load list available	Documentation gap	Raise as finding; capture ammeter readings

★ **The demand estimate sets the size of the entire electrical plant. Every other calculation in Part 2 is downstream of this number. If the load list is wrong or absent, all downstream ratings are unverifiable.**

2 — Transformer Sizing & N-1 Check Core

Goal: verify the 2×1600 kVA selection and confirm that one transformer (TX-1 or TX-2) can carry the essential load via the bus-tie (BT) when the other is lost.

Normal loading

WORKED EXAMPLE — NORMAL PER-UNIT LOADING

With 2.35 MVA shared equally across both transformers:

$$S_{\text{per tx}} = 2.35 \text{ MVA} / 2 = 1.175 \text{ MVA}$$

$$\text{Loading}_{\text{normal}} = 1.175 / 1.6 = \mathbf{0.73 \text{ pu}} \text{ } (\approx 73 \text{ \%})$$

Each transformer runs at roughly 73 % of nameplate — a comfortable margin.

N-1 check (one transformer lost)

WORKED EXAMPLE — N-1 LOADING WITH ESSENTIAL-LOAD SHEDDING

Worst case (no shedding):

$$\text{Loading}_{\text{N-1, full}} = 2.35 / 1.6 = \mathbf{1.47 \text{ pu}} \text{ } (\mathbf{147 \text{ \%}}) \text{ — not acceptable for continuous operation.}$$

With non-essential load shed (essential fraction $\approx 55 \text{ \%}$, $\approx 0.65 \text{ MW}$ / $1.0\text{--}1.3 \text{ MVA}$):

$$S_{\text{essential}} = 0.55 \times 2.35 = \mathbf{1.29 \text{ MVA}} \text{ } (\approx 0.65 \text{ MW active at pf } 0.8 \rightarrow 0.81 \text{ MVA apparent running; includes } 1.25 \times \text{step-load factor} \rightarrow 1.01 \text{ MVA for DG sizing — see } \text{\textcolor{blue}{§c-genset}})$$

$$\text{Loading}_{\text{N-1, essential}} = 1.29 / 1.6 = \mathbf{0.81 \text{ pu}} \text{ } (\approx \mathbf{81 \text{ \%}}) \text{ — within continuous rating. } \checkmark$$

Result: the 2×1600 kVA + bus-tie design is N-1 secure for essential load ($\sim 55 \text{ \%}$ of total, $\approx 0.65 \text{ MW}$ / 1.29 MVA), provided non-essential load is shed automatically. This is the load-shedding requirement on LV-MSB . The same 0.65 MW essential-load basis is used for DG sizing in §c-genset.

 **Assessment check:** confirm the bus-tie (BT) auto-transfer scheme is functional and the load-shedding sequence is current. A bus-tie that cannot close automatically, or load shedding that has never been tested, makes the N-1 claim invalid.

What to check on site

Progress: **0 / 5**

- ☐ Transformer nameplates: confirm kVA, voltage ratio, vector group, Z %, off-load tap positions and current tap setting.
- ☐ Transformer loading (incomer ammeter or SCADA trend): compare against 73 % normal / 81 % N-1 limits.
- ☐ Bus-tie (BT): confirm auto-transfer is enabled and last test date.
- ☐ Load-shedding relay/PMS: confirm essential/non-essential split list is documented and relays are active.
- ☐ Transformer cooling: ONAN or ONAF? Are fans/pumps serviceable?

On the forms →

- (a) *Document inventory:* populate the Transformer record row — enter nameplate kVA, Z %, vector group, tap position, and calculate loading % from the incomer ammeter reading.
- (b) *AS-FOUND site data required:* transformer nameplate (kVA, Z %, voltage ratio, vector group, tap range and current position); incomer ACB ammeter reading (A) or SCADA kVA trend at peak load; bus-tie status (normally-open confirmed Y/N); load-shedding relay type and last-test date; cooling type (ONAN/ONAF) and fan status.

 **Two transformers are only as good as their bus-tie and load-shedding scheme. Verify all three on site before claiming N-1 security.**

3 — Full-Load Current (LV & MV) Core

Goal: calculate the 400 V full-load current (FLC) of a 1600 kVA transformer — the basis for the incomer ACB frame size and busbar rating — and the corresponding MV primary current.

Formula

$$I_{fl} = S / (\sqrt{3} \times V)$$

WORKED EXAMPLE — LV FLC (1600 KVA AT 400 V)

$$I_{fl} = 1\,600\,000\text{ VA} / (1.732 \times 400\text{ V})$$

$$I_{fl} = 1\,600\,000 / 692.8 = \mathbf{2\,310\text{ A}}$$

This sets the LV-MSB incomer ACB frame ($\geq 2\,500\text{ A}$) and the $\approx 2\,500\text{--}3\,200\text{ A}$ busbar rating.

WORKED EXAMPLE — MV FLC (1600 KVA AT 11 KV)

$$I_{fl,MV} = 1\,600\,000\text{ VA} / (1.732 \times 11\,000\text{ V})$$

$$I_{fl,MV} = 1\,600\,000 / 19\,052 = \mathbf{84\text{ A}}$$

This sizes the MV transformer-feeder breakers [52-T1] / [52-T2] in MV-SWGR and the CT primary ratio for the feeder protection relays.

✓ **Quick sanity check on site:** read the incomer ammeter or clamp-meter on the LV bus incomer cable. Divide by 2 310 A to get per-unit loading. Values consistently above 0.85 pu should be investigated.

General 3-phase FLC formula

For any three-phase equipment rated S (kVA) at line voltage V (V):

$$I = S \times 1000 / (\sqrt{3} \times V) \quad [\text{A}]$$

Rating (kVA)	Voltage	FLC (A) indicative
1 600	400 V	2 310
1 600	11 kV	84
1 250	400 V	1 804
1 000	400 V	1 443
630	400 V	909

i **On the forms →**
(a) *Document inventory:* populate the LV-MSB Incomer row — record ACB frame rating (A), current trip-unit setting (A), and measured FLC from ammeter (A); compare with calculated FLC to flag under- or over-rating.
(b) *AS-FOUND site data required:* transformer nameplate kVA; LV line voltage (measure with meter, do not rely on drawings); incomer ACB frame size and trip-unit In setting; busbar rated current (from board label or drawings); MV CT primary ratio on transformer feeder.

★ **LV FLC $\approx 2\,310\text{ A}$ per 1 600 kVA transformer at 400 V. Every conductor, breaker, and busbar on the LV incomer must be rated above this figure, with appropriate derating for installation conditions.**

4 — Short-Circuit Current & Switchgear Withstand Core

Goal: calculate the LV prospective short-circuit current from the transformer impedance, check the bus-tie parallel case, and verify that the switchgear breaking capacity and busbar bracing are adequate.

Formula

$$I_{sc} = I_{fl} / Z_{(pu)}$$

Assumption: $Z = 6\%$; infinite upstream MV source (conservative — ignores MV source impedance, so the real value is slightly lower).

WORKED EXAMPLE — SINGLE-TRANSFORMER FAULT CONTRIBUTION AT 400 V

$$I_{sc} = 2\,310\text{ A} / 0.06 = 38\,500\text{ A} \approx 38.5\text{ kA}$$

This is the prospective symmetrical short-circuit current at the LV-MSB busbars from one transformer alone.

WORKED EXAMPLE — BOTH TRANSFORMERS IN PARALLEL (BUS-TIE CLOSED)

$$I_{sc,parallel} \approx 2 \times 38.5\text{ kA} = 77\text{ kA} \text{ (upper bound; ignores source and cable impedance)}$$

This is the bounding case when the bus-tie is closed and both transformers feed the same bus simultaneously.

Why the bus-tie is normally open

Operating the bus-tie (BT) normally-open keeps each bus half fed from only one transformer, limiting the fault duty on each section to $\approx 38.5\text{ kA}$. Selecting the LV-MSB and MCC bus at **50–65 kA** covers this single-transformer fault with margin. If the bus-tie is closed continuously (or accidentally left closed), the fault duty doubles to $\approx 77\text{ kA}$ and may exceed the board rating — a safety-critical finding.

 **Critical assessment check:** if you find the bus-tie (BT) closed under normal running conditions, verify the switchboard fault rating immediately. A 50 kA-rated board exposed to 77 kA cannot safely interrupt a fault — this is a Category A finding requiring immediate action.

Site verification checklist

Progress: 0 / 5

- ☐ Confirm BT is normally-open and electrically interlocked (both incomers cannot close simultaneously onto the same bus).
- ☐ Read ACB and MCCB nameplates: confirm rated breaking capacity (I_{cu}) $\geq 38.5\text{ kA}$ (single-tx scenario).
- ☐ Check busbar bracing rating on LV-MSB label or type-test certificate: should be $\geq 50\text{ kA} / 1\text{ s}$ (or equivalent peak).
- ☐ Confirm a short-circuit/fault-level study exists and is dated within ≤ 5 years (or after any network change).
- ☐ Check that MCCs fed from LV-MSB are rated for the same fault level.

On the forms →

(a) *Document inventory / recommendations register:* populate the Short-circuit Assessment row — record calculated I_{sc} (kA), rated I_{cu} on each ACB/MCCB (kA from nameplate), busbar bracing (kA from label), and the result (adequate / marginal / inadequate).

(b) *AS-FOUND site data required:* transformer nameplate $Z\%$ (each unit); ACB and MCCB breaking capacity (I_{cu}) from nameplate or door label (kA); LV-MSB busbar short-time withstand rating (kA / duration) from board label or type-test certificate; bus-tie status and interlock scheme description; date of last formal fault-level study.

-  **38.5 kA from one transformer; up to 77 kA with both in parallel. The normally-open bus-tie is a protective operating rule, not just a design convention — verify the interlock is active and the board rating has not been quietly exceeded.**

5 — Motor FLC, Starting Methods & LV/MV Feed Threshold Reference

Goal: calculate the full-load current and starting current of a representative LV motor, explain why large motors use soft-start or VFD, and identify the LV-to-MV feed decision threshold.

Full-load current formula (motor)

$$I_{fl} = P / (\sqrt{3} \times V \times pf \times \eta)$$

WORKED EXAMPLE — 110 KW LV MOTOR FLC AND DOL STARTING CURRENT

Given: P = 110 kW, V = 400 V, pf = 0.86, η = 0.95.

FLC:

$$I_{fl} = 110\,000 / (1.732 \times 400 \times 0.86 \times 0.95)$$

$$I_{fl} = 110\,000 / 566 = \mathbf{194\,A}$$

DOL starting current ($\approx 6.5 \times \text{FLC}$):

$$I_{start} = 6.5 \times 194 = \mathbf{1\,260\,A}$$

A 1 260 A inrush causes a voltage dip on the bus, mechanical/electrical stress on the motor windings, and possible nuisance trips on other feeders. A soft-starter limits this to $\approx 3 \times \text{FLC}$ ($\approx 580\,A$); a VFD limits it to near FLC *and* provides speed control.

Starting method selection guide

Motor size (LV)	Typical starting method	Approx. start current	Notes
Up to $\approx 7.5\,kW$	Direct-on-line (DOL)	$\approx 6\text{--}7 \times \text{FLC}$	Bus impact negligible at this size
$\approx 7.5\text{--}55\,kW$	DOL / star-delta / soft-starter	Star-delta: $\approx 2 \times \text{FLC}$	Star-delta suits constant-torque loads; soft-starter preferred for pumps
$\approx 55\text{--}200\,kW$	Soft-starter or VFD	Soft-starter: $\approx 3 \times \text{FLC}$	Soft-start/VFD reduces bus dip and extends motor life
Speed-controlled loads	VFD (any size)	$\approx 1 \times \text{FLC}$	Energy saving on pumps/fans (affinity laws)

LV-to-MV feed threshold

Individual motors larger than $\approx 150\text{--}200\,kW$ are often more economically fed at **MV (3.3 / 6.6 kV)** via a dedicated MV motor starter rather than the 400 V MCC. At 200 kW, the LV current reaches $\approx 350\,A$ — cable and starter costs rise steeply. For a $\approx 2\,MW$ plant the largest drives may sit near this boundary; check the motor schedule.

✓ **Assessment tip:** if you find a large motor ($> 150\,kW$) on the LV MCC with no soft-start or VFD, check the voltage-dip impact on adjacent motors and instruments. Also check whether the MCC starter contactor and cable are correctly rated — the starting current can be 6–7 times the FLC nameplate.

★ **For a 110 kW motor at 400 V: FLC $\approx 194\,A$, DOL inrush $\approx 1\,260\,A$. Motors above $\approx 55\,kW$ should have soft-start or VFD; above $\approx 150\text{--}200\,kW$, consider MV feed.**

6 — Cable Sizing & Voltage Drop Reference

Goal: size a motor feeder cable and verify the voltage drop is within acceptable limits (typically $\leq 5\%$ for motor feeders).

Four-step method

WORKED EXAMPLE — 110 KW MOTOR FEEDER (FROM §C-MOTORS)

Step 1 — Design (base) current:

$$I_B = \text{motor FLC} = 194 \text{ A}$$

Step 2 — Derating factors (IEC 60364):

$$k_{\text{grouping}} = 0.80 \text{ (cables grouped on tray)}$$

$$k_{\text{temp}} = 0.94 \text{ (40 °C ambient)}$$

$$k_{\text{total}} = 0.80 \times 0.94 = 0.752$$

$$I_2(\text{req}) = I_B / k_{\text{total}} = 194 / 0.752 = 258 \text{ A}$$

Step 3 — Cable selection:

Select **95 mm² Cu XLPE** with tabulated rating $\approx 264\text{--}300$ A on tray (per manufacturer / IEC tables). $I_2 \geq 258$ A. ✓

Step 4 — Voltage-drop check:

Run length $L = 120$ m; for 95 mm² Cu: $R \approx 0.23 \text{ } \Omega/\text{km}$, $X \approx 0.08 \text{ } \Omega/\text{km}$; $\text{pf} = 0.86 \rightarrow \cos\phi = 0.86$, $\sin\phi = 0.51$.

$$\Delta V = \sqrt{3} \times I \times (R \cdot \cos\phi + X \cdot \sin\phi) \times L$$

$$\Delta V = 1.732 \times 194 \times (0.23 \times 0.86 + 0.08 \times 0.51) \times 0.120$$

$$\Delta V = 1.732 \times 194 \times (0.1978 + 0.0408) \times 0.120$$

$$\Delta V = 1.732 \times 194 \times 0.2386 \times 0.120 = \mathbf{9.6 \text{ V}}$$

$$\%\Delta V = 9.6 / 400 \times 100 = \mathbf{2.4\%}$$

Within the $\leq 5\%$ limit (and $\leq 3\%$ is comfortable). **Selection accepted.**

Assessment guidance

In the field you cannot always measure cable resistance directly. Use the cable size stamped on the gland tag or tray label, the feeder length from the cable schedule, and the above formula to estimate voltage drop. Then compare with the measured voltage at the motor terminal (meter at the motor DB) versus the LV-MSB bus voltage.

 **Undersized cable signs:** discoloured insulation, hot gland or termination, cable tray showing signs of heat distortion, motor terminal voltage significantly below bus voltage. Any of these should be flagged immediately.

Common derating factors (indicative — confirm against project installation tables)

Factor	Condition	Indicative k value
Grouping (k_g)	3 cables on tray	0.90
Grouping (k_g)	6 cables on tray	0.80
Grouping (k_g)	9 cables on tray	0.73
Ambient (k_t)	35 °C	0.97
Ambient (k_t)	40 °C	0.94
Ambient (k_t)	45 °C	0.91
Burial depth	Standard 0.7 m direct burial	1.00
Soil thermal resistivity	High (dry/sandy)	0.80–0.85

 **Cable sizing = thermal capacity (derating) + voltage drop. Both checks are mandatory. A cable that is thermally adequate but drops too much voltage leaves motors unable to start or run at rated torque.**

7 — Power-Factor Correction (PFC) Reference

Goal: size the automatic capacitor bank (PFC) to raise the plant power factor from 0.85 to 0.95, and understand why detuning reactors are mandatory when VFDs are present.

Formula

$$Q_c = P \times (\tan\phi_1 - \tan\phi_2)$$

WORKED EXAMPLE — PFC FROM 0.85 TO 0.95 (P = 2.0 MW)

$$\text{pf}_1 = 0.85 \rightarrow \phi_1 = 31.8^\circ, \tan\phi_1 = 0.620$$

$$\text{pf}_2 = 0.95 \rightarrow \phi_2 = 18.2^\circ, \tan\phi_2 = 0.329$$

$$Q_c = 2\,000\text{ kW} \times (0.620 - 0.329) = 2\,000 \times 0.291 = \mathbf{582\text{ kVAr}}$$

Select a standard $\approx \mathbf{600\text{ kVAr}}$ automatic bank in steps (e.g. $6 \times 100\text{ kVAr}$).

Correcting to 0.95 also reduces apparent power: $S = 2.0 / 0.95 = 2.11\text{ MVA}$ (from 2.35 MVA), relieving transformer and cable loading by $\approx 10\%$.

Detuning reactors

VFDs and non-linear loads inject harmonic currents (5th, 7th, 11th... harmonics). An un-detuned capacitor bank presents a very low impedance at harmonic frequencies and can resonate with the supply inductance, causing catastrophic capacitor failure and high harmonic distortion. For this plant, with multiple VFDs, the bank must be fitted with **7 % detuning reactors** (series inductors that raise the bank's resonant frequency above the 5th harmonic).

 **Assessment check:** if you find a capacitor bank without detuning reactors on a plant with VFDs, this is a reliability finding. Check for capacitor over-temperature, bulging, or blown fuses in the bank — signs that resonance has already caused damage.

What to record on site

Progress: 0 / 5

- ☐ Confirm PFC bank kVAr rating matches $\approx 600\text{ kVAr}$ ($\pm 20\%$ acceptable).
- ☐ Confirm detuning reactors (7 % or 14 %) are present and intact.
- ☐ Read power-factor meter at LV-MSB (or from the SCADA historian trend): should be ≥ 0.93 with bank in service.
- ☐ Check power-factor penalty charges on the utility invoice — evidence of consistently low pf even with the bank in service suggests under-sizing or damaged capacitors.
- ☐ Note the automatic controller step count and switching threshold settings.

 **$\sim 580\text{ kVAr}$ of correction raises pf from 0.85 to 0.95, reducing apparent power by $\approx 10\%$. On a VFD-heavy plant, detuning reactors are not optional — they are the difference between a working bank and an exploding one.**

8 — Standby Generator Sizing Rationale Reference

Goal: justify the indicative DG rating ($\approx 1\,000\text{ kVA}$) on an essential-load basis and explain the constraints that really drive the size.

Sizing calculation

WORKED EXAMPLE — ESSENTIAL-LOAD GENERATOR SIZING

Assumptions (indicative): the DG covers essential loads only (not full 2 MW plant); essential active load $\approx 0.65\text{ MW}$ at pf 0.8.

Running (steady) apparent demand:

$$S_{\text{essential}} = P / \text{pf} = 0.65\text{ MW} / 0.8 = 0.81\text{ MVA}$$

Step-load / motor-start allowance (the largest essential motor starting against the generator's limited transient capacity):

$$S_{\text{DG}} \geq S_{\text{essential}} \times 1.25 = 0.81 \times 1.25 = \mathbf{1.01\text{ MVA}}$$

Result: select a $\approx 1\,000\text{ kVA}$ standby DG (indicative band: 800–1250 kVA per [equipment schedule](#); 1000 kVA is the worked point value) — consistent with the N-1 essential-load basis of $0.65\text{ MW} / \sim 1.29\text{ MVA}$ in [§c-transformer](#). All values indicative.

The governing constraint: voltage dip on motor start

For a generator, the largest single motor start is often the *governing* constraint, not the steady running load. A generator has a much higher source impedance than the utility, so a large DOL motor start causes a severe transient voltage dip that can stall other motors already running. The remedy is:

- Soft-start or VFD all large essential motors (reduces starting kVA to $\approx 1.0\text{--}1.5 \times \text{FLC}$).
- Sequence essential motor starts after DG reaches rated voltage/frequency (ATS timer).
- Size the DG AVR for fast voltage recovery ($< 2\text{ s}$ to within $\pm 3\%$).

✓ **De-rating reminders:** genset ratings are typically at ISO conditions ($25\text{ }^{\circ}\text{C}$, sea level). For hot climates or elevated sites, apply the manufacturer's de-rating factor (commonly -1% per $5\text{ }^{\circ}\text{C}$ above $25\text{ }^{\circ}\text{C}$, -1% per 100 m above sea level). A $1\,000\text{ kVA}$ unit at $45\text{ }^{\circ}\text{C}$ and 300 m may deliver only $\approx 940\text{ kVA}$ — check the site data sheet.

Site assessment checklist

Progress: 0 / 7

- ☐ Confirm DG nameplate kVA rating and compare with essential-load estimate.
- ☐ Check de-rating data sheet for site altitude and ambient temperature.
- ☐ Confirm ATS (ATS) transfer time setting and load-shed sequence at DG start.
- ☐ Check fuel day-tank level and fuel consumption log (evidence of recent tests).
- ☐ Review last monthly/annual test run report: confirm the DG accepted the full essential load at rated voltage and frequency.
- ☐ Confirm AVR and governor are within calibration.
- ☐ Check battery/charger for engine starting: voltage, last replacement date.

★ **The genset is not sized for full plant load — only the essential fraction ($\approx 55\%$, $\approx 1\,000\text{ kVA}$). The motor-start transient is usually the binding constraint, not the steady-state load. A genset that has never been tested under full essential load cannot be claimed as reliable backup.**

9 — Reading & Validating a Load-Flow Study Reference

Goal: confirm the voltage profile, power flows and losses at peak load — that every bus stays within $\pm 5\%$ of nominal and no branch is thermally overloaded.

Transformer voltage regulation (hand check)

WORKED EXAMPLE — VOLTAGE REGULATION AT 73 % LOADING

Split $Z = 6\%$ into $R \approx 1.2\%$, $X \approx 5.9\%$. At normal $\approx 73\%$ loading (§c-transformer), $pf \approx 0.90$ ($\cos\phi = 0.90$, $\sin\phi = 0.436$):

$$\Delta V\% \approx \text{load}_{pu} \times (R\% \cdot \cos\phi + X\% \cdot \sin\phi)$$

$$\Delta V\% \approx 0.73 \times (1.2 \times 0.90 + 5.9 \times 0.436)$$

$$\Delta V\% \approx 0.73 \times (1.08 + 2.57) = 0.73 \times 3.65 = 2.7\%$$

With a +2.5 % off-load tap, the LV-MSB bus sits at $\approx 400\text{ V}$ at full load. Adding the worst feeder drop of 2.4 % (§c-cables) puts the motor terminal at $\approx 97\text{--}98\%$ of nominal — inside the $\pm 5\%$ limit. ✓

Indicative transformer losses

Transformer copper (load) loss scales with loading². For a unit with full-load copper loss $\approx 17\text{ kW}$ at 73 % loading:

$$P_{Cu} = 0.73^2 \times 17\text{ kW} = 0.533 \times 17 \approx 9\text{ kW} \text{ per transformer at normal load.}$$

Iron (no-load) loss is constant; for a 1 600 kVA unit, typically $\approx 2\text{--}3\text{ kW}$.

How to read a load-flow report

Report output	What to check	Acceptable limit
Bus voltage (pu or V)	All buses at peak load	$\pm 5\%$ of nominal (IEC 60364 / utility spec)
Transformer loading (%)	Each unit in normal / N-1	$\leq 100\%$ continuous; $\leq 120\%$ emergency (IEC 60076 short-time)
Feeder branch loading (%)	All cables and busbars	$\leq 100\%$ of rated ampacity
Reactive flow (MVar)	Power factor at PCC	Target ≥ 0.95 with PFC in service
Losses (kW)	Transformer Cu + Fe losses	Flag if $> 2\%$ of transformer rating

✓ **If no load-flow study exists:** this is itself a finding. A plant with no documented power-system study has no formal basis for its tap settings, cable ratings, or transformer loading limits. Recommend a load-flow study as part of your remediation programme.

i On the forms →

- (a) *Document inventory:* record date and revision of the load-flow study; confirm worst-case bus voltage (pu), peak transformer loading (%), and whether any branch exceeded rated ampacity. If no study exists, raise as a documentation gap.
- (b) *AS-FOUND site data required:* latest load-flow study report (or SCADA trend data for bus voltages and feeder amps at peak); transformer tap position (from nameplate tap switch / HV tap changer setting); measured LV bus voltage (V) at peak shift; power-factor reading at main incomer; any voltage complaints from process.

★ **A load-flow study validates the sizing from §1–§8 under realistic load conditions. Without it you cannot formally claim that voltages, flows, and taps are correct. Its absence is a finding, not just a gap.**

10 — Protection Coordination & Selectivity Core

Goal: set protective devices so the device *closest to the fault trips first*, isolating the smallest possible zone (selectivity/discrimination), while still clearing every fault within the required time.

The protection chain

Tracing upstream from a fault on a motor feeder:

Motor MCCB → MCC busbar → LV outgoing MCCB → LV incomer ACB (LV-MSB) → MV transformer-feeder relay [52-T] (MV-SWGR) → MV incomer relay [52-I]

Time grading

WORKED EXAMPLE — TIME-GRADED RELAY SETTINGS (INDICATIVE)

Between two graded relays, leave a margin Δt for breaker opening time + relay overshoot + CT error + safety ≈ 0.3 s:

$$t_{\text{upstream}} = t_{\text{downstream}} + \Delta t$$

Indicative ladder (from feeder to incomer):

Device / relay	Setting (indicative)	Notes
Motor feeder MCCB	≈ 0.05 s (instantaneous/short-time)	Closest to fault; fastest clear
LV incomer ACB short-time delay	$t = 0.20$ s	$0.05 + \Delta t \approx 0.20$ s
MV feeder relay [52-T] (IDMT 51)	$t = 0.50$ s	$0.20 + 0.30 = 0.50$ s
MV incomer relay [52-I]	$t = 0.80$ s	$0.50 + 0.30 = 0.80$ s

WORKED EXAMPLE — MV RELAY PICKUP SETTING (CT 100/1 ON 84 A PRIMARY FLC)

$$I_{\text{pickup}} = 1.25 \times 84 \text{ A} = 105 \text{ A} \rightarrow 1.05 \times \text{CT (set } \approx 1.05 \text{ on a 100/1 CT)}$$

Transformer internal faults are cleared instantaneously by the **87T differential relay**, independent of the time grading above.

Assessment: what to verify

Progress: 0 / 6

- ☐ Request the coordination study / time-current coordination curves (TCC). If none exists, raise as a finding.
- ☐ Confirm relay settings in the IED configuration files match the approved coordination study values.
- ☐ Confirm each protective device (ACB, MCCB, relay) has been tested within the required period (typically every 1–3 years for MV relays; 5 years for LV trip units).
- ☐ Check that the 87T differential relay (if fitted) is in service and its CTs are correctly wired and matched.
- ☐ Confirm IDMT curve type and time-multiplier setting are consistent across all upstream relays.
- ☐ Verify zone-selective interlocking (ZSI) or bus-differential is active (if the design relies on it for arc-flash mitigation).

 **Loss of selectivity** means an upstream breaker trips for a downstream fault, blacking out a much larger zone than the faulty feeder. This is a high-impact finding — it multiplies the production loss from any fault. It is also invisible until a fault occurs unless the coordination curves are checked.

 **On the forms** →
(a) *Document inventory / recommendations register*: record whether a coordination study (TCC) exists and its date; note the LV incomer ACB short-time setting (s) and MV feeder relay time-dial / time-multiplier setting; flag any device whose as-found setting does not match the study.
(b) *AS-FOUND site data required*: relay settings reports or IED configuration printouts for each MV feeder relay, MV incomer relay, and transformer differential relay; ACB trip-unit settings (Long-time, Short-time, Instantaneous bands and delays in amps and seconds); CT ratios on each MV feeder; last relay/trip-unit test date and result.

 **A coordination study tells you that relay A trips before relay B for a fault between them. Without documented, verified settings, you cannot make that claim — and the next fault will reveal the gap.**

11 — Arc-Flash / Incident Energy (IEEE 1584) & the Clearing-Time Lever Reference

Goal: estimate the incident energy and arc-flash boundary at the LV-MSB , label the gear and select PPE — and show why *clearing time* is the dominant lever for reducing arc-flash hazard.

 Arc-flash is a **life-safety calculation**. The indicative values below illustrate method only. Actual PPE selection, labelling, and arc mitigation **must** be based on a full IEEE 1584-2018 study with site-specific data. Never rely on these figures for live-work decisions.

Inputs for the LV-MSB

Input parameter	Indicative value	Source
Bolted fault current I_{bf}	38.5 kA	§c-shortcircuit
System voltage	400 V	Basis of design
Enclosure type	Switchboard (MCC-type)	IEEE 1584-2018
Working distance D	600 mm	IEEE 1584 default LV switchboard
Upstream device clearing time t	0.20 s (LV incomer short-time delay)	§c-protection

WORKED EXAMPLE — INDICATIVE ARC-FLASH CALCULATION AT LV-MSB (400 V)

Step 1 — Arcing current (LV arcing/bolted ratio ≈ 0.58 per IEEE 1584):

$I_a \approx 0.58 \times 38.5 \text{ kA} \approx \mathbf{22 \text{ kA}}$

Step 2 — Incident energy at D = 600 mm, t = 0.20 s (indicative from IEEE 1584-2018 empirical model):

$E \approx \mathbf{10 \text{ cal/cm}^2}$ (assumed illustrative value — not computed from the full six-parameter IEEE 1584-2018 model; use recognised power-system software for labelling studies)

(This is an illustration only — the IEEE 1584-2018 model uses six parameters and electrode configuration corrections that cannot be reduced to a simple formula here.)

Step 3 — Arc-flash boundary (distance where E falls to $E_B = 1.2 \text{ cal/cm}^2$, inverse-square scaling):

$AFB \approx D \times \sqrt{(E / E_B)}$

$AFB \approx 0.6 \text{ m} \times \sqrt{(10 / 1.2)} = 0.6 \times 2.89 \approx \mathbf{1.7 \text{ m}}$

Result: $\approx 10 \text{ cal/cm}^2$ at 600 mm working distance. Arc-rated PPE must exceed this (e.g. a 12–25 cal/cm² arc-rated kit); arc-flash boundary $\approx 1.7 \text{ m}$.

The clearing-time lever

Incident energy is approximately proportional to clearing time. Halving the clearing time roughly halves the energy. This is the cheapest and most effective arc-mitigation tool:

Clearing time (s)	Indicative E (cal/cm²)	Mechanism
0.20	≈ 10	Normal LV incomer short-time delay
0.05	≈ 2.5	Arc-reduction maintenance switch / ZSI / arc-flash relay
0.02	≈ 1.0	High-speed arc-flash detector (optical) — near threshold level

Adding an **arc-flash reduction maintenance switch**, **zone-selective interlocking (ZSI)**, or an optical **arc-flash relay** to achieve t $\approx 0.05 \text{ s}$ reduces E to $\approx 2.5 \text{ cal/cm}^2$ and shrinks the boundary substantially — at some tension with the selectivity grading of §c-protection, which is why bus differential / ZSI is the preferred solution (it achieves fast clearing without losing selectivity).

Assessment: what to check

Progress: 0 / 5

- ☐ Confirm an IEEE 1584-compliant arc-flash study exists and is dated (≤ 5 years or since last network change).
- ☐ Check that arc-flash labels are posted on all switchgear panels including LV-MSB , MCCs, and MV-SWGR .
- ☐ Confirm PPE kits available on site match or exceed the arc-flash incident energy class shown on the labels.
- ☐ Verify the maintenance switch (if fitted) is functional and its activation procedure is documented in the site safety rules.

☐ Check whether ZSI or bus-differential is active and tested.

i On the forms →

(a) *Document inventory / recommendations register*: record whether an IEEE 1584 arc-flash study exists and its date; note incident energy (cal/cm²) and arc-flash boundary from the study for each assessed panel; record PPE category required; flag missing or outdated labels.

(b) *AS-FOUND site data required*: bolted fault current (from short-circuit study, §c-shortcircuit); upstream device clearing time (relay setting or ACB short-time delay in seconds); electrode/enclosure configuration (from drawings or physical inspection); working distance (measure or per IEEE 1584 default); arc-flash label (photograph it — capture cal/cm², boundary, and PPE category); PPE available on site (category rating on kit label).

★ Arc-flash hazard is driven by fault current and clearing time. The clearing time is the most cost-effective lever. A maintenance switch or ZSI that cuts clearing from 0.20 s to 0.05 s reduces incident energy by 75 % — without replacing any switchgear.

12 — Earthing & Lightning-Protection Assessment Reference

Goal: verify the MV neutral earthing resistor (**MV-NER**), the LV earth system, the earth-conductor cross-sections, and the lightning-protection classification (IEC 62305).

MV neutral earthing resistor (NER)

The MV system is **resistance-earthed** to limit earth-fault current to ≈ 300 A, so faults are detectable but arc energy and damage are limited.

WORKED EXAMPLE — NER RESISTANCE (11 KV, 300 A EARTH FAULT)

$$R_{\text{NER}} = (V_{\text{LL}} / \sqrt{3}) / I_{\text{ef}}$$

$$R_{\text{NER}} = (11\,000 / 1.732) / 300 = 6\,351 / 300 \approx 21\, \Omega$$

The NER must be rated for its short-time dissipation: $P = I^2 \times R = 300^2 \times 21 = 90\,000 \times 21 = 1.89$ MW briefly (for the fault-clearing duration, e.g. 10 s).

LV earthing (TN-S)

The 400 V system is **TN-S, solidly earthed**. An LV earth fault is effectively a line fault and is cleared quickly by the feeder protection. No NER is used on the LV side.

Earth-conductor sizing (adiabatic)

WORKED EXAMPLE — MINIMUM EARTH-CONDUCTOR CROSS-SECTION (LV FAULT)

Formula (IEC 60364-5-54, adiabatic): $A = I \times \sqrt{t} / k$

$I = 38.5$ kA (LV fault from **§c-shortcircuit**); $t = 0.2$ s (LV incomer clearing time from **§c-protection**); $k = 143$ (Cu conductor with PVC/XLPE insulation per IEC 60364).

$$A = 38\,500 \times \sqrt{0.2} / 143 = 38\,500 \times 0.447 / 143 \approx 120\, \text{mm}^2 \text{ Cu}$$

Earth conductors at the LV-MSB and main equipment earthing conductors must be $\geq 120\, \text{mm}^2$ Cu (or equivalent aluminium per the standard).

Earth grid & lightning protection

Earth grid: target resistance typically $\leq 1\, \Omega$ for the substation (confirm against soil resistivity ρ and IEEE 80 touch/step voltage calculations for the specific site). The earth-grid design must ensure touch voltages < 50 V (step voltages lower) during the maximum earth-fault condition.

Lightning protection (IEC 62305): the lightning-protection level (LPL I to IV) is set by a risk assessment per IEC 62305-2, considering structure dimensions, location, use, and annual flash density. LPL I (most stringent) applies to process plants with flammable/explosive content; LPL III–IV for standard industrial buildings. The LPS comprises air terminals (rods/mesh), down-conductors, and earth termination, all bonded to the main earth grid.

Site assessment checklist

Progress: 0 / 7

- ☐ Confirm **MV-NER** nameplate resistance (Ω) and short-time rating — compare with calculated $21\, \Omega$ / 1.89 MW (indicative).
- ☐ Check the NER neutral CT is wired to the earth-fault relay and the relay is in service.
- ☐ Confirm main earth-conductor sizes at **LV-MSB** busbars $\geq 120\, \text{mm}^2$ Cu (or aluminium equivalent).
- ☐ Measure (or request certificate for) earth-grid resistance: target $\leq 1\, \Omega$ (or as specified).
- ☐ Confirm all MV and LV equipment bodies, cable armouring, and cable trays are bonded to the earth grid.
- ☐ Check the IEC 62305 lightning-protection risk assessment and LPS installation certificate are available.
- ☐ Inspect air terminals and down-conductors for corrosion, mechanical damage, or missing/loose connections.



On the forms →

(a) *Document inventory / recommendations register:* record NER nameplate resistance (Ω) and short-time rating; main earth-conductor sizes (mm^2); earth-grid resistance measurement (date and value in Ω); LPS certificate date and LPL class; note any earth-bonding deficiencies found.

(b) *AS-FOUND site data required:* NER nameplate (Ω , short-time kW / kVA, rated duration); neutral CT ratio and earth-fault relay trip threshold; measured main earth-conductor size (mm^2) from gland tags or cable schedule; earth-grid resistance test certificate; soil resistivity test data (if available); IEC 62305 risk assessment and LPS installation drawings/certificate; photographs of air-terminal and down-conductor condition.

★ Earthing is the last line of defence against electrocution, equipment damage, and lightning. An NER at the wrong resistance, an under-sized earth conductor, or an unbonded cable tray can all be lethal. Verify with measurements, not just drawings.

13 — Calculations → Equipment → Cost-Impact Summary

The table below maps each calculation to the equipment it sizes, the consequence of skipping it, and the approximate capital-cost impact. Use it to prioritise which calculations to verify first in a time-limited assessment.

#	Calculation	Key inputs	Key outputs	Equipment sized	If skipped	Cost impact
1	Load & demand	Connected kW, diversity factor, pf	P (MW), S (MVA)	Base for TX-1/TX-2, MV-SWGR, LV-MSB, feeders	Whole plant mis-sized; costly retrofit	High — sets entire plant size
2	Transformer sizing & N-1	MVA, redundancy concept, essential fraction	Unit rating, normal/N-1 loading (pu)	TX-1/TX-2, BT scheme	Overload on N-1 or oversized switchgear	High — transformers + frames scale with it
3	FLC (LV & MV)	Transformer kVA, voltage	LV FLC $\approx$ 2 310 A, MV FLC $\approx$ 84 A	LV-MSB incomer ACB, busbars; MV feeder breakers	Under-rated busbar/breaker overheats	High — ACB frame and busbar ampacity
4	Short-circuit / fault level	LV FLC, Z %, source assumptions	$I_{sc} \approx$ 38.5 kA	Breaking capacity & bracing: LV-MSB, MCC, ACB/MCCB	Breaker cannot clear fault — safety failure	High — drives I_{cu} and busbar bracing
5	Motor FLC & starting	Motor kW, V, pf, η , start multiple	FLC, start current, start method	MCC starter/overload, soft-starter/VFD	Bus dip/nuisance trip; under-rated starter fails	Medium — soft-starter/VFD vs DOL cost
6	Cable sizing & volt-drop	I_B , derating, length, R/X	I_z (req), cross-section, % ΔV	Feeder cables throughout plant	Cable overheats (fire) or volt-drop too high	Medium — copper/aluminium volume
7	Power-factor correction	P, present pf, target pf, harmonics	$\approx$ 580 kVAR, detuning required	PFC bank + 7 % detuning reactors	Reactive penalties; capacity wasted; bank fails	Medium — avoids ongoing penalties
8	Standby generator	Essential load, pf, step load, derating	DG rating $\approx$ 1 000 kVA	DG set, ATS, essential distribution	Genset stalls on outage, or grossly oversized	High — gensets are costly per kVA
9	Load-flow	Loads, source V, Z, taps, pf	Bus voltages, flows, losses, tap position	Transformer tap; validates cable sizes & PFC	Hidden under/over-voltage or overloads	Medium — mainly validates; flags reinforcement
10	Protection coordination	Fault levels, TCC curves, CT ratios	Relay pickup & time settings	MV-PROT relays; ACB/MCCB trip units	Loss of selectivity; wider outage per fault	Low-Med — mostly settings engineering
11	Arc-flash (IEEE 1584)	Arcing current, clearing time, distance	E (cal/cm ²), AFB, PPE category	PPE, labels, arc-reduction measures	Burn risk; non-compliant labels	Medium — PPE + arc-reduction relays; safety-critical
12	Earthing & lightning	System earthing, I_{ef} , soil ρ	$R_{NER} \approx$ 21 Ω ; earth csa $\geq$ 120 mm ²	MV-NER; earth grid; conductors; LPS	Touch/step hazard; undetected earth faults	Medium — earth copper + NER + LPS; safety-critical

Where the money is

The calculations that **dominate capital cost** are load/transformer sizing (§1–§3) and short-circuit rating (§4): between them they fix ACB frame sizes, breaking capacities, busbar bracing, cable cross-sections, and the genset — the most expensive items on the plant. By contrast, PFC (§7) and voltage-drop/cable losses (§6) mainly affect **operating cost and utility penalties** rather than headline capital. Motor starting (§5) sits in between. The power-system studies (§9–§12) are mostly **engineering and settings plus modest hardware**, but their value is measured in avoided injuries, equipment loss, and unplanned downtime.

★ **Prioritise verification of §1–§4 first in any assessment: if the transformer/breaker/ busbar ratings are wrong, everything downstream is at risk. Then verify §11–§12 for life-safety compliance. All others follow.**

Formula Reference Card

Printable quick-reference. All values indicative — confirm for your specific plant. Constants: $\sqrt{3} = 1.732$; $V_{LV} = 400$ V; $V_{MV} = 11$ kV; 50 Hz.

Quantity	Formula	This-plant example	Units
Apparent power	$S = P / \text{pf}$	$2.0 \text{ MW} / 0.85 = 2.35 \text{ MVA}$	MVA, kVA
3-phase current (equipment)	$I = S / (\sqrt{3} \times V)$	$1\,600 \text{ kVA} @ 400 \text{ V} \rightarrow 2\,310 \text{ A}$	A
Motor FLC	$I = P / (\sqrt{3} \times V \times \text{pf} \times \eta)$	$110 \text{ kW} \rightarrow 194 \text{ A}$	A
Transformer loading	$\text{Loading} = S_{\text{load}} / S_{\text{rated}}$	$1.175 / 1.6 = 0.73 \text{ pu}$	pu or %
Short-circuit current	$I_{\text{sc}} = I_{\text{fl}} / Z_{(\text{pu})}$	$2\,310 / 0.06 = 38.5 \text{ kA}$	A, kA
DOL starting current	$I_{\text{start}} \approx 6\text{--}7 \times I_{\text{fl}}$	$6.5 \times 194 = 1\,260 \text{ A}$	A
Cable derating	$I_{\text{z(req)}} = I_{\text{B}} / (k_{\text{g}} \times k_{\text{t}})$	$194 / 0.752 = 258 \text{ A}$	A
Cable voltage drop (3-phase)	$\Delta V = \sqrt{3} \times I \times (R \cdot \cos\phi + X \cdot \sin\phi) \times L$	$= 9.6 \text{ V} = 2.4 \%$	V, %
PFC reactive power	$Q_{\text{c}} = P \times (\tan\phi_1 - \tan\phi_2)$	$2\,000 \times 0.291 = 582 \text{ kVAr}$	kVAr
Generator sizing (essential)	$S_{\text{DG}} = (P_{\text{ess}} / \text{pf}) \times \text{step factor}$	$0.81 \times 1.25 = 1.01 \text{ MVA}$	kVA, MVA
Transformer voltage regulation	$\Delta V\% \approx \text{load}_{\text{pu}} \times (R\% \cdot \cos\phi + X\% \cdot \sin\phi)$	$0.73 \times 3.65 = 2.7 \%$	%
Relay grading margin	$t_{\text{up}} = t_{\text{down}} + \Delta t$	$0.20 + 0.30 = 0.50 \text{ s}$	s
Arc-flash boundary	$\text{AFB} \approx D \times \sqrt{(E / E_{\text{B}})}$	$0.6 \times \sqrt{(10 / 1.2)} = 1.7 \text{ m}$	m
NER resistance	$R_{\text{NER}} = (V_{\text{LL}} / \sqrt{3}) / I_{\text{ef}}$	$6\,351 / 300 = 21 \Omega$	Ω
Earth conductor (adiabatic)	$A = I \times \sqrt{t} / k$	$38\,500 \times \sqrt{0.2} / 143 = 120 \text{ mm}^2$	mm ²
Transformer Cu loss at load	$P_{\text{Cu}} = \text{loading}_{\text{pu}}^2 \times P_{\text{Cu,fl}}$	$0.73^2 \times 17 \text{ kW} = 9 \text{ kW}$	kW
N-1 essential loading	$\text{Loading}_{\text{N-1}} = S_{\text{ess}} / S_{\text{rated}}$	$1.29 / 1.6 = 0.81 \text{ pu}$	pu

- i Handy approximations for field checks:**
- 1 kVA per ampere per kV (3-phase): $I \approx \text{kVA} / \text{kV}$ (rough; $\div 1.73$ for exact)
 - At 400 V: 1 kW $\approx$ 1.6 A (pf 0.85, η 0.95)
 - At 11 kV: 1 MW $\approx$ 60 A (pf 0.85)
 - Short-circuit level in kA $\approx (\text{TX kVA} / 1000) / (Z\% \times V_{\text{LV,kV}} \times 1.73)$
 - Quick voltage-drop check: $\% \Delta V \approx 2 \times I(\text{A}) \times L(\text{km}) \times R(\Omega/\text{km}) / V(\text{kV})$

★ **Keep this card on you during switchroom inspections. Cross-reference with the nameplate data you photograph on site and you can verify most ratings without a laptop.**

Self-Check Problems

Work each problem before opening the answer. All values are *indicative*; use $\sqrt{3} = 1.732$.

Interpretation questions

These questions test your ability to draw conclusions from numbers — the skill the PM will assess when reviewing your FMECA entries. Attempt each before opening the answer.

1. The LV-MSB incomer ammeter reads **2 650 A** on a 1 600 kVA transformer. The transformer FLC at 400 V is calculated as 2 310 A. What do you conclude, and what is your immediate action?

▼ Show answer

The transformer is loaded at $2\,650 / 2\,310 = \mathbf{1.15\ pu\ (115\ \%)}$ — significantly above its rated FLC. This is an overload condition.

Immediate FMECA implication: The transformer is unable to carry full current without exceeding its thermal rating (continuous overload accelerates insulation ageing). If this is normal operating load, the transformer is undersized for its duty. If one transformer is lost and the bus-tie closes to back-feed from this overloaded unit, it will trip on overcurrent — the N-1 redundancy claim fails.

Action: Record the reading as an observation in SF-04 (OBS-xx); open an FMECA row for TX overloading (S = 4 — loss of N-1 capability; O = depends on whether overload is sustained; D = 2 if transformer monitoring with alarms is live). Add a recommendation to revalidate the essential-load list and consider demand-side reduction or transformer uprating.

2. The LV prospective short-circuit current at the LV-MSB is calculated at **38.5 kA** for a single transformer. The LV-MSB bus-tie ACB nameplate shows a rated breaking capacity of **36 kA**. The two-transformer bus-tie-closed fault level is **77 kA**. What are the implications?

▼ Show answer

Single-transformer case: The BT ACB at 36 kA is below the calculated 38.5 kA fault level — even in the normal single-transformer condition the bus-tie breaker appears under-rated by ~7 %. This is a potential P1 safety SPOF. The ACB must not be closed if it cannot interrupt the prospective fault current.

Bus-tie-closed parallel case: If both transformers are in parallel (e.g. during a transfer), the fault level rises to ~77 kA — more than double the ACB rating. This is a critical finding: operating with the bus-tie closed and no ACB upgrade is a life-safety hazard.

FMECA row: S = 5 (catastrophic — ACB fails to interrupt fault, potential catastrophic switchboard failure); P1 regardless of RPN. Recommended action: confirm fault level with a detailed short-circuit study; specify a replacement ACB with $I_{cu} \geq 77\text{ kA}$ (or maintain strict NOP bus-tie operation and confirm single-TX I_{sc} value with utility data).

3. The arc-flash label on the LV-MSB Bus A incomer cubicle shows an incident energy of **22 cal/cm²**, derived from a study conducted in 2014. Since that study, a second transformer (TX-2) has been connected, roughly doubling the available fault current to the bus. How do you treat this finding?

▼ Show answer

The 2014 study is **Obsolete** — log it as such in SF-01. The actual incident energy at the bus is now substantially higher than 22 cal/cm² because higher fault current means faster energy delivery and longer clearing times may no longer hold. Worse, the PPE category assigned in 2014 is likely wrong and may be unsafe.

Immediate QHSE action (SF-04 Section 6): Flag the out-of-date arc-flash label and the connected TX-2 as a QHSE finding. No work requiring arc-flash PPE should be performed against the 2014 label values until a restudied.

FMECA row: FM for "arc-flash boundary unverified / label obsolete" — S = 5 (personnel injury risk); O = 2 (switching events occasional); D = 4 (no current study; undetectable error in PPE assignment). RPN = 40 → P2; but S = 5 with $S \geq 4$ rule should be flagged for immediate escalation at next switching event.

Recommendation: commission updated IEEE 1584 study and update all cubicle labels before any live switching maintenance.

Numerical problems

1. A second LV motor is rated **75 kW**, 400 V, pf 0.85, η 0.94. Find its FLC and its DOL starting current (use 6.5×).

▼ Show answer

$FLC = 75\,000 / (1.732 \times 400 \times 0.85 \times 0.94) = 75\,000 / 553 = \mathbf{136\ A}$

$DOL = 6.5 \times 136 = \mathbf{882\ A}$

2. A **1 250 kVA** transformer has Z = 6 %. Find its 400 V FLC and secondary short-circuit current.

▼ Show answer

$FLC = 1\,250\,000 / (1.732 \times 400) = 1\,250\,000 / 692.8 = \mathbf{1\,804\ A}$

$$I_{sc} = 1\,804 / 0.06 = \mathbf{30.1\,kA}$$

3. Re-correct the plant from **0.85** → **0.93** (instead of 0.95). How much kVAr is needed (P = 2.0 MW)?

▼ Show answer

$$pf_2 = 0.93 \rightarrow \varphi_2 = 21.6^\circ, \tan\varphi_2 = 0.396; \tan\varphi_1 = 0.620.$$

$$Q_c = 2\,000 \times (0.620 - 0.396) = 2\,000 \times 0.224 = \mathbf{448\,kVAr} \text{ (}\approx 450\text{ kVAr bank).}$$

4. With one transformer out, the bus-tie carries **1.3 MVA** of essential load on a 1 600 kVA unit. What is the per-unit loading — is it acceptable?

▼ Show answer

$$\text{Loading} = 1.30 / 1.60 = \mathbf{0.81\,pu\, (81\,\%)}. \quad \checkmark$$

Acceptable for the essential-load N-1 case (non-essential load shed). Within continuous rating. ✓

5. A feeder carries **I_B = 160 A** with grouping derating $k_g = 0.85$ and temperature derating $k_t = 0.91$. What tabulated cable rating I_z is required?

▼ Show answer

$$I_{z(\text{req})} = 160 / (0.85 \times 0.91) = 160 / 0.7735 = \mathbf{207\,A}.$$

Select a cable with tabulated rating $\geq 207\,A$ (e.g. 70 mm² Cu XLPE on tray ≈ 220 –240 A, confirm against manufacturer table).

6. The MV system is resistance-earthed to limit earth-fault current to **400 A** at 11 kV. What NER resistance is required?

▼ Show answer

$$R_{NER} = (11\,000 / 1.732) / 400 = 6\,351 / 400 = \mathbf{15.9\,\Omega}$$

7. An arc-flash study gives **6 cal/cm²** at a 600 mm working distance. Estimate the arc-flash boundary using $E_B = 1.2\text{ cal/cm}^2$ and inverse-square scaling.

▼ Show answer

$$AFB = 0.6\text{ m} \times \sqrt{(6 / 1.2)} = 0.6 \times \sqrt{5} = 0.6 \times 2.236 = \mathbf{1.34\text{ m}}$$

8. A cable feeder is **150 m** long, carries **220 A** at pf 0.90 ($\sin\varphi = 0.436$). The cable is **120 mm² Cu XLPE**: R = 0.196 Ω/km , X = 0.079 Ω/km (on tray). Calculate the voltage drop and state whether it is within the 5 % motor-feeder limit at 400 V.

▼ Show answer

$$\Delta V = 1.732 \times 220 \times (0.196 \times 0.90 + 0.079 \times 0.436) \times 0.150$$

$$\Delta V = 1.732 \times 220 \times (0.1764 + 0.0344) \times 0.150$$

$$\Delta V = 1.732 \times 220 \times 0.2108 \times 0.150 = \mathbf{12.1\text{ V}}$$

$$\% \Delta V = 12.1 / 400 \times 100 = \mathbf{3.0\,\%} \text{ — within the 5\,\% limit. } \checkmark$$

9. The upstream MV relay [52-T] trips at $t = 0.50\text{ s}$. The MV incomer relay [52-I] must grade above it with a 0.3 s margin. What is the minimum MV incomer relay trip time?

▼ Show answer

$$t_{52-I} = t_{52-T} + \Delta t = 0.50 + 0.30 = \mathbf{0.80\text{ s}}$$

10. A DG set is rated 1 000 kVA at ISO conditions. The site is at **400 m altitude** and **42 °C** ambient. Apply de-rating: –1 % per 100 m above sea level; –1 % per 5 °C above 25 °C. What is the effective site rating?

▼ Show answer

$$\text{Altitude de-rating: } -1\,\% \times (400 / 100) = -4\,\%$$

$$\text{Temperature de-rating: } -1\,\% \times (42 - 25) / 5 = -1\,\% \times 3.4 = -3.4\,\%$$

$$\text{Total de-rating} = 4\,\% + 3.4\,\% = 7.4\,\%$$

$$\text{Site rating} = 1\,000 \times (1 - 0.074) = \approx \mathbf{926\,kVA} \text{ (indicative; confirm with manufacturer data).}$$

★ You have now verified that the equipment on the SLD is correctly rated — or found where it is not. Part 3 (SPOF Analysis & FMECA) takes those findings and asks: *what happens when this item fails?* The calculations in Part 2 are your evidence base for the criticality ratings and recommendations you will write in Part 3.

i **Where you are in the workflow:** You have identified the equipment (Part 1) and worked through the key calculations (Part 2 — see [fault levels](#) and [protection co-ordination](#)). Part 3 turns that knowledge into a structured single-point-of-failure analysis and a formal reliability assessment. The findings will drive the recommendations in [Part 4](#).

What is a SPOF — and Why It Matters

A **single point of failure (SPOF)** is any one component or shared element whose failure, on its own, disables a function or de-energises a load that must stay energised. It does not matter how much redundancy exists downstream: if a common element upstream can black it all out, the function has a SPOF.

Why SPOFs matter in a process plant

Driver	What happens when a SPOF trips	Typical consequence
Availability / production loss	Critical load loses supply without warning	Process shutdown, spoiled batch, forced restart — hours to days of lost production
Safety	Safety system, cooling pump, agitator or ventilation fan de-energises	Hazardous condition: thermal runaway, toxic release, loss of fire suppression
Cost	Emergency repair, overtime, expedited spares, regulatory notification	Often 10-100× the cost of the redundancy that would have prevented it

★ A SPOF is a *system property*, not just a component property. Two pumps in parallel give no protection if they share a single supply cable. Always trace from source to load — the SPOF is wherever there is no alternative path.

The SPOF-Hunt Method

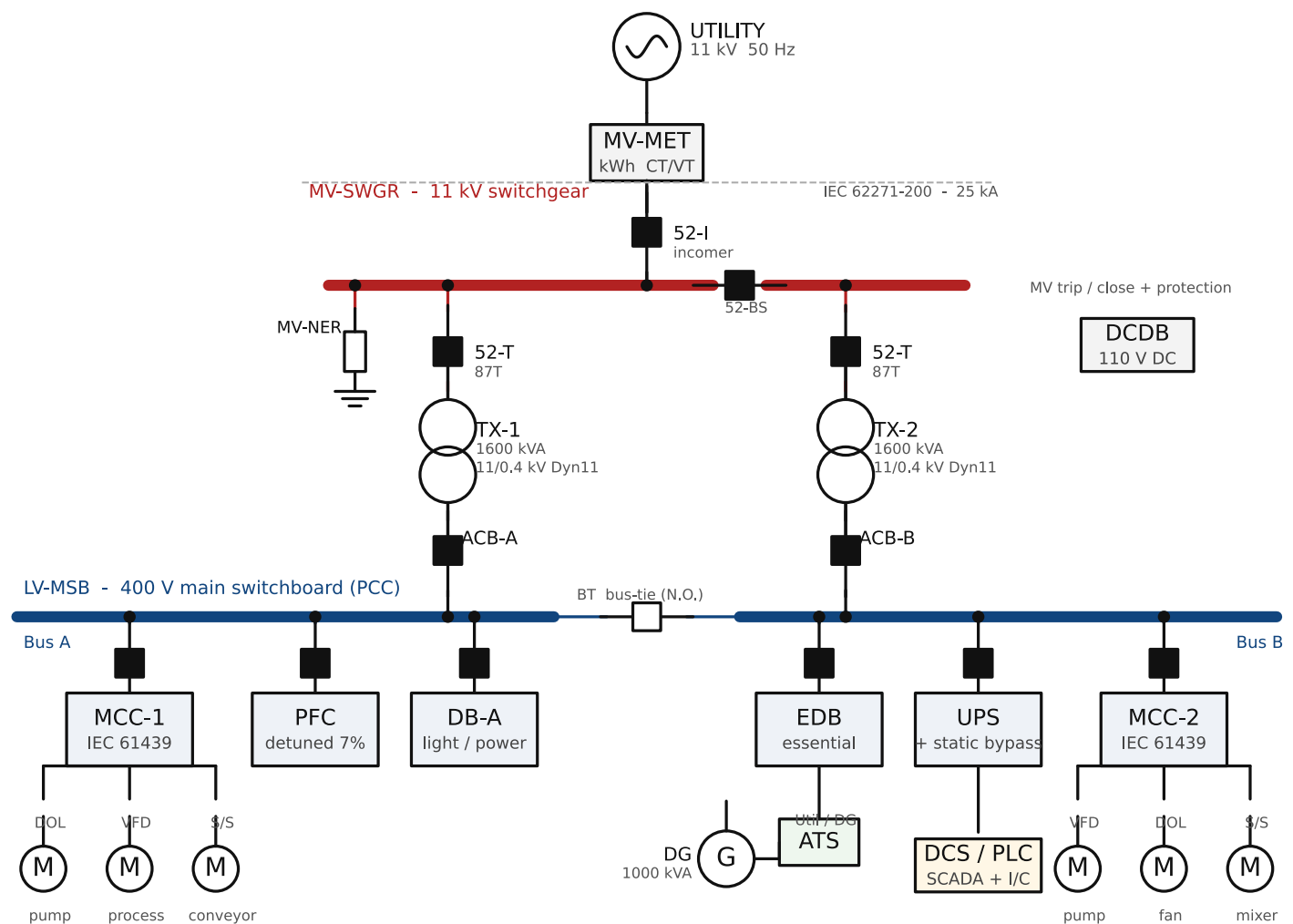
Work the single-line diagram systematically, starting at the point of supply and moving toward the loads. Use five repeating steps:

1. **Trace** — follow every power path from the utility intake (MV-MET / MV-SWGR) through each transformer (TX-1, TX-2), along each bus section of LV-MSB, through every MCC and feeder, to each final load.
2. **Ask** — at every element: *"If this one item fails or is isolated for maintenance, what loses supply?"*
3. **Flag** — if the answer is "a load that must stay up" and there is *no alternative path*, mark it as a SPOF candidate.
4. **Check common elements** — look especially at shared items that sit upstream of apparent redundancy: a single utility incomer, a single MV bus, an un-tied LV bus, a shared outgoing cable or MCC, a single ATS. Redundancy downstream is defeated if these are common.
5. **Mitigate** — for each flagged item, propose an alternative path: N+1 equipment, dual feed, auto bus-tie, separate routing, DG/ATS standby, or UPS/DCDB.

✓ **Practical tip:** Print the master SLD on a large sheet and use a highlighter to trace each path. Mark every element that is a common ancestor of two or more critical loads. Those are your prime SPOF candidates.

Demonstration on the master SLD

Apply the method to the indicative 2 MW plant master SLD below. Starting from the top: utility supply → MV-MET → MV-SWGR → (TX-1 to Bus A, TX-2 to Bus B) → LV-MSB with bus-tie (BT) → MCCs → loads / DG / ATS / UPS.



MASTER SINGLE-LINE DIAGRAM - ~2 MW MV/LV PROCESS PLANT
11 kV / 400 V - 50 Hz - 2 x 1600 kVA (Dyn11, Z~6%) - IEC 60617 - indicative

Fig 3-0 — Master SLD, indicative 2 MW MV/LV process plant (IEC 60617 symbols, indicative). Trace each element from source to load; flag any with no alternative path.

Walking through the method on this diagram:

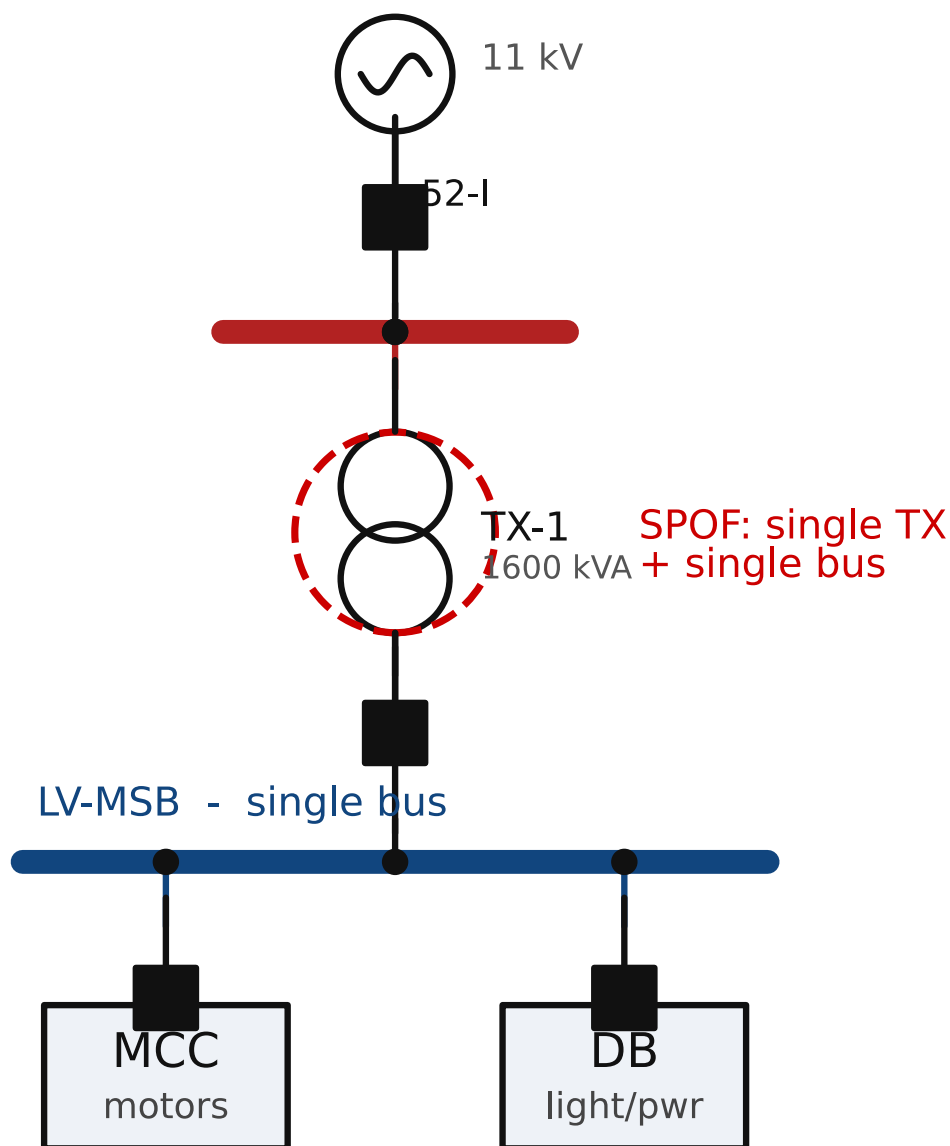
Element	Tag(s)	Common element upstream of...	SPOF?	Alternative path present?
Utility incomer(s)	MV-MET	Entire plant	Potentially yes — depends on whether one or two incomers	DG + ATS covers essential loads on utility loss
MV switchgear / RMU	MV-SWGR	Both transformers	If single MV bus — yes (see SLD C)	Bus-section breaker gives partial mitigation
Transformer 1	TX-1	LV-MSB Bus A + all MCC-1 (Bus A) loads	No — BT auto-closes, TX-2 backs up	TX-2 via auto-close bus-tie BT
Transformer 2	TX-2	LV-MSB Bus B + all MCC-2 (Bus B) loads	No — BT auto-closes, TX-1 backs up	TX-1 via auto-close bus-tie BT
LV main switchboard bus-tie	BT	Cross-bus essential loads	Yes — if BT fails to close, N-1 benefit is lost	Manual close (slow); periodic test essential
MCC incomer / shared feeder	MCC-1 ... n	All loads on that MCC	Yes — if duty + standby share same MCC (see SLD D)	Split duty/standby across Bus A and Bus B MCCs
ATS control	ATS	Essential bus on DG transfer	Yes — if ATS fails to operate	Redundant control; monthly on-load test
UPS / DCDB	UPS , DCDB	DCS / PLC / protection supplies	Yes — if UPS battery fails during disturbance	Redundant UPS; battery monitoring

 **Warning — the "two of everything" trap:** Installing two transformers does not give N-1 resilience unless the bus-tie is present *and* functional. Verify the BT auto-transfer scheme is tested and operational before claiming N-1 credit.

SPOF Library — Variants A to E

The five example SLDs below form a teaching library. SLDs **A through D are flawed configurations** — each isolates one type of SPOF. SLD **E is the resilient reference**. Study each in sequence: the flaw in A is corrected in B, B's flaw in C, and so on until E carries all the mitigations.

Variant A — Single Transformer, Single LV Bus



SPOF-A - Single transformer, single bus

Any loss of TX-1, its feeder, the incomer or the bus blacks out the whole plant.

Fig 3-A — SPOF variant A: single transformer TX-1, single un-split LV bus (indicative).

Configuration: One utility feed → TX-1 (1600 kVA indicative) → single un-split LV bus (LV-MSB) feeding all loads. No second source, no bus-tie.

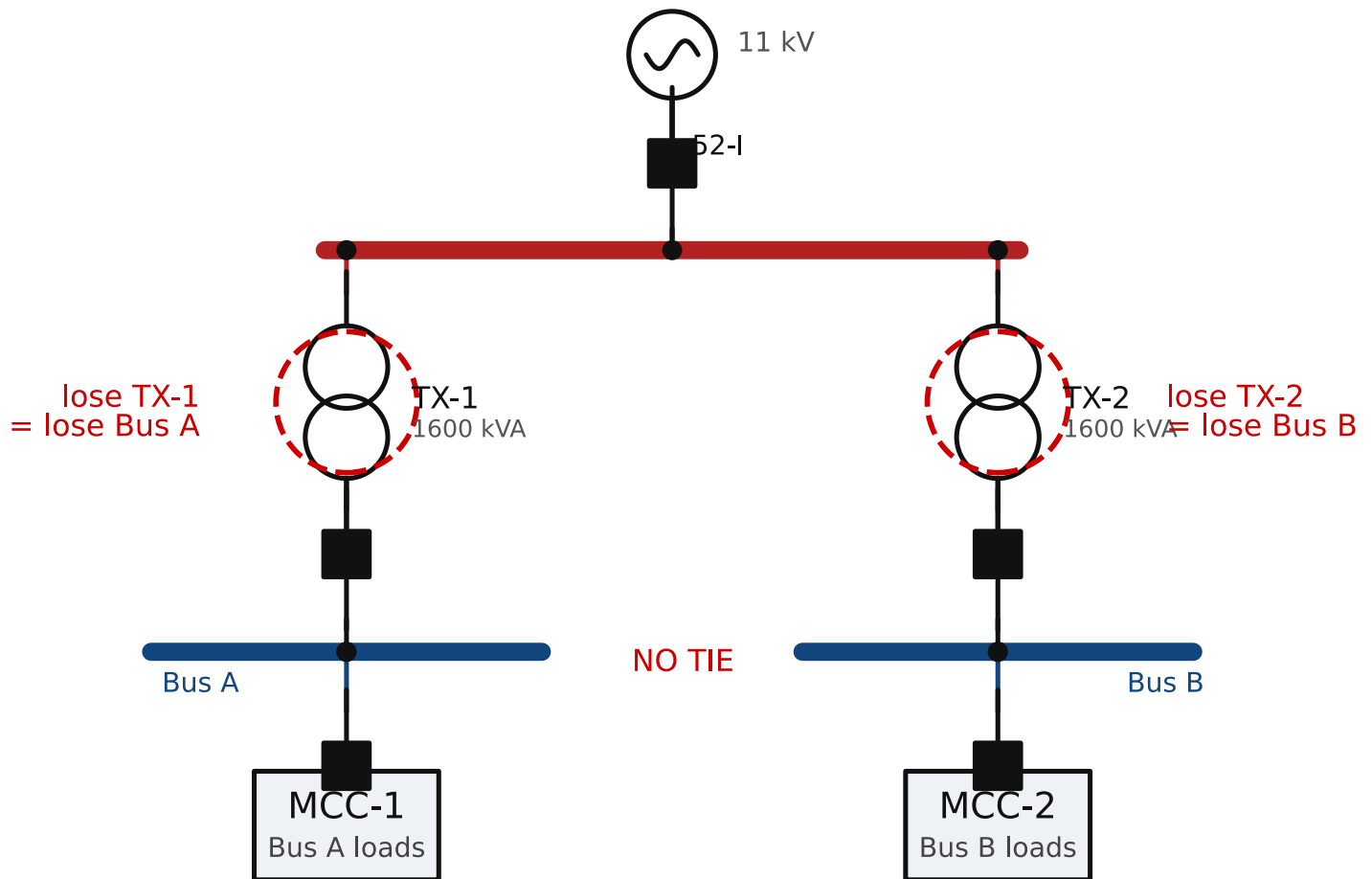
SPOF(s) identified: The single transformer TX-1 and the single un-split LV bus are both whole-plant SPOFs. Also: the MV feeder to TX-1 and the LV incomer breaker.

Consequence: Loss of TX-1, its MV feeder, the LV incomer breaker, or any LV bus fault **de-energises the entire plant**. Total blackout. No load segregation limits the affected zone.

⚠ This configuration is common in small or legacy installations. Any single component failure produces a total plant blackout — there is no time to transfer loads because there is nowhere to transfer to.

Mitigation: Move to the master-SLD arrangement — add TX-2 on a split bus with auto-close bus-tie BT (N-1 capability), and provide DG + ATS for essential loads. This is exactly the reference design shown in Variant E.

Variant B — Two Transformers, Two Buses, No Bus-Tie



SPOF-B - Two transformers, no bus-tie

Redundant sources, but no tie: a single source loss drops its entire half of the plant.

Fig 3-B — SPOF variant B: two transformers, two buses, no bus-tie between them (indicative).

Configuration: TX-1 feeds Bus A, TX-2 feeds Bus B — two independent buses with **no bus-tie**. Appears redundant at first glance.

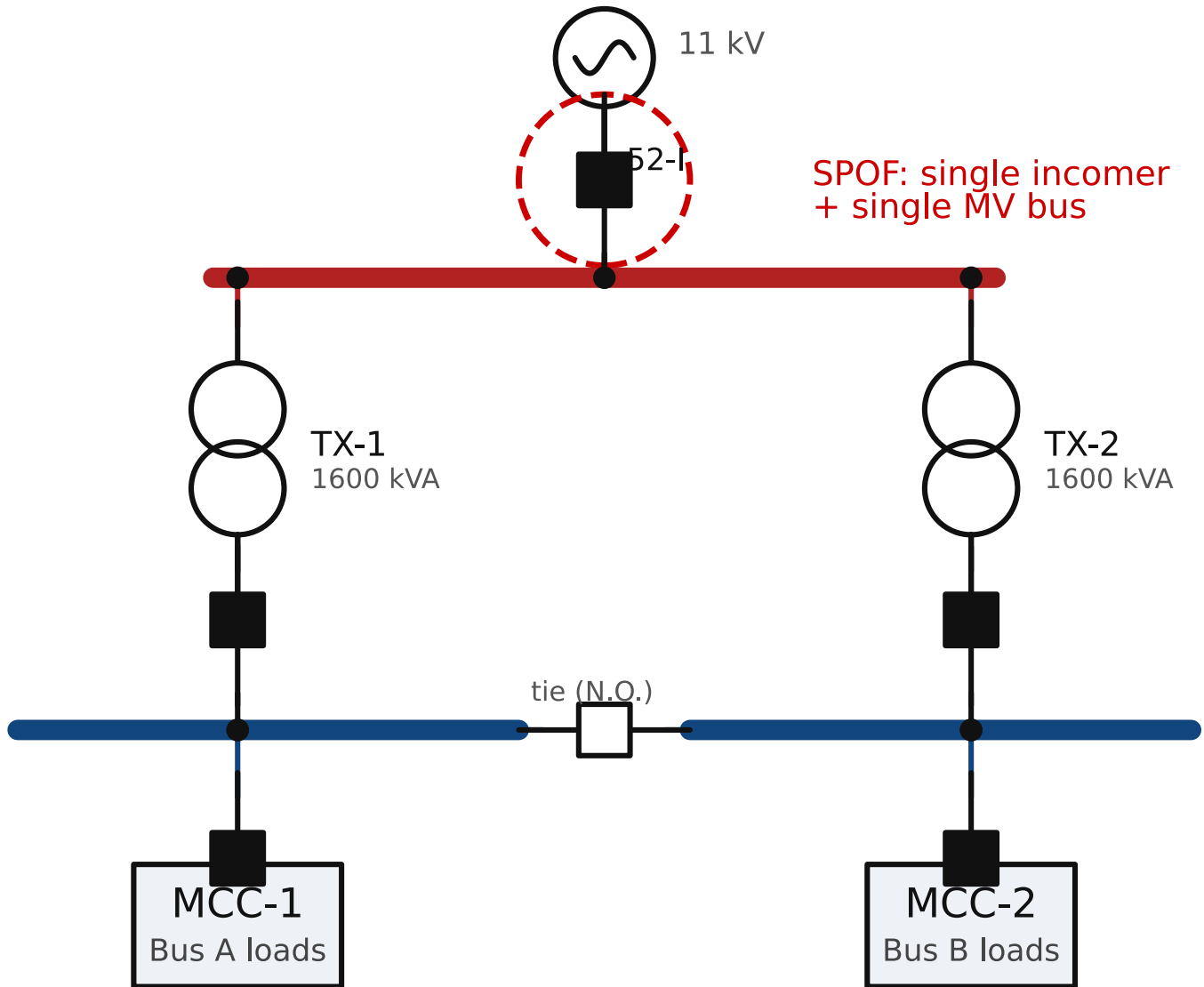
SPOF(s) identified: Each transformer is a SPOF for its own bus. The redundancy is illusory: the healthy transformer has no path to the dead bus.

Consequence: Loss of TX-1 drops **all of Bus A**; loss of TX-2 drops **all of Bus B**. Half the plant is lost on any single transformer or feeder failure — even though two transformers are installed.

i The "two transformers but still not N-1" lesson: The number of transformers is irrelevant if there is no means to transfer load between buses. N-1 requires both the equipment *and* the interconnection.

Mitigation: Install a normally-open auto-close bus-tie (BT) between Bus A and Bus B. The bus-tie must close automatically on loss of either incomer. Verify that either transformer can carry the essential plant load in N-1 condition (see transformer loading calculations in Part 2).

Variant C — Single Utility Incomer / Single MV Breaker



SPOF-C - Single utility incomer / MV bus

LV redundancy is fine, but one incomer or MV-bus fault still blacks out everything.

Fig 3-C — SPOF variant C: good LV architecture but single MV incomer and bus (indicative).

Configuration: The LV side is well-arranged (TX-1, TX-2, bus-tie), but **all power flows through a single utility feeder, single MV incomer breaker [52-I], and single MV bus.**

SPOF(s) identified: Single utility feeder, single MV incomer CB, and the single MV busbar. The SPOF has **moved upstream** — it is now at the MV intake, not the LV distribution level.

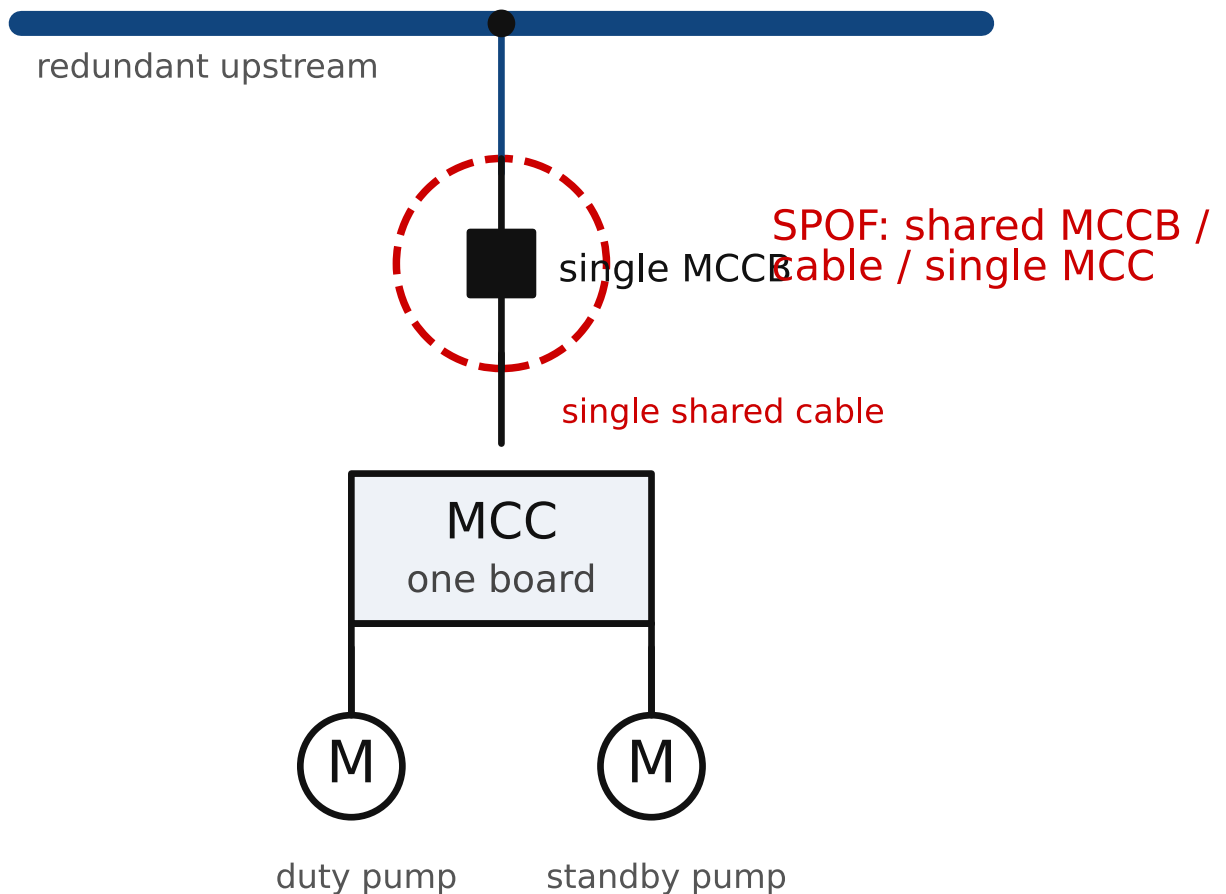
Consequence: Loss of the utility supply, the incomer breaker, or an MV bus fault **blacks out the whole plant** regardless of LV redundancy. The LV bus-tie cannot help — both transformers lose their source simultaneously.

⚠ This is the most common "hidden SPOF" found in existing plants. Engineers who only examine the LV distribution miss it entirely. Always trace the *full* power path from the utility boundary.

Mitigation: Provide a second independent utility feeder with a closeable MV bus-section (as in Variant E). Where a second feeder is not feasible, rely on DG + ATS to ride through utility loss for all essential loads. Strengthen MV incomer protection (differential, IDMT O/C, earth fault) to minimise fault duration.

Variant D — Shared Outgoing Cable / Single MCC for Duty + Standby

LV-MSB (dual-fed, healthy)



Duty + standby share one board -> the standby gives no real redundancy.

SPOF-D - Shared cable / single MCC for duty + standby

Fig 3-D — SPOF variant D: upstream redundancy defeated by a shared MCC feeder cable (indicative).

Configuration: Sources are redundant upstream (TX-1, TX-2, bus-tie), but a **single shared outgoing feeder and single MCC (MCC-1)** supplies both a critical duty pump *and* its standby pump from the same bus section.

SPOF(s) identified: The shared outgoing cable, the MCC-1 bus, and the MCC-1 incomer breaker are all common elements serving *both* the duty and standby machines.

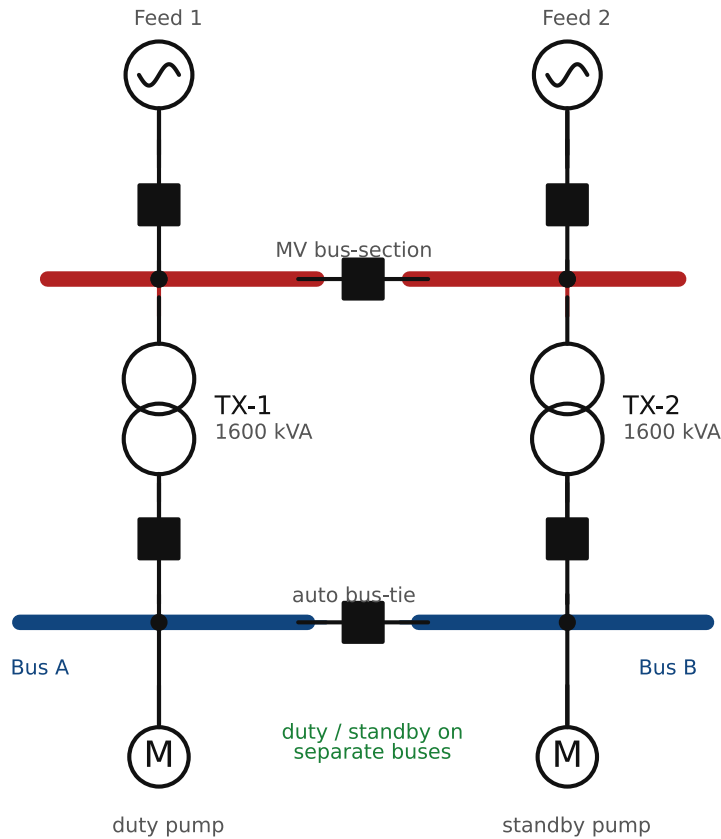
Consequence: One feeder cable fault, MCC bus fault, or MCC incomer trip takes **both duty and standby pumps offline simultaneously**. The standby machine provides no protection against the common supply element. The upstream redundancy is wasted.

★ **Variant D demonstrates the general principle: redundancy of equipment is only as effective as the independence of its supply. Always check whether duty and standby loads share any cable, MCC, bus section, or protection device upstream.**

Mitigation: Feed duty pump from MCC-1 on Bus A; feed standby pump from MCC-2 on Bus B. Route the two supply cables via physically separated routes. No single cable, MCC, or bus now disables the function.

Variant E — Resilient Reference Layout

RESILIENT REFERENCE - no single point of failure



SPOF-E - Resilient layout (for contrast)

Dual feeds, two transformers, bus-section + auto tie, segregated buses: every common element has an alternative path.

Fig 3-E — Resilient reference layout: all A-D SPOFs mitigated (indicative).

Configuration: Dual independent utility incomers, closeable MV bus-section, two full-rated transformers on segregated LV buses with a closed/auto bus-tie, duty and standby pumps split across Bus A and Bus B with separately routed cables, DG + ATS for essential loads, and UPS + DCDB for control and protection supplies.

Why there is no dominant SPOF: Every element that was a SPOF in A–D now has an alternative path:

Element that was a SPOF in A-D	Variant	Alternative path in E
Single transformer + single un-split LV bus	A	Two full-rated transformers on segregated buses; N-1 load transfer
Two transformers but no bus-tie	B	Closed / auto bus-tie BT — fast automatic transfer on loss of either incomer
Single utility incomer / single MV bus	C	Dual independent incomers + closeable MV bus-section
Shared cable / single MCC for duty + standby	D	Duty on Bus A (MCC-1), standby on Bus B (MCC-2), separately routed cables
Total utility loss leaving essential loads dark	A, B, C	DG + ATS — automatic start and transfer to essential bus
Control / protection supply loss during disturbance	All	UPS (AC no-break) + DCDB (DC trip / close) — independent of LV bus

Summary comparison — Variants A to E

Variant	Description	Main SPOF	Worst-case consequence	Primary mitigation
A	Single TX, single un-split bus	TX-1, LV-MSB bus, MV feeder	Whole-plant blackout on any single failure	Add TX-2 + split bus + auto bus-tie BT; DG for essential
B	Two TXs, two buses, no bus-tie	Each TX/incomer for its own bus	Loss of one source drops half the plant	Install normally-open auto-close bus-tie BT; confirm N-1 loading
C	Good LV, single MV incomer + bus	Single utility feeder, single MV incomer CB, single MV bus	Whole-plant blackout despite LV redundancy	Dual independent incomers + MV bus-section; DG + ATS for essential
D	Good upstream, shared MCC for duty + standby	Shared cable / MCC-1 incomer	Single common-element fault loses both duty and standby simultaneously	Duty on MCC-1 / Bus A; standby on MCC-2 / Bus B; separate cable routes

Variant	Description	Main SPOF	Worst-case consequence	Primary mitigation
E	Resilient reference — all A-D mitigations applied	<i>No dominant single SPOF</i>	Any single failure survivable by automatic transfer or standby	— (reference target; all common elements have an alternative path)

SPOF-Hardening Principles

The A–E library illustrates every mitigation type in practice. The seven underlying principles below apply to any MV/LV plant. Apply them in priority order: first eliminate the highest-consequence SPOFs, then work down the risk register.

N+1 Redundancy

Install at least one more capacity unit than the minimum required to carry the load. For transformers this means **two transformers each rated to carry the essential load in N-1 condition** (i.e. each $\geq$ essential kVA, not total kVA $\div$ 2). For motors, one duty + one standby. N+1 is worthless without the means to transfer — see bus-tie below.

Segregation

Split the LV switchboard into **Bus A and Bus B** with separate incomer CBs and separate LV cable compartments. Physically separate cable routes (different trenches or cable ladders), separate protection panels, and — where practicable — separate room or bay allocations. Segregation limits the zone of any single fault: a bus fault, fire, or arc-flash event on Bus A cannot propagate to Bus B.

Bus-Tie / Auto-Transfer

A **normally-open auto-close bus-tie (BT)** between Bus A and Bus B lets the surviving transformer back up the dead bus within seconds (typical auto-transfer time: 100–500 ms, indicative). Without the bus-tie, N+1 transformers give no N-1 benefit (Variant B). The bus-tie scheme must be tested periodically (at minimum annually on-load or as defined by the site maintenance plan). The BT protection relay must co-ordinate with the transformer incomers to prevent mal-operation.

Dual Feed

Two **independent utility incomers** (physically separate cables from separate substations or ring points) with a closeable MV bus-section remove the upstream intake SPOF (Variant C). Where a second utility feeder is not economically or geographically feasible, the DG + ATS combination must cover all loads that cannot tolerate blackout. Document the gap explicitly in the risk register.

Separate Routing of Duty vs Standby

Run the supply cables for a duty load and its standby load via **separate MCCs and physically separate cable routes** — different cable trenches, ducts or ladder routes that cannot be simultaneously damaged by a single cable fire, flood, or excavation strike. The separation should extend from the LV bus section right back to the MCC incomer (Variant D lesson).

Standby Generation (DG)

A **diesel generator (DG)** sized for the essential load (typically 30–50 % of installed kVA, indicative) with an automatic transfer switch (ATS) provides a completely independent source independent of the utility supply. The DG covers: total utility failure, MV incomer fault, and transformer outage that would otherwise leave essential loads dark. Weekly no-load test and monthly on-load test are minimum recommended practice. Fuel, coolant, and battery condition are critical.

UPS for Control and Protection

The **uninterruptible power supply (UPS)** provides AC no-break power to DCS, PLC, and marshalling panels; the **DC distribution board (DCDB)** provides battery-backed DC (typically 110 V DC or 24 V DC, indicative) for protection relay trip coils, remote indication, and emergency lighting. Without these, a supply disturbance triggers a protection relay loss-of-supply condition, preventing controlled re-energisation. UPS battery condition, bypass functionality, and charger health must be routinely verified.

i Proportionality: Every mitigation has a cost, complexity, and footprint trade-off. Apply redundancy where the consequence of failure justifies it. Use the FMECA risk ranking (Section [FMECA](#)) to prioritise spend. Essential and safety-critical loads always take precedence.

Typical-SPOF Field Checklist

Use this checklist during a site walkdown or drawing review. Work through each system zone in order. Tick each item when you have inspected it and are satisfied, or annotate with a finding reference. The checklist is generic and applies to any MV/LV plant of similar configuration; not every item will be present on every site.

Field tip: Bring a printed copy of the master SLD and mark each SPOF candidate directly on the drawing as you work through the zones below. Cross-reference finding numbers in the margin.

Site / plant name	_____
Assessment reference	_____
Engineer (assessor)	_____
Date of walkdown	_____
SLD revision reviewed	_____

Zone 1 — MV Intake / Utility Incomer (MV-MET)

Typical consequence if SPOF present: Total plant blackout, including any downstream LV redundancy. Severity: Critical. Typical mitigation: dual independent incomers, DG + ATS for essential loads.

Progress: 0 / 5

- ☐ Confirm whether the site has ONE or TWO independent utility incomers (physically separate cables from separate utility substations or ring-main points).
- ☐ If a single incomer: verify that DG + ATS coverage for essential loads is sized and commissioned — document the unprotected load gap.
- ☐ Check the MV metering cubicle (MV-MET) — confirm protection relay is present, set, and test records are current.
- ☐ Verify the incoming MV cable route: is the cable protected from physical damage (buried and marked, armoured, away from excavation zones)?
- ☐ Confirm interlock/key scheme between utility incomer and any generator incomer — an accidental parallel condition is a SPOF for the protection system.

Zone 2 — MV Switchgear (MV-SWGR, MV-NER)

Typical consequence if SPOF present: Total plant blackout or loss of earth fault detection. Severity: Critical to High. Typical mitigation: MV bus-section breaker, dual-incomer MV switchgear, proven NER scheme.

Progress: 0 / 6

- ☐ Is the MV switchgear a single bus or a split bus (with a bus-section breaker)? A single MV bus is a whole-plant SPOF.
- ☐ Check that the MV incomer CB(s) have functional spring-charge / charged indicator and trip coil continuity — a failed trip coil is a hidden SPOF for fault clearance.
- ☐ Verify the MV protection relay (IDMT O/C and earth fault) is active, in-service, and last tested within the maintenance interval.
- ☐ Confirm the neutral earthing resistor (MV-NER) is connected, resistance value within tolerance, and has thermal overload monitoring.
- ☐ Check arc-flash / pressure-relief arrangement on the MV switchgear — a single MV arc event that damages the bus will cause a prolonged outage unless the design contains it.
- ☐ Confirm that MV switchgear control / trip power supply is from the UPS or DCDB — a supply disturbance must not disable protection.

Zone 3 — Transformers TX-1 / TX-2

Typical consequence if SPOF present: Loss of one transformer — half the plant if no bus-tie, or full plant if only one transformer installed. Severity: High. Typical mitigation: N+1 sizing, auto bus-tie, differential protection, oil/temperature monitoring.

Progress: 0 / 6

- ☐ Confirm two transformers are installed (TX-1 and TX-2). If only one transformer: the entire LV plant is a SPOF for that transformer — document and assess DG coverage.
- ☐ Verify each transformer nameplate (kVA, voltage ratio, vector group Dyn11, impedance) matches the design basis; mismatched impedances affect load sharing after N-1 transfer.

- ☐ Check that each transformer has **independent** MV and LV protection (IDMT O/C, earth fault, winding temperature, oil level / Buchholz gas — as applicable) — shared protection is a common-element SPOF.
- ☐ Confirm the LV incomer CB of TX-1 and the LV incomer CB of TX-2 are **separate circuit breakers** — not a single shared LV incomer.
- ☐ Verify transformer loading in N-1 condition: is either transformer rated to carry the full essential load if the other is out of service?
- ☐ Check physical separation between TX-1 and TX-2 — are they in separate compartments or bays so a fire or catastrophic failure of one cannot damage the other?

Zone 4 — LV Main Switchboard & Bus-Tie (LV-MSB, BT)

i **Typical consequence if SPOF present:** Loss of bus-tie = loss of N-1 capability; single LV bus = whole-plant SPOF. Severity: High to Critical. Typical mitigation: split bus with functional auto-close BT, Form 4 compartmentalisation, arc detection.

Progress: 0 / 7

- ☐ Confirm the LV-MSB is a **split-bus arrangement** (Bus A and Bus B as separate bus sections with separate incomer CBs and separate busbars).
- ☐ Verify the bus-tie CB (BT) is present and **in service** (not permanently open with no auto-transfer scheme).
- ☐ Confirm the bus-tie auto-transfer scheme is commissioned: on loss of Bus A incomer, BT closes automatically to feed Bus A from Bus B (and vice versa). Test records?
- ☐ Check the interlocking logic: both LV incomers and the bus-tie should not be able to close simultaneously (prevents dangerous parallel operation).
- ☐ Verify the LV-MSB construction standard (Form 4 compartmentalisation preferred) — a low-form panel allows a fault to spread across bus sections, defeating the split.
- ☐ Check for arc-flash detection (optical or pressure) on the LV-MSB — a bus arc event in a high-fault-level plant is a high-consequence SPOF for the LV distribution.
- ☐ Confirm that Bus A and Bus B feeders to critical loads are balanced — a heavily unbalanced split means N-1 transfer may overload the surviving transformer.

Zone 5 — MCCs and Outgoing Feeders (MCC-1...n)

i **Typical consequence if SPOF present:** Loss of a critical duty and standby pair simultaneously (Variant D pattern). Severity: High. Typical mitigation: duty and standby on separate MCCs / bus sections, separately routed cables.

Progress: 0 / 6

- ☐ For every critical process function with a duty + standby machine: confirm the duty machine is fed from one MCC / bus section and the standby from a **different** MCC / bus section.
- ☐ Trace the cable routes for duty and standby pairs — do they share a common cable tray, duct bank, or trench? If yes, a single cable fire defeats the redundancy.
- ☐ Check each MCC incomer protection: is it set to clear faults within the cable rating and not to trip the whole MCC on a single outgoing feeder fault?
- ☐ Confirm the MCC bus bar rating and construction (Form type) — a low-form or under-rated MCC bus is a common SPOF for all loads on that MCC.
- ☐ Verify VFD bypass arrangements where provided — a VFD failure with no bypass on a critical load is a SPOF for the driven process function.
- ☐ Check PFC (power factor correction) connection point — shunt PFC banks failing open can cause voltage/reactive power issues that trip loads indirectly.

Zone 6 — Standby / Essential Supply (DG, ATS)

i **Typical consequence if SPOF present:** Essential loads dead on utility failure; ATS or DG unable to start. Severity: Critical (for safety-critical loads). Typical mitigation: regular tested auto-start, redundant ATS control, essential bus clearly defined.

Progress: 0 / 7

- ☐ Confirm the DG rating is sufficient for the defined essential load — if the essential load list has grown since commissioning, the DG may now be undersized.
- ☐ Verify the essential bus is clearly defined and segregated: only loads that must run on generator should be connected. Non-essential loads left on the essential bus overload the DG and cause it to trip — a SPOF for all essential functions.
- ☐ Check ATS operation: is the transfer from mains to DG **automatic** on mains failure detection? What is the auto-start to load time (indicative: <15–30 s)?
- ☐ Confirm ATS transfer logic — does a failed ATS cause the essential bus to stay dead? Is there a manual transfer bypass?
- ☐ Verify DG auto-start test records (weekly no-load, monthly on-load indicative). A DG that has not been tested recently is a high-probability SPOF.
- ☐ Check DG fuel level, fuel storage capacity (hours at full essential load), and fuel delivery contract — fuel exhaustion is a simple and avoidable SPOF.
- ☐ Confirm DG battery (starter battery) condition and last replacement date — a failed battery is a very common reason for a DG not to start.

Zone 7 — Control / Protection Supplies (UPS, DCDB)

i **Typical consequence if SPOF present:** DCS / PLC brownout, loss of protection relay trip power, inability to operate circuit breakers. Severity: High (cascades into all other systems). Typical mitigation: redundant UPS, battery monitoring, bypass capability.

Progress: 0 / 6

- ☐ Confirm a UPS is installed for DCS / PLC / relay panels. Check the rated hold-up time (battery autonomy at full load, **indicative minimum 15–30 min for UPS**; confirm per plant) — is it current or degraded?
- ☐ Verify the UPS battery has been tested (discharge test) within the maintenance interval and that battery string health is monitored (cell voltage, temperature).
- ☐ Check UPS bypass path — can the UPS be bypassed for maintenance without a supply interruption to critical control panels? A UPS with no bypass is itself a SPOF during UPS maintenance or failure.
- ☐ Confirm the DCDB (DC distribution board) is fed from a dedicated battery charger, and that DC supplies reach all protection relay trip / close coils. Check the rated DCDB battery hold-up time (**indicative minimum 2 h**; confirm per plant requirement).
- ☐ Verify DCDB battery charger AC supply — if it is fed from the same bus as the equipment it is meant to back up, a bus fault defeats the DC backup.
- ☐ Check there is no single MCB or fuse in the DC distribution that, if it operates, cuts power to all relay trip coils simultaneously — that is a common-element SPOF for protection.

Zone 8 — Earthing and Neutral (MV-NER, LV neutral)

i **Typical consequence if SPOF present:** Loss of earth fault protection selectivity, dangerous touch potentials, or undetected earth fault developing into a phase fault. Severity: High (safety). Typical mitigation: periodic resistance check, thermal monitoring, continuity testing of earth conductors.

Progress: 0 / 5

- ☐ Confirm MV-NER is connected and resistance value is within design tolerance (measured at last maintenance, with certificate).
- ☐ Verify only one NER is connected at a time when the MV bus-section is closed — two parallel NERs alter the earth fault current and can defeat protection settings.
- ☐ Check LV transformer neutral earthing (solidly earthed at one point per bus section) — multiple neutral earthing points or a broken neutral are both SPOFs for LV earth fault protection.
- ☐ Confirm the main earth bar continuity: a broken earth conductor is invisible under normal conditions but becomes a dangerous SPOF during a fault event.
- ☐ Verify LV earth fault relay settings match the NER impedance and cable capacitance — a mis-set relay that fails to detect an earth fault is a protection SPOF.

★ **Complete all eight zones before closing the field inspection. Annotate findings directly on the SLD print. Assign a finding number to each SPOF identified; these numbers flow directly into the risk register and recommendations in Part 4.**

Zone-to-SPOF-type mapping

The eight zones above align to the SPOF library variants (§ SPOF Library). Use this cross-reference to connect a zone finding to the appropriate modification concept:

Checklist zone	SPOF type (library)	Typical modification
Zone 1 — MV intake / utility incomer	Type C — single utility incomer / single MV bus (§ Variant C)	MOD-02 (second MV incomer); DG + ATS (MOD-04)
Zone 2 — MV switchgear	Type C — single MV bus (§ Variant C)	MOD-02 (MV bus-section); MOD-09 (arc-flash detection)
Zone 3 — Transformers TX-1/TX-2	Type A — single transformer / single LV bus (§ Variant A)	MOD-04 (add TX-2 + split bus); MOD-01 (bus-tie)
Zone 4 — LV main switchboard & bus-tie	Type B — two TXs, no bus-tie (§ Variant B)	MOD-01 (auto-close bus-tie)
Zone 5 — MCCs and outgoing feeders	Type D — shared cable / MCC for duty + standby (§ Variant D)	MOD-03 (split MCCs); MOD-07 (separate cable routes)
Zone 6 — Standby / essential supply (DG, ATS)	Type A/C — no backup for utility loss (§ Variant A, § Variant C)	MOD-04 (DG + ATS); MOD-05 (upgrade to auto ATS)
Zone 7 — Control / protection supplies (UPS, DCDB)	Control SPOF — control/protection supply loss (all variants, see § Variant E)	MOD-08 (UPS + DCDB installation)
Zone 8 — Earthing and neutral	Earthing SPOF — no standard variant; cross-reference P-8 earthing philosophy	MOD-10 (NER resize / earth-grid extension)

Reliability Metrics

SPOF analysis answers *where* the weak points are. Reliability metrics answer *how bad* each weak point is, in quantitative terms. These four quantities are the common language of reliability engineering.

Quantity	Symbol	Formula	Meaning
Failure rate	λ	failures / hour	How often a component fails on average
Mean time between failures	MTBF	1 / λ	Average operating hours between successive failures
Mean time to repair / restore	MTTR	hours	Average time from failure detection to restoration (includes spares, labour, access)
Reliability	R(t)	$e^{-\lambda t}$	Probability of zero failures in operating time t
Availability	A	MTBF / (MTBF + MTTR)	Fraction of time the function is actually available
Unavailability	U	1 – A	Fraction of time down; multiply by 8760 h/yr for annual downtime estimate

Worked example — single transformer availability

WORKED EXAMPLE — SINGLE TRANSFORMER (TX-1) AVAILABILITY (INDICATIVE)

Given indicative data: **MTBF = 200,000 h** (transformer major failure, oil-filled distribution type), **MTTR = 168 h** (one week, including spare logistics).

Step 1 — Availability:

A = MTBF / (MTBF + MTTR) = 200,000 / (200,000 + 168) = 0.99916

Step 2 — Unavailability:

U = 1 – 0.99916 = 8.4 × 10⁻⁴

Step 3 — Estimated annual downtime:

Annual downtime = U × 8760 h/yr = 8.4 × 10⁻⁴ × 8760 ≈ 7.4 h/yr

Result: A single transformer is unavailable for approximately 7.4 hours per year on average (indicative). This sets the baseline for comparison with redundant configurations in the RBD section.

All values indicative — use project-specific or published reliability data for real studies.

✓ **Practical implication of MTTR:** MTTR is strongly influenced by spares holding, access, and contractor response time — all factors within the plant operator's control. A critical spare transformer held on-site can cut MTTR from 168 h to perhaps 24–48 h (indicative), which roughly doubles the annual availability. Reducing MTTR is often the most cost-effective lever available.

Reliability Block Diagrams (RBD)

A reliability block diagram (RBD) models how individual component availabilities combine at the system level. Two fundamental topologies:

Series and Parallel Topologies

Topology	Block representation	Formula	System availability vs components
Series (all must work)	[A ₁] — [A ₂] — [A ₃] — ...	A _{sys} = A ₁ × A ₂ × A ₃ × ...	Always <i>lower</i> than the worst individual component
Parallel (1-out-of-2)	[A ₁] in parallel with [A ₂]	A _{par} = 1 – (1 – A ₁)(1 – A ₂)	Always <i>higher</i> than either component alone

WORKED EXAMPLE — ESSENTIAL PATH AVAILABILITY WITH AND WITHOUT BUS-TIE

Using the indicative transformer availability derived in [Section Reliability Metrics](#): A_{TX} = 0.99916 each.

Case 1 — Single transformer (SPOF — Variant A):

A = 0.99916 → estimated downtime ≈ 7.4 h/yr

Case 2 — Two transformers, no bus-tie (Variant B) — each bus still a SPOF:

Each bus: A = 0.99916 (unchanged – the two transformers are in series with their own buses, not in parallel with each other without a tie)

Case 3 — Two transformers with auto-close bus-tie (master SLD / Variant E):

With the bus-tie, TX-1 and TX-2 are effectively in a 1-out-of-2 parallel arrangement for the essential path:

A_{par} = 1 – (1 – 0.99916)² = 1 – (8.4 × 10^{–4})² = 1 – 7.1 × 10^{–7} ≈ 0.9999929

Annual downtime ≈ (1 – 0.9999929) × 8760 ≈ 0.006 h/yr (about 22 seconds per year – indicative)

Comparison: Adding the bus-tie reduces estimated annual downtime of the essential path from ~7.4 h/yr to ~0.006 h/yr — a factor of approximately 1 200× improvement (indicative), for the cost of one auto-transfer scheme.

Series elements on the essential path (the common-element catch):

The parallel TX benefit is capped by the *series* elements that both transformers share. For the essential path: MV-SWGR (series) → (TX-1 || TX-2) → LV-MSB bus → BT (series) → MCC → load. Each series element reduces system availability:

A_{essential} ≈ A_{MV-SWGR} × A_{par}(TX) × A_{LV-MSB} × A_{BT} × A_{MCC}

Using indicative values (MV-SWGR: MTBF 500,000 h, MTTR 48 h → A ≈ 0.99990; LV-MSB busbar: MTBF 1,000,000 h, MTTR 72 h → A ≈ 0.99993; BT breaker: MTBF 200,000 h, MTTR 24 h → A ≈ 0.99988; MCC incomer: MTBF 300,000 h, MTTR 8 h → A ≈ 0.99997):

A_{essential} ≈ 0.99990 × 0.9999929 × 0.99993 × 0.99988 × 0.99997 ≈ 0.99968 → ~28 h/yr estimated downtime (dominated by series elements, not the transformers)

Lesson: The transformer redundancy is so effective that the system availability is now dominated by the *series* common elements — MV switchgear and bus-tie. This is exactly the Module 3 insight, now quantified. Improving MTTR (faster repair, on-site spares) on these series elements gives more benefit than trying to add further transformer redundancy.

All values indicative — use project failure data for real studies.

MTTR as a Design Lever

An **automatic bus-tie (BT)** or **ATS** reduces MTTR for the *essential load* from hours (manual switching, call-out, access) to seconds (automatic transfer). This single change typically raises availability far more cheaply than chasing a lower component failure rate:

Transfer method	MTTR for essential load	Estimated annual downtime (illustrative)
Manual switching (call-out, access, operate)	~4–8 h (indicative)	~3–7 h/yr per transformer outage event
Auto-close bus-tie or ATS (100–500 ms)	<1 s (indicative)	<0.01 h/yr equivalent (effectively zero for most processes)

★ Improving MTTR through automation (auto-transfer, ATS) is usually the most cost-effective reliability lever. It does not require new hardware on the supply side — only a correctly designed and tested control scheme.

FMECA — Failure Mode, Effects and Criticality Analysis

FMECA (IEC 60812) is the formal tool for converting a SPOF list into a scored, ranked risk register. For each component, you identify its **failure modes**, trace the **local and system effects**, and score **criticality**.

S / O / D Scale Definitions (1-5)

The table below defines what each score value *means* for Severity, Occurrence and Detection. Use these definitions consistently across all FMECA rows — calibrate to project data (IEEE 493 / ISO 14224) for formal studies. *All descriptors are indicative; adapt thresholds to the plant's specific process criticality.*

Score	Severity (S) — worst system-level consequence	Occurrence (O) — failure frequency / likelihood	Detection (D) — how easy to find
1	Negligible — minor local disturbance; no process impact; corrected in minutes	Very rare — once per 20+ years; MTBF > 150,000 h (e.g. NER resistor)	Continuously monitored with alarm; certainly detected immediately
2	Minor — brief interruption of non-critical sub-system; process dip <5 min; no safety event	Unlikely — once per 10-20 years; MTBF 60,000–150,000 h (e.g. MV switchgear panel)	Periodic inspection / routine testing at the maintenance interval
3	Significant — partial process shutdown or sub-area loss; production loss >1 h; no safety impact	Occasional — once per 3-10 years; MTBF 10,000–60,000 h (e.g. transformer, LV ACB)	Intermittent monitoring or testing between scheduled maintenance
4	High — major production loss or half-plant outage; or a significant safety concern (arc-flash PPE boundary violated)	Moderate — once per 1-3 years; MTBF 3,000–10,000 h (e.g. MCC complete, ATS, UPS)	No routine monitoring; failure only detected after event (standby equipment, latent failure)
5	Catastrophic — whole-plant blackout; safety-critical load lost; personnel injury risk or major equipment damage	Frequent — more than once per year; MTBF < 3,000 h (e.g. ageing VFD power electronics, corroded battery)	Effectively undetectable without testing; fault event reveals it (D = 5)

- ✓ **Using these anchors:** Start with Severity. If loss of this one item causes a whole-plant blackout or a safety event, $S \geq 4$ regardless of probability. Then set Occurrence from MTBF data (IEEE 493 / Appendix G). Finally set Detection: if there is no live monitoring and no regular test record, $D \geq 4$. A $D = 5$ row (hidden failure) almost always reaches P1 or P2 RPN.

Scoring method — RPN

The most common scoring scheme is the **Risk Priority Number (RPN)**:

Factor	Symbol	Scale	Meaning (higher = worse)
Severity	S	1-5	How bad is the worst system-level consequence of this failure mode? (5 = catastrophic / safety-critical)
Occurrence	O	1-5	How likely / frequent is this failure mode? (5 = frequent / high failure rate)
Detection	D	1-5	How difficult is it to detect before it becomes a system event? (5 = effectively undetectable)
RPN	$S \times O \times D$	1-125	Overall risk priority — mitigate highest numbers first
Criticality (Crit)	$S \times O$	1-25	Severity $\times$ Occurrence, used alongside SPOF flag for ranking when detection data is uncertain

- ✓ **Scale guidance (1-5):** S=5 = safety-critical or whole-plant blackout; S=4 = essential loads lost; S=3 = significant process impact; S=2 = minor/local disruption; S=1 = negligible. O=5 = frequent (annual or more often); O=1 = very rare (<once per 20 yr). D=5 = failure is latent / undetectable without testing; D=1 = continuously monitored with alarm.

Priority bands

After scoring, assign each row a priority band. These bands are consistent with the SF-03 priority rule in [Part 5 — Post-Site Analysis](#):

Band	Condition	Action
P1	SPOF with $S \geq 4$, or $RPN \geq 60$	Immediate action — safety-critical or highest-risk; act before next scheduled outage
P2	$RPN 36-59$	Short-term — plan and implement within 6-12 months
P3	$RPN 16-35$	Medium-term — address in next planned maintenance window (12-24 months)
P4	$RPN < 16$	Monitor — review at next FMECA update; no immediate modification required

- ✓ **Detection (D) is the key lever.** Adding condition monitoring (winding temperature, oil level, battery discharge test, relay self-test) lowers D and therefore RPN without changing the hardware or its failure rate. This is often the cheapest mitigation available.

Worked FMECA extract — master SLD key components

The table below is a representative FMECA extract for the principal components of the indicative 2 MW plant. Scores are illustrative; calibrate the S/O/D scales and use project-specific or published failure data on a real study.

Scale reminder: S, O, D each scored 1–5 (5 = worst). RPN = S × O × D (max 125). All scores are illustrative training values — calibrate with project-specific data for real studies. Priority bands: P1 = SPOF with S ≥ 4 or RPN ≥ 60; P2 = RPN 36–59; P3 = RPN 16–35; P4 < 16.

☐	Tag	Function	Failure mode	Local effect	System effect	S	O	D	RPN	Priority	Recommended mitigation	
☐	MV - SWGR	MV distribution; isolation	MV busbar fault / arc	MV bus de-energised; incomer CB trips	Total plant blackout (both TXs lose source); DG/ATS essential path only	5	1	3	15	P3	MV bus-section breaker; arc flash detection; SF ₆ or sealed technology; strict access control	
☐	MV - SWGR	MV incomer switching / protection	Incomer CB fails to trip on fault	Fault not cleared; upstream protection operates	Upstream supply lost — utility outage; whole plant affected until utility restores	5	2	4	40	P2	Annual CB trip-coil continuity test; protection relay secondary injection test; trip circuit supervision alarm	
☐	TX-1	Step MV → LV; feed Bus A	Winding fault (internal)	TX-1 trips on differential / Buchholz; Bus A dead	Half plant loses supply; BT must close — if BT fails, Bus A stays dark	4	2	3	24	P3	Auto bus-tie BT + N-1 loading verified; 87T differential protection; winding temperature and oil-level monitoring	
☐	TX-2	Step MV → LV; feed Bus B	Winding fault (internal)	TX-2 trips on differential / Buchholz; Bus B dead	Half plant loses supply; BT must close — if BT fails, Bus B stays dark	4	2	3	24	P3	Auto bus-tie BT + N-1 loading verified; 87T differential protection; winding temperature and oil-level monitoring	
☐	BT	Interconnect Bus A and Bus B; N-1 back-feed	Fails to close on bus-tie demand	BT remains open; dead bus stays dark	Essential loads on dead bus lost for duration of repair — N-1 credit eliminated	5	2	4	40	P2	Auto-transfer scheme with redundant control; annual on-load transfer test; BT position monitoring and alarm	
☐	LV - MSB	Distribute LV to all MCCs	Busbar fault / arc-flash event	Bus section trips; all outgoing feeders on that section de-energised	All loads on affected bus section lost; extended outage during repair/inspection	5	1	4	20	P3	Form 4 Type 7 compartmentalisation (contains arc to one functional unit); arc-flash detection; planned arc-flash study; routine PM	
☐	☐	DG	Standby power for essential bus	Fails to start on mains loss	DG does not reach rated voltage / frequency within transfer time	Essential bus stays dark during mains loss event; all essential loads lost	5	4	3	60	P1	
☐	ATS	Transfer essential bus from mains to DG and back	Fails to transfer (control or contactor fault)	ATS remains on dead mains; DG runs unloaded or not at all	Essential bus dead during mains outage; safety-critical loads lost for duration	5	3	4	60	P1	Redundant ATS control relay; monthly on-load transfer test; alarm to PMS on ATS fault; manual bypass provision	
☐	UPS	AC no-break supply for DCS / PLC / relays	Battery failure / inverter fault	UPS output interrupted or degraded	DCS / PLC loss-of-power trip; relay panels de-energised; protection may mal-operate or fail to operate	4	3	3	36	P2	Redundant UPS (N+1 modules or 2 × 50 % UPS with static switch); battery discharge test every 6 months; bypass verified; battery monitoring alarm	
☐	DCDB	Battery-backed DC supply for trip / close coils	Charger fault or battery depletion	DC rail voltage falls below trip coil pick-up level	All protection relay trip coils lose power; circuit breakers cannot be tripped — a SPOF for the entire protection system	5	2	4	40	P2	DC bus voltage monitoring with alarm; periodic battery discharge test; redundant charger; trip circuit supervision on all critical CBs	
☐	MV - NER	Limit MV earth fault	Open-circuit fault (resistor)	MV system becomes	Earth fault overvoltages;	4	2	5	40	P2	Thermal overload monitoring; annual	

☐	Tag	Function	Failure mode	Local effect	System effect	S	O	D	RPN	Priority	Recommended mitigation
		current; enable selective earth fault protection	or connection fails open)	effectively unearthed	phase-to-phase fault escalation; earth fault relay loses sensitivity — hidden hazard						resistance check; NER connected alarm; verify NE in circuit whenever MV bus section is closed (parallel NERs)

RPN scores are illustrative training values — calibrate S/O/D scales with real plant failure data before use in any project study. Indicative only.

Reading the FMECA output

Sort the FMECA table by RPN descending. The top P1 entries — ATS (RPN 60) and DG (RPN 60) — are the immediate-action priority items. P2 items (RPN 36–59) include BT, MV-SWGR incomer CB (both RPN 40), DCDB (RPN 40), MV-NER (RPN 40) and UPS (RPN 36). Note how **detection (D)** drives a large share of the risk scores: adding monitoring lowers D and cuts RPN without changing hardware. This is consistently the most cost-effective first step.

Risk Ranking and Recommendations Register

The FMECA RPN scores are converted to a ranked recommendations register — the deliverable that drives plant improvements. The process has three steps:

Step 1 — Plot on a risk matrix

Map each FMECA finding to a **Severity × Probability** matrix and assign a risk colour. Act on findings top-down:

Severity ↓ / Probability →	Low (rare)	Medium (occasional)	High (frequent)
High (Critical consequence)	Medium	High	CRITICAL
Medium	Low	Medium	High
Low	Low	Low	Medium

Step 2 — Build the recommendations register

Each finding becomes a row in the recommendations register. A minimum register structure:

☐	Ref	Tag / Location	Finding / SPOF	Risk (before)	Recommended action	Risk (after)	Priority	Owner	Target date
☐	REC-01	ATS	ATS fails to transfer on utility loss — essential bus stays dark	High (RPN 60)	Install redundant ATS control relay; commission monthly on-load transfer test	Low (RPN est. 15)	1 — Immediate	—	—
☐	REC-02	MV-NER	NER open-circuit fault — undetected MV earth fault risk	Medium (RPN 40)	Add NER thermal monitor with alarm; annual resistance measurement programme	Low (RPN est. 16)	2 — Short term	—	—
☐	REC-03	TX-1 , TX-2	Transformer winding fault with BT failing to close — half-plant extended outage	Medium (RPN 24)	Commission and test BT auto-transfer scheme; add 87T differential protection; winding temperature alarm	Low (RPN est. 12)	3 — Medium term	—	—
☐	REC-04	DG	DG fails to start on mains loss — essential bus stays dark	High (RPN 60)	Implement weekly no-load / monthly on-load test regime; add fuel and battery monitoring	Low (RPN est. 15)	1 — Immediate	—	—
☐	REC-05	BT	Bus-tie fails to close — N-1 benefit lost on transformer outage	Medium (RPN 40)	Annual on-load BT transfer test; BT position monitoring and alarm; BT trip coil continuity supervision	Low (RPN est. 16)	2 — Short term	—	—
☐	REC-06	DCDB	DC rail collapses — all protection trip coils lose power	Medium (RPN 40)	Add DC bus voltage monitoring alarm; redundant charger; trip circuit supervision on all critical CBs	Low (RPN est. 16)	2 — Short term	—	—
☐	REC-07	UPS	UPS battery or inverter failure — DCS/PLC unplanned trip	Medium (RPN 36)	6-monthly battery discharge test; add redundant UPS module; verify bypass path	Low (RPN est. 12)	2 — Short term	—	—
☐	REC-08	LV-MSB	Busbar arc-flash — extended section outage pending inspection and repair	Low (RPN 20)	Upgrade to Form 4 Type 7 if not already; add arc-flash detection; commission arc-flash hazard study	Low (RPN est. 8)	3 — Medium term	—	—

Step 3 — Link findings forward

Each recommendation in the register flows directly into the modifications catalogue and the site visit action list. See [Part 4 — Modifications Catalogue](#) for the detailed implementation guidance on each item, and [the Jubail case study](#) for an example of how this register was applied in practice on a similar plant.

- ✓ **State risk before and after:** Every recommendation should include the estimated RPN (or risk colour) both before and after the mitigation is applied. This makes the value of each action explicit and enables prioritisation when budget is constrained.

Self-Check Questions

Work through these questions using the SPOF library and reliability sections above. Reveal the answer only after attempting each question.

Q1 — Identifying a whole-plant SPOF

In Variant A (single transformer, single bus), name **two distinct elements** that are each individually capable of causing a whole-plant blackout.

▼ Show answer

Any two of the following are correct: **TX-1** (the single transformer); the **single un-split LV bus (LV-MSB)**; the **MV feeder cable to TX-1**; the **LV incomer circuit breaker**. Any single one of these failing removes supply from the entire plant — there is no alternative path and no load segregation.

Q2 — The bus-tie and N-1

Variant B installs two transformers, yet it is not N-1 secure. Explain why — and state exactly what single addition gives N-1 capability (subject to transformer loading).

▼ Show answer

Without a bus-tie, the two transformers are electrically isolated — the healthy transformer has no path to the dead bus. Loss of TX-1 drops *all* of Bus A; loss of TX-2 drops *all* of Bus B. Half the plant is lost on any single transformer/feeder failure.

The single addition that restores N-1 capability is a **normally-open auto-close bus-tie (BT)** between Bus A and Bus B, with an automatic transfer scheme that closes BT within seconds on loss of either incomer — provided the surviving transformer is rated to carry the combined essential load in N-1 condition.

Q3 — Hidden SPOF upstream

Variant C has two transformers, a split bus, and a functional bus-tie. Why can a total plant blackout still occur?

▼ Show answer

Because the SPOF is *upstream* of the transformers. All power flows through a **single utility feeder** and a **single MV incomer CB** on a **single MV bus**. If the utility supply is lost, the incomer CB fails, or an MV bus fault occurs, both transformers simultaneously lose their source. The LV bus-tie cannot help — it interconnects two dead buses. The lesson: always trace the full path from utility boundary to load, not just the LV distribution.

Q4 — Common element behind apparent redundancy

In Variant D, the upstream supply is fully redundant (two transformers, two buses, bus-tie). Where is the SPOF, and how do you remove it?

▼ Show answer

The SPOF is the **shared outgoing cable and single MCC-1 incomer** that supply both the duty pump *and* its standby pump from the same bus section. One cable fault, MCC bus fault, or MCC incomer trip takes both pumps offline at once — the standby gives no protection against this common element.

Remove it by feeding the **duty pump from MCC-1 on Bus A** and the **standby pump from MCC-2 on Bus B**, with separately routed supply cables. No single cable, MCC, or bus section now disables both the duty and standby function.

Q5 — Why "two of everything" can still be wrong

A colleague says: "We have two transformers, two buses, and two pumps — we're N-1 secure." Using the lessons from Variants B and D, explain why this statement can still be wrong.

▼ Show answer

Variant B lesson: Two transformers and two buses without a bus-tie means each bus still has only one source. Lose one transformer, lose half the plant. The "redundancy" of the second transformer is inaccessible to the dead bus.

Variant D lesson: Two pumps supplied by the same cable, same MCC, and same bus section — a common element — means a single cable or MCC fault disables both duty and standby simultaneously. The "redundancy" of the second pump is defeated by the shared supply.

Conclusion: N-1 security requires not just N+1 equipment but also that *no single shared element* can disable the function. Always check for common elements that bridge the apparent redundancy.

Q6 — Reliability metrics calculation

A component has **MTBF = 150,000 h** and **MTTR = 120 h**. Calculate its availability and estimated annual downtime.

▼ Show answer

Availability:

$$A = \text{MTBF} / (\text{MTBF} + \text{MTTR}) = 150,000 / (150,000 + 120) = 0.99920$$

Unavailability:

$$U = 1 - 0.99920 = 7.99 \times 10^{-4}$$

Estimated annual downtime:

$$U \times 8760 \text{ h/yr} = 7.99 \times 10^{-4} \times 8760 \approx 7.0 \text{ h/yr (indicative)}$$

Q7 — Series vs parallel availability

Two identical units, each with **A = 0.99**. Find the system availability in: (a) series; (b) 1-out-of-2 parallel. Comment on the difference.

▼ Show answer

(a) Series:

$$A_{\text{ser}} = 0.99 \times 0.99 = 0.9801 \rightarrow \sim 174 \text{ h/yr downtime}$$

(b) Parallel (1-out-of-2):

$$A_{\text{par}} = 1 - (1 - 0.99)^2 = 1 - (0.01)^2 = 0.9999 \rightarrow \sim 0.9 \text{ h/yr downtime}$$

Comment: The parallel arrangement is approximately 200× better than series for the same two components. This is the fundamental quantitative argument for N+1 with a transfer path (bus-tie). Series elements (such as the MV switchgear) cap the system availability; they must be addressed through MTTR reduction (faster repair, auto-transfer) rather than adding parallel components.

Q8 — FMECA scoring and detection

An FMECA entry for the ATS initially scores **S = 5, O = 3, D = 4** (1–5 scale). (a) Calculate the RPN. (b) If monthly on-load testing improves detection to **D = 2**, what is the new RPN? (c) What does this illustrate?

▼ Show answer

(a) Initial RPN:

$$\text{RPN} = S \times O \times D = 5 \times 3 \times 4 = 60$$

(b) After improving detection:

$$\text{RPN} = 5 \times 3 \times 2 = 30 \text{ (reduction of 50 \%)}$$

(c) Illustration: Improving *detection* through regular testing cuts the RPN by 50 % without any change to the hardware or its failure rate. This is often the lowest-cost mitigation available — a well-structured maintenance and testing programme is a direct reliability investment.

★ **Bridge to Part 4:** You have now identified the SPOFs, quantified their availability impact, scored their criticality in the FMECA, and ranked them in a recommendations register. Part 4 — [Modifications Catalogue](#) — takes the highest-priority findings from this register and develops the detailed implementation guidance: what to specify, how to verify, and what to document. Move there next to close the loop from analysis to action.

Continuity-of-Supply Philosophy Templates

Where you are in the workflow: you have identified the SPOFs and calculated the numbers (Parts 2–3); now apply the standard philosophies to judge whether each plant's design is sound, and to specify what must change.

i **First assignment?** On day one you mainly need the [Modifications Catalogue](#) (below); it gives you the ready-made rows to copy into your Recommendations Register. The eight philosophy templates in this section are reference material for deeper assessments and peer review — return to them once you have completed your first plant cycle.

The eight templates below form a reusable reference. Each one can be applied directly to a plant you are assessing: scan the philosophy, locate the matching equipment on the SLD, and use the **How to verify** and **What a deficiency looks like** checklists to produce a rapid, defensible finding. Templates P-1 to P-8 are cumulative layers of the same goal — **no single event interrupts the process**.

i **Note — indicative figures:** All ratings, times and settings in this section are indicative and to be confirmed by project-specific studies, vendor data and the governing grid code.

P-1 — N+1 Transformers, Split Bus & Normally-Open Bus-Tie (Baseline)

Principle. Provide two full-rated MV/LV transformers (**TX-1** , **TX-2**) feeding a *split* LV main switchboard (**LV-MSB**) with a normally-open bus-tie (**BT**). Each transformer is rated to carry the entire essential plant load alone — N-1 capability. Under normal conditions, both incomers are closed and the bus-tie is open (split-bus, approximately equal loading). On loss of one incomer, the bus-tie closes to back-feed the dead bus from the surviving transformer.

When it applies. Every MV/LV industrial plant with a continuous or semi-continuous process. It is the baseline — all other philosophies build on top of it.

How to verify on a site visit.

Progress: 0 / 6

- ☐ Confirm two separate transformers (**TX-1** , **TX-2**) are installed and both in service.
- ☐ Check **LV-MSB** is genuinely split — two bus sections, each fed by one transformer incomer.
- ☐ Verify the bus-tie (**BT**) is normally *open* (breaker in "OPEN" position, padlocked or interlocked in NOP state).
- ☐ Confirm nameplate kVA of each transformer ≥ full essential load (N-1 loading check — see [Part 2 — Load Flow](#)).
- ☐ Check that the bus-tie has an *auto-close* relay or PMS function enabled (not just manual).
- ☐ Review protection study to confirm auto-close interlocks prevent parallel running during fault (bus-tie must not close onto a fed fault).

What a deficiency looks like.

- Single transformer feeding an un-split bus (SPOF type A — see [Part 3 — Variant A](#)).
- Two transformers present but bus-tie has no auto-close function — bus-tie is a manual-only switch.
- N-1 loading check fails: one transformer cannot carry the essential load; loss of one source will shed critical process motors.
- Bus-tie permanently closed (running in parallel) without differential protection — a fault on one bus propagates to both.
- Single main incoming breaker at LV-MSB feeding an un-sectioned bus (common to small plants).

P-2 — Automatic Bus Transfer (Fast / In-Phase / Residual-Voltage)

Principle. On loss of an LV incomer (voltage collapse, transformer trip, MV feeder loss), the **BT** bus-tie closes *automatically* to restore supply to the dead bus without operator intervention. Three transfer modes exist — select the fastest that the plant process tolerates:

Mode	Transfer time	How it works	Process impact
Fast transfer	< ~100 ms	Bus-tie closes before residual motor voltage drifts out of phase with the incoming source. Requires synchronism-check relay or PMS pre-authorisation.	Motors keep spinning; minimal process dip. Preferred where possible.
In-phase transfer	~100 ms - 1 s	Waits for the instant the residual bus voltage and incoming voltage are in phase (synchro-check). Slower but avoids high re-acceleration inrush.	Moderate speed dip on large motors; process may dip but typically not trip.
Residual-voltage (time-delayed) transfer	~2 - 10 s	Waits for residual motor voltage to decay below a threshold (e.g. < 25 % rated) before closing. Avoids out-of-phase closing but allows motors to coast to near-stop.	Large process dip; motors will require re-acceleration (see P-6). Least preferred.

When it applies. Any plant that has a split bus with a bus-tie. The mode selection depends on motor inertia, process sensitivity and relay capability.

How to verify on a site visit.

Progress: 0 / 5

- ☐ Confirm the bus-tie protection relay (or PMS) has an auto-transfer function enabled.
- ☐ Ask the operations team or consult the relay settings: what triggers the auto-transfer (UV relay pick-up level and time delay)?
- ☐ Confirm transfer mode is documented (fast / in-phase / residual-voltage) and matches the relay configuration.
- ☐ Check that the transfer logic includes a check that the receiving bus has a live healthy source before closing the bus-tie.
- ☐ Review event logs or test records: has the auto-transfer been tested in the last 12 months?

What a deficiency looks like.

- Bus-tie auto-transfer function is *disabled* in relay settings — only manual transfer possible (SPOF type B — see [Part 3 — Variant B](#)).
- Residual-voltage transfer configured when fast transfer is achievable — unnecessarily slow, causing motor stalls.
- No UV relay on the incomer — bus-tie has no trigger condition and will not operate automatically on loss of incomer.
- Transfer logic closes the bus-tie regardless of whether the surviving source is healthy — risk of cascading a fault onto the remaining bus.
- No evidence of regular test records for the auto-transfer function.

P-3 — Standby Generation & ATS Load Build-Up

Principle. On total utility supply loss, a diesel generator set (**DG**) starts automatically, runs up to rated voltage and frequency (typically 10–20 s), and the automatic transfer switch (**ATS**) connects it to the essential bus (**EDB** / **DB**). Loads are reconnected in a *staged priority sequence* (black-start load build-up) to avoid collapsing the DG bus with simultaneous motor starting inrush. The DG is sized for the essential load — not the full plant load.

When it applies. All plants with safety-critical or continuous-process loads that cannot tolerate a total utility blackout for more than the DG start-up period.

How to verify on a site visit.

Progress: 0 / 7

- ☐ Confirm **DG** nameplate kVA ≥ essential load (confirm essential load list exists and is current).
- ☐ Check the **ATS** is set to auto-mode and confirm the utility-loss trigger setpoint (UV and time delay).
- ☐ Verify the DG auto-start sequence: is the start signal hardwired from the **ATS** or via the PMS?
- ☐ Locate the load build-up sequence document — priority groups, inter-step time delays, and total reconnect time. Does the DG have adequate kVA headroom for each step?
- ☐ Check fuel level indicator and fuel tank capacity (days of autonomy at full essential load).
- ☐ Review DG test log: monthly no-load test and quarterly on-load test are minimum good practice.
- ☐ Confirm the DG output breaker interlock prevents parallel operation with the utility (anti-paralleling relay or mechanical interlock on the ATS).

What a deficiency looks like.

- DG present but **ATS** is in manual mode — operator must manually transfer during an emergency.
- No documented essential load list or load build-up sequence; DG may be overloaded on reconnect.
- DG kVA less than the verified essential load — DG will trip on overload when all essential motors reconnect.
- No anti-paralleling interlock — DG can be connected in parallel with a live utility supply without a synchroniser, risking DG damage or backfeed.
- DG is the only source for the bus-tie control power — loss of utility before DG starts blacks out the bus-tie relay itself.

P-4 — Load Shedding / Load Management Philosophy

Principle. When available generation is less than demand (utility loss, N-1 transformer loading, running on DG), a *load-shedding* scheme automatically disconnects **non-essential loads** in a defined priority order to protect the power supply to the essential process. Load shedding is triggered by under-frequency (UF), under-voltage (UV), or source overload detection, and acts *before* the supply collapses — typically within 1–5 cycles on a GOOSE-based system or 2–5 s on a traditional under-frequency relay scheme.

Priority tiers (indicative — confirm per plant):

Tier	Examples	Action on shed trigger
Essential (never shed)	Safety systems, fire pump, emergency lighting, DCS, UPS , DCDB , ventilation for hazardous areas	Never disconnected by automatic shedding

Tier	Examples	Action on shed trigger
Priority 1 — shed last	Critical process pumps, compressors, main product line drives	Shed only on deepest overload (Stage 3+)
Priority 2	Utility services (cooling towers, HVAC compressors, secondary pumps)	Shed at Stage 2 of the scheme
Priority 3 — shed first	Non-essential loads, office buildings, workshops, site lighting (non-safety)	Shed immediately at Stage 1 (first trigger)

When it applies. Any plant with more load than a single source can carry (N-1 condition) or with a standby DG that is smaller than the total load.

How to verify on a site visit.

Progress: 0 / 6

- ☐ Confirm a written load-shedding philosophy document exists and is current.
- ☐ Verify priority assignments match the actual process criticality — ask the plant engineer.
- ☐ Check relay/PMS settings: UF setpoints (e.g. 49.0 / 48.5 Hz staged), UV setpoints and time delays.
- ☐ Confirm load-shedding actuators (trip coils on MCC feeders or contactors) are wired and commissioned.
- ☐ Review test records showing the scheme has been function-tested within the last year.
- ☐ Check that critical **VFD** -driven loads are not shed before the drives have had time to attempt ride-through (coordinate with P-6).

What a deficiency looks like.

- No load-shedding scheme — on N-1 transformer loading the surviving transformer simply overloads and trips, causing a total blackout.
- Priority list is notional only (written on a diagram note) but not wired to any relay output.
- Essential loads (fire pump, DCS) included in the auto-shed trip chain — risk of shutting down safety systems automatically.
- Single-stage shedding only — all non-essential loads shed simultaneously, causing a large load step that overshoots and sheds essential loads too.
- UF/UV setpoints not coordinated with the generator AVR/governor response time — scheme sheds before the generator has had time to respond.

P-5 — No-Break Supplies: UPS & DC Control Battery (DCDB)

Principle. The control system, protection relays, DCS/PLC I/O and critical instrumentation must remain energised through *any* supply disturbance — bus transfer, DG start-up, voltage dip, or total blackout. Two independent no-break systems serve this:

- **UPS** — online double-conversion UPS providing clean 230/400 V AC with zero-interruption transfer to battery. Feeds DCS/SCADA servers, PMS, operator consoles, critical instrumentation, communications, and emergency lighting.
- **DCDB** — 110 V DC (or 48/24 V DC) battery-backed charger panel feeding MV and LV protection relay trip/close coils, circuit-breaker charging motors, and switchgear indication. Remains live even when the AC bus is dead.

Both systems must have sufficient battery autonomy to ride through the longest credible outage before DG pick-up and reconnection (minimum 30 min autonomy for DCS; 2 h or more for DCDB is common good practice — confirm per plant requirement).

When it applies. All plants — without this layer, protection relays cannot operate during a blackout and the DG/ATS sequence cannot be supervised.

How to verify on a site visit.

Progress: 0 / 6

- ☐ Locate **UPS** and **DCDB** panels — confirm they are energised and alarms clear.
- ☐ Check **UPS** battery age against manufacturer's recommended replacement interval (typically 5 years for VRLA).
- ☐ Read the **DCDB** charger panel: float voltage, battery condition indicator, last discharge test date.
- ☐ Confirm **UPS** and **DCDB** are fed from the essential bus (**EDB**) or the **ATS** output, not from the bus that goes dark on utility loss.
- ☐ Review battery discharge test records — a full autonomy test (simulate mains loss, run on battery, confirm hold-up time) should be completed annually.
- ☐ Verify that protection relay DC supplies are from **DCDB**, not from a 230 V AC adapter that dies with the bus.

What a deficiency looks like.

- No **DCDB** — protection relay trip coils fed from AC 230 V adapters; on total blackout, relays cannot operate and breakers cannot be closed.
- **UPS** present but fed from the non-essential bus — goes dark precisely when it is needed most.
- Battery age > 7 years and no recent discharge test — capacity may be well below nameplate; hold-up time is unknown.

- **UPS** in static-bypass mode (common after a fault) — inverter is not running; next mains dip causes an outage to the "protected" DCS loads.
- **DCDB** charger alarm active (e.g. "battery fault" LED) but left unattended — plant is unaware the DC backup is compromised.

P-6 — Motor Re-Acceleration After Transfer or Voltage Dip

Principle. When supply is restored after a bus transfer or a voltage dip, multiple large motors will attempt to accelerate simultaneously. Their combined locked-rotor inrush (up to $6\text{--}7 \times \text{F.L.C.}$ per motor) can collapse the restored bus voltage and prevent any motor from accelerating — a re-acceleration failure. The remedy is a *sequenced re-acceleration* scheme: motors restart in priority groups with time delays between groups, so each group reaches running current before the next group is connected.

Variable-frequency drives (**VFD**) offer additional protection: *kinetic buffering* (regenerating energy from the spinning load into the DC bus) and *flying-restart* (re-synchronising to the spinning motor) allow VFDs to ride through short dips (100–500 ms) and reconnect without a process stop.

When it applies. Any plant with multiple large motors ($> \sim 30 \text{ kW}$ individually, or significant aggregate DOL starting current) and a bus transfer or DG start-up sequence.

How to verify on a site visit.

Progress: 0 / 5

- ☐ Confirm a re-acceleration sequence document exists listing motor groups and inter-step time delays.
- ☐ Check MCC starters: are auto-restart relay modules fitted (e.g. time-delay undervoltage restart relays)?
- ☐ For **VFD** -driven loads, confirm flying-restart and kinetic-buffering functions are enabled in the drive parameter settings.
- ☐ Verify that the DG's voltage and frequency stabilisation time is accounted for before the first motor group reconnects.
- ☐ Review motor control philosophy: "auto-restart enabled / disabled" status for each motor — safety-critical loads (e.g. agitators in vessels) may require manual restart.

What a deficiency looks like.

- No re-acceleration sequence — all motors attempt to restart simultaneously after a transfer, collapsing the restored bus and tripping the bus-tie or DG on overload.
- Auto-restart disabled globally as a blanket safety measure — operators must manually restart every motor after any supply event, causing extended process downtime.
- VFD kinetic-buffering disabled or trip-on-undervoltage set to $< 50 \text{ ms}$ — drives trip on every short utility dip, stopping the process when there was no real fault in the plant.
- Re-acceleration time between groups too short — bus voltage collapses during step 2 and starves subsequent groups.

P-7 — Protection Coordination Philosophy (Selectivity & Clearing Time)

Principle. Protection must be *selective*: the device closest to the fault (the "discriminating" relay or fuse) clears before any upstream device operates. This isolates only the faulted zone and leaves the rest of the plant energised. Without selectivity, a fault on one MCC feeder can cascade to trip the whole LV bus or even the MV incomer. Equally, *clearing time* must be fast enough to limit fault damage and arc-flash incident energy (see [Part 2 — Arc Flash](#)) while remaining slow enough to discriminate from the zone below.

The protection coordination study must cover: MV fuses / IDMT relays (at **MV-SWGR**), LV ACBs (incomers, bus-tie and main feeders at **LV-MSB**), MCC MCCBs and fuses, and motor overload/short-circuit protection at every starter — full top-to-bottom cascade.

For a full treatment of protection, refer to [Part 2 — Protection Coordination](#).

How to verify on a site visit.

Progress: 0 / 6

- ☐ Confirm a current protection coordination study exists and has been updated after any major plant modification.
- ☐ Check relay settings against the study: IDMT curves, pick-up current, time-multiplier settings. Do they match?
- ☐ Verify grading margins: upstream relay should have $\geq 0.25\text{--}0.4 \text{ s}$ margin over the downstream relay at the maximum fault current (confirm with the study).
- ☐ Check that bus-tie relay settings do not conflict with the incomer relay — bus-tie overcurrent must grade with both incomers.
- ☐ Review incident energy calculations at key buses — if incident energy $> 40 \text{ cal/cm}^2$ at any working position, this is a critical finding. See [Part 2](#).
- ☐ Confirm relay test records: IEC 60255 relay testing at commissioning and periodic functional tests (at least every 3 years for numerical relays).

What a deficiency looks like.

- Protection settings have not been reviewed since commissioning; plant load has grown and the original study is no longer valid.
- Upstream relay grading margin $< 0.2 \text{ s}$ — risk of sympathetic tripping of the whole bus on a downstream feeder fault.
- All LV MCCBs rated the same as the incomer ACB — no discrimination; the incomer trips before the MCCB on a feeder fault.

- MV incomer IDMT relay time-multiplier set to minimum (fast) — clears every disturbance including downstream faults before the LV protection has time to operate.
- No record of relay test since commissioning — relay may be mal-set or defective.

P-8 — Earthing Philosophy (MV Resistance Earth via MV-NER; LV TN-S)

Principle. The earthing philosophy governs fault current magnitudes, touch voltages, EPR (earth potential rise) and protection sensitivity — it underpins all ground-fault detection. This plant uses two distinct earthing regimes:

- **MV system (11 kV) — resistance earthed via MV-NER** : the neutral of each MV/LV transformer (TX-1 , TX-2) star winding is connected to earth through a neutral earthing resistor. The NER limits the single-phase earth-fault current to a defined level (indicatively 200–400 A for 30 s) — high enough for sensitive earth-fault relays to detect, low enough to limit damage and EPR. A running earth fault is detectable and the system can remain in service for a short period while the fault is located and isolated.
- **LV system (400/415 V) — TN-S (solid neutral earth)**: the LV transformer neutral is solidly earthed at the transformer; the protective earth (PE) and neutral (N) conductors are separate throughout the installation. Earth-fault current is high (limited only by loop impedance), enabling fast fuse/MCB/RCCB operation. See [Part 2 — Earthing](#) for calculation method.

When it applies. All plants — earthing regime affects protection relay settings, cable insulation requirements, touch-voltage compliance and arc-flash severity.

How to verify on a site visit.

Progress: 0 / 6

☐

Locate MV-NER : confirm resistance value (stamped on nameplate), voltage rating, continuous and timed current rating.

☐

Check that the MV earth-fault relay (51G/64) pickup current is correctly set for the NER fault current (typically $\approx 10\text{--}20\%$ of NER rated current for sensitive detection).

☐

At the LV transformer neutral, verify the neutral earthing point is accessible, bolted solid to the earth electrode, and clearly labelled. Check earth electrode condition if visible.

☐

Confirm PE and N conductors are kept separate (TN-S): check at the main earth terminal in LV-MSB — PEN conductors (combined PE+N, TN-C) must not appear downstream of the MSB.

☐

Check earth grid continuity: main earth bar, building structural earth, MV cable screen earths, and equipment frames all bonded to the same grid.

☐

Review most recent earth electrode resistance test value (target typically $< 1\ \Omega$ for a large plant; confirm per plant specification).

What a deficiency looks like.

- MV-NER absent or bypassed — MV system is either unearthed (high transient overvoltages on earth fault) or solidly earthed (very high fault currents, equipment damage).
- NER resistance value incorrect for the system voltage: too high → earth-fault protection insensitive; too low → defeats the purpose of resistance earthing.
- LV neutral not earthed (floating neutral) — TN-S system undefined; touch voltages uncontrolled; MCBs and RCCBs may not trip on earth fault.
- TN-C wiring present downstream of the MSB (PEN conductors in sub-distribution) — earth-fault protection sensitivity is reduced; shock risk is increased.
- Earth electrode resistance $> 10\ \Omega$ — substation EPR on fault will exceed 430 V touch-voltage limit; personnel hazard.
- MV cable screen not bonded at both ends — circulating currents and touch voltage on screen during a cable fault.

★

Key takeaway — the eight templates form a stack: P-1 (redundancy) makes the topology robust; P-2 (auto-transfer) and P-3 (DG/ATS) restore supply on a source failure; P-4 (load shedding) prevents an overloaded source from collapsing; P-5 (UPS/DCDB) keeps the brain alive through any disturbance; P-6 (motor re-acceleration) reconnects the process without overloading the restored source; P-7 and P-8 underpin all of the above with selective protection and a safe earth reference. A plant that fully implements all eight templates has no credible single event that interrupts the essential process.

DCS / SCADA / PMS Integration & IEC 61850

Where you are in the workflow: having assessed the power-system hardware, now evaluate the intelligence layer that automates transfers, sheds loads and monitors the plant in real time.

The electrical system is not an island. It is supervised by a layered control architecture linking field devices through a Power Management System (PMS) to the plant DCS and SCADA. Understanding this architecture is essential because a SPOF in the control layer can neutralise expensive hardware redundancy — a bus-tie that cannot close automatically because the PMS has failed is no different from a bus-tie that was never installed.

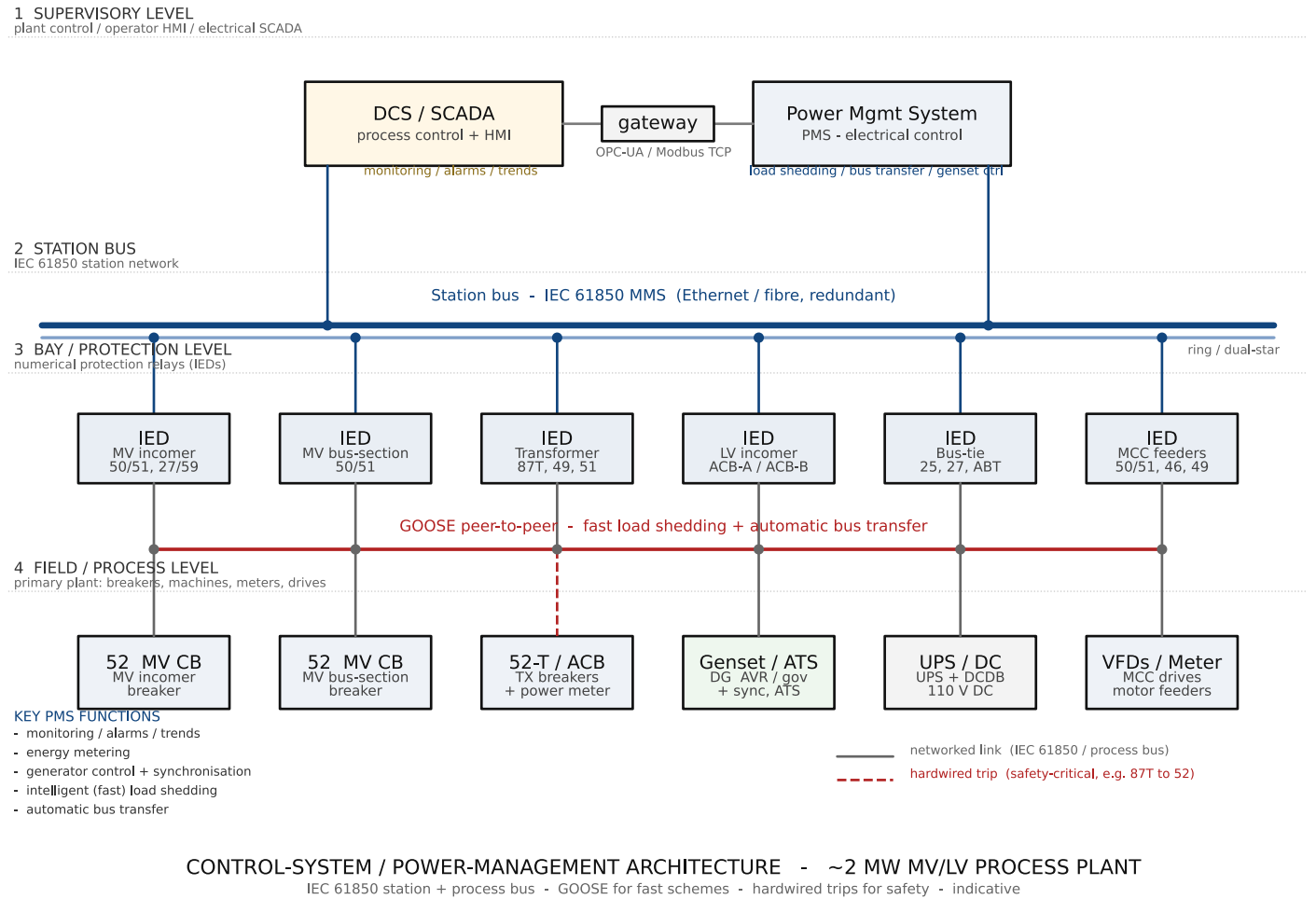


Fig 4-1 — Layered control architecture: field IEDs and devices, IEC 61850 station bus, Power Management System (PMS), and DCS/SCADA supervisory level (indicative)

The Four Layers

Layer	Equipment / function	Relevant tags
Layer 4 — DCS / SCADA	Process control and operator HMI. Receives electrical status, alarms and metering from the PMS. Issues high-level commands ("start DG", "shed non-essential bus") — does not directly operate individual breakers in normal operation.	Plant-wide; communicates via OPC-UA or Modbus TCP gateway from the PMS
Layer 3 — Power Management System (PMS)	Dedicated electrical "brain": load shedding logic, automatic bus transfer, generator control and synchronisation, source priority management, energy metering aggregation, power-quality alarming. Acts on GOOSE messages from IEDs in sub-10 ms for fast functions.	LV-MSB (BT control), DG, ATS, MCC-1..n shed outputs
Layer 2 — Bay / protection IEDs	Numerical multifunction relays at each breaker — protection, measurement, control and event recording. Each IED is both a protection device and a source of real-time data (current, voltage, power, energy, fault records, event logs). IEC 61850 IEDs report directly onto the station bus.	MV-SWGR IEDs, LV-MSB incomer / bus-tie relays, MCC feeder relays
Layer 1 — Field / process devices	Breakers (device number 52), generator controller (AVR + governor + synchroniser), ATS logic, UPS and DCDB monitors, VFD controllers, power quality meters at intake.	MV-MET, TX-1/2 winding monitors, DG, VFD, PFC controller

Communications Protocols

A redundant Ethernet / fibre station bus carries multiple protocols simultaneously:

- **IEC 61850 MMS (Manufacturing Message Specification):** client-server reporting for monitoring, settings changes, commands and historian feeds. Used between IEDs and the PMS/SCADA. Bandwidth-efficient; latency typically 50–500 ms acceptable for this role.
- **IEC 61850 GOOSE (Generic Object Oriented Substation Events):** fast multicast peer-to-peer messaging *between IEDs* — typical latency < 4 ms. Enables fast bus transfer, fast load shedding interlocks and protection interlocks without routing through the PMS server. This is the key enabler of sub-cycle electrical automation.
- **Modbus TCP / DNP3 / OPC-UA:** gateways to the DCS and third-party devices (energy meters, power quality monitors). Standard for legacy integration.
- **PROFIBUS / PROFINET / EtherNet/IP:** to VFD controllers (**VFD**) and motor starters in the MCCs.

i **Note — IEC 61850 GOOSE vs polled Modbus:** Traditional Modbus polling cycles are typically 100–1000 ms. A load-shedding scheme that relies on polled Modbus may take hundreds of milliseconds to respond to a generation shortfall — fast enough for some schemes, too slow for fast bus-transfer or short-circuit interlocks. GOOSE operates on a publish/subscribe model with sub-4 ms latency and self-supervision (heartbeat), making it reliable for safety-critical interlocks.

Hardwired vs Networked: The Right Philosophy

The choice between hardwired and networked signals is a risk decision, not a technology preference. The rule of thumb is:

Signal type	Preferred path	Reason
Master trip, emergency stop, safety interlock (e.g. transformer Buchholz trip, fire suppression trip)	Hardwired — dedicated copper to trip coil	Deterministic, independent of IED / network health, fail-safe on wire break (if normally energised-to-hold)
Fast bus transfer authorisation, fast load-shed trip to feeder breaker	GOOSE (IEC 61850) or hardwired	Sub-4 ms; GOOSE is acceptable if the station bus is redundant and GOOSE retransmission is configured; hardwired provides ultimate independence
Control commands (open/close breaker) from PMS or DCS	Networked (IEC 61850 MMS or Modbus)	Operator-issued; latency of hundreds of ms acceptable; network path is auditable and logged
Status / measurement / alarms / metering	Networked (MMS / Modbus / OPC-UA)	Not time-critical; rich data, historian integration, no hardwiring cost

⚠ Warning — network SPOF: If the PMS server is the single path for the bus-tie auto-close command, a PMS server failure becomes a SPOF in the control layer. Ensure that IED-to-IED GOOSE or hardwired control wiring provides a *backup* path for the bus-tie transfer that does not depend on the PMS server being alive.

How to Verify the Control Architecture on Site

Progress: 0 / 6

- ☐ Confirm the PMS or IED-based auto-transfer function is enabled and tested (not just installed).
- ☐ Ask for the GOOSE configuration or hardwired logic diagram for bus-tie auto-close — trace the signal path end-to-end.
- ☐ Verify redundancy of the station bus: dual-ring or dual-star Ethernet, with managed switches and alarming on link failure.
- ☐ Check that the DCS receives live electrical alarms (transformer temperature, UPS fault, DC battery fault) — not just a generic "electrical fault" point.
- ☐ Confirm cybersecurity: control LAN is segmented from the corporate LAN (firewall / DMZ); IED firmware versions are current; remote access (if any) uses a VPN with multi-factor authentication.
- ☐ Review the PMS UPS supply: the PMS server must itself be fed from **UPS** — a PMS that dies with the bus cannot execute the transfer scheme.

★ Key takeaway: the PMS and IEC 61850 GOOSE layer is the nervous system of the electrical system. A SPOF in this layer can prevent all the hardware redundancy from operating. Always verify that the auto-transfer signal path is independent of the supply it is meant to restore, and that hardwired back-up exists for all safety-critical trips.

Best Practice & Emerging Technology

Where you are in the workflow: use this section when writing the Recommendations Register — these are the technologies and approaches that should be cited when proposing enhancements beyond the baseline philosophy.

Digital Substation (IEC 61850 Process Bus)

In a conventional substation, copper wiring runs from every CT and VT secondary terminal to every relay panel. In a digital substation, *merging units* (MUs) sit at the primary equipment and digitise CT/VT samples, transmitting *Sampled Values* (SV) over fibre to the IED bay units. Benefits: dramatic reduction in copper wiring and termination points (a major source of wiring errors and maintenance burden), self-supervision of the measurement chain, simpler future expansion, and a unified data model (IEC 61850-9-2) that allows any IED from any vendor to consume the same SV stream. For an existing brownfield plant, a selective retrofit (new IEDs with process-bus capability at the point of planned relay replacement) is more practical than a full digital conversion.

Intelligent / Fast Load Shedding

Traditional under-frequency load shedding responds in 2–5 s. Modern systems using **IEC 61850 GOOSE** messaging and a PMS with a real-time power balance model can shed the correct load block in **under 50 ms** after a generation event. The scheme knows the exact MW of each feeder (from IED metering), calculates the deficit in real time, and sheds only what is necessary — no over-shedding, no under-shedding. This eliminates the risk of the DG tripping on overload during a utility loss event because non-essential loads were not shed quickly enough.

✓ **Tip — specify the response time:** When writing a recommendation for a load-shedding upgrade, specify the required response time (e.g. "shed non-essential load within 100 ms of utility loss detection") — this drives the vendor to propose a GOOSE-based solution rather than a relay-timer scheme.

Arc-Flash Detection & Zone-Selective Interlocking (ZSI)

Arc-flash incident energy is proportional to *arcing time*. Two technologies reduce clearing time without sacrificing selectivity:

- **Arc-flash light + current detection:** optical sensors in the switchgear busbar compartment detect the arc flash signature (intense light pulse + overcurrent) and trip the upstream breaker within 8–12 ms — an order of magnitude faster than IDMT relay operation. Dramatically reduces incident energy at the bus.
- **Zone-Selective Interlocking (ZSI):** IEDs exchange ZSI signals: if both the downstream IED and the upstream IED see overcurrent simultaneously, it is a through-fault and the upstream IED stays slow (maintaining discrimination). If only the upstream IED sees current, it is a bus fault and the upstream IED trips instantaneously. ZSI achieves fast clearing for bus faults while maintaining full selectivity for feeder faults — without relaxing grading margins.

Reference: [Part 2 — Arc-Flash Assessment](#) for the incident energy calculation methodology.

Condition Monitoring & Predictive Maintenance

Unplanned failures of transformers, switchgear and motors are a primary cause of SPOF events — even well-designed redundant systems are defeated by simultaneous or sequential failures of components under stress. Condition monitoring moves maintenance from time-based (change everything at fixed intervals) to condition-based (change it when data says it is degrading). Key technologies:

- **Partial discharge (PD) monitoring** on MV cables, switchgear and transformer bushings — detects insulation degradation before it becomes a fault.
- **Transformer winding temperature (WTI / OTI) and DGA** (dissolved gas analysis for oil-filled transformers) — detects thermal stress and incipient faults.
- **Breaker health monitoring** — contact resistance, spring energy, operating time — detects wear before a mechanical failure during a fault.
- **Motor winding temperature and vibration sensors** integrated into the MCC IED — early detection of overload or mechanical bearing failure.
- **Digital twin / analytics platform** — aggregates sensor data, applies ML models, and alerts the maintenance team to anomalies days or weeks before failure.

Battery Energy Storage System (BESS) & Grid-Forming Inverters

A BESS connected to the LV-MSB or MV bus via a bidirectional inverter can fulfil multiple roles simultaneously, making it one of the highest-value additions to a modern plant:

Function	How BESS delivers it
Ride-through & bridging power	Supplies full plant load for the DG start-up period (10–30 s) — eliminates the process dip between utility loss and DG pick-up; replaces the ATS dead time.
Synthetic inertia	Grid-forming inverter emulates the inertia of a synchronous machine — slows ROCOF on loss of generation, giving protection schemes time to respond.
Voltage / frequency support	Fast active and reactive power injection within milliseconds — stabilises voltage and frequency during disturbances before other controls respond.
Peak shaving	Reduces peak demand charges by discharging during high-demand periods; charges during low-tariff periods.
Islanding capability	With a grid-forming inverter and adequate energy capacity, BESS + DG can form a stable microgrid, disconnected from a disturbed utility, and reconnect seamlessly when the grid is stable.

i **Note — BESS sizing:** BESS sizing for ride-through is driven by the DG start-up time and the essential load (e.g. 500 kW essential load × 30 s start-up time = ~4.2 kWh minimum, including inverter efficiency and state-of-charge margins). All figures are indicative; detailed sizing requires a dynamic simulation study.

★ **Key takeaway:** the six technology areas above (digital substation, intelligent load shedding, arc-flash detection/ZSI, condition monitoring, and BESS) represent the difference between a plant that handles its first SPOF event by luck and one that handles it by design. When writing the Recommendations Register, reference these as concept-level enhancements beyond the P-1 to P-8 baseline.

Grid Voltage Instability from Renewables

Where you are in the workflow: this section helps you identify whether the plant's exposure to grid-quality disturbances is a SPOF in its own right — and what mitigations to recommend.

As wind and solar generation displace synchronous plant on the transmission network, the grid becomes "weaker": less stored kinetic energy (inertia), a lower short-circuit level, and more volatile voltage and frequency. An industrial plant connected to such a grid will experience **more frequent and more severe disturbances at its point of connection** — dips, swells, fast rate-of-change-of-frequency (ROCOF) events, harmonics, and flicker — even when no fault occurs within the plant. On these 21 plants in the Central and Eastern sector, utility grid mix and renewable penetration should be confirmed per plant, as exposure varies.

How Grid Instability Hurts the Plant

Grid disturbance	Plant effect	Example trigger
Voltage dip (short interruption < 1 s)	Contactors drop-out (contactors typically release at 60–70 % Vn), VFD DC-bus undervoltage trip, motor stall	Distant fault clearance on the transmission network; switching of a large inverter-connected source
Fast ROCOF event	Loss-of-mains (LOM) relay at DG trips the generator offline; protection schemes operating faster than intended	Sudden disconnection of large renewable block from the grid
Voltage swell (> 110 % Vn)	Overvoltage trip on sensitive electronics, accelerated insulation ageing, capacitor bank overload	Load rejection on a weak grid or tap-changer hunting
Harmonic distortion (THD rising)	Overheating of capacitor banks (PFC resonance), neutral conductor overload, interference with protection relays	Large inverter-based resources connected to the MV network near the plant
Voltage flicker	Visible flicker on lighting, interference with sensitive process instrumentation	Wind generation output variability, arc furnaces on the same feeder

Ride-Through: Don't Trip on a Transient

The first line of defence is ensuring the plant does **not** trip on disturbances it can survive. This requires coordinated settings across multiple device types:

- Under-voltage ride-through (UVRT) — time-delayed UV trips:** set the UV trip relay time delay so the plant rides through dips that conform to the applicable grid-code LVRT curve (e.g. a 500 ms dip to 50 % Vn should not trip the plant). Coordinate with the upstream utility protection so genuine faults in the plant still clear within the required time.
- Contactor ride-through:** DC-held coils or time-delay undervoltage relays on contactors hold them closed through a short dip (typically 0.5–1 s hold-up). Alternatively, specify contactors with a minimum hold-in voltage of 45–50 % Vn rather than the standard 70 %.
- VFD kinetic buffering / flying-restart:** enable these functions in drive parameters. Kinetic buffering allows the drive to regenerate motor inertia energy into the DC bus, extending ride-through. Flying restart re-synchronises to a spinning motor after a supply restoration without requiring a coast-to-stop cycle.
- Loss-of-mains (LOM) relay sensitivity:** if the DG is fitted with a ROCOF-based LOM relay, confirm the ROCOF setting is not over-sensitive. An overly sensitive LOM relay trips the DG on a grid event even when the DG is supplying essential loads correctly — converting a grid disturbance into a plant blackout.

Dynamic Voltage Support: Hold the Voltage Up

The second line of defence is active voltage support at the plant's point of connection. For plants where grid instability is identified as a material risk:

- STATCOM / SVC:** fast dynamic reactive power compensation (response in 1–4 cycles); stabilises voltage, reduces flicker, and supports the plant voltage through deep dips. Far faster than switched capacitor banks.
- Dynamic Voltage Restorer (DVR):** series compensation device that injects voltage in series with the supply, correcting dips and swells in real time for sensitive downstream loads. Typical response: < 1 ms.
- Thyristor-switched or fast-switched detuned PFC banks:** faster reactive support than contactor-switched **PFC** banks; ensure banks are detuned (7 % reactor) to avoid resonance with rising background harmonic levels.
- OLTC / AVR coordination:** confirm transformer on-load tap-changer (OLTC) deadband and response time are set to maintain LV bus voltage within ±5 % at all load conditions. Over-hunting of the tap-changer can worsen voltage instability.

Storage and Local Stability

For severe grid environments or plants with critical processes that cannot tolerate any interruption, storage-based solutions provide the highest level of protection:

- BESS with grid-forming inverter:** provides synthetic inertia, fast frequency response, and can island the essential plant automatically during a severe grid event. On grid restoration, the BESS microgrid controller checks synchronisation and reconnects the plant seamlessly. See [Best Practice — BESS](#).
- UPS and flywheel:** keep control systems and critical instrument loads firm regardless of any external disturbance. Essential baseline (P-5).

See It Before It Trips You — Power Quality Monitoring

Continuous power quality (PQ) metering at the MV intake (**MV-MET**) and on critical LV buses provides the data to:

- Characterise the actual grid disturbance environment (how many dips per month, minimum voltage, ROCOF histogram).
- Set ride-through relay margins correctly based on measured data rather than assumptions.
- Detect rising harmonic background and proactively adjust **PFC** tuning before resonance occurs.
- Support claims against the utility if grid-quality events cause plant damage or downtime.

✓ **Tip — field check:** During the site visit, ask whether the plant has experienced unexplained process trips that coincided with apparent voltage events. Check the DCS alarm historian for clusters of "undervoltage" or "frequency alarm" events — these are evidence of grid instability affecting the plant, and they justify a recommendation for ride-through improvements or PQ monitoring.

★ **Key takeaway:** a plant that trips on every grid disturbance has an effective SPOF in the utility connection, regardless of how much internal redundancy it has. Ride-through settings, contactor hold-in improvement, VFD flying-restart, and PQ monitoring are low-cost first steps; STATCOM, DVR or BESS are the higher-cost solutions for severe grid environments. Always check the PQ history before concluding the plant is reliable.

Typical Modifications Catalogue

Where you are in the workflow: use this catalogue when populating the Recommendations Register. Each row maps one modification to the SPOF it resolves, the affected tags, a concept description, and an indicative cost band. All costs are indicative H/M/L bands only — confirm scope and cost by market enquiry.

The catalogue below covers the most common electrical modifications identified on MV/LV industrial process plants. Every modification cross-references the SPOF type from the [Part 3 — SPOF Library](#) (variants A–E). Use these rows as the starting point for concept modification packages — each package must include a marked-up SLD, an equipment outline specification, a scheme description / sequence of operation, and a high-level bill of materials.

i Note — concept level only: All modifications described here are at concept / feasibility level. Detailed design, equipment specification, procurement, and construction are out of scope for the SPOF assessment campaign. Indicative cost bands: **H** = high (typically > USD 200 k installed); **M** = medium (USD 30 k – 200 k); **L** = low (< USD 30 k). All figures are indicative and to be confirmed by market enquiry. *Cost band currency shown is USD (catalogue order-of-magnitude basis); actual project costs must be confirmed by local market enquiry — the Jubail case study in Part 5 uses SAR as the client's currency. Indicative bands are currency-agnostic at H/M/L level.*

Mod ID	Modification	SPOF type addressed (see)	Affected tag(s)	Concept description	Typical scope to spec
MOD-01	Add normally-open auto-close bus-tie to LV-MSB	Type B — two transformers, no bus-tie	LV-MSB , BT	Install a 4-pole ACB bus-tie breaker (rated to carry full essential load) between Bus A and Bus B of the LV-MSB. Auto-close relay (UV-triggered on either incomer failure) with synchro-check and interlock to prevent closing onto a fault. Hardwired or GOOSE signal path independent of PMS server.	ACB rating and frame size; protection relay specification (auto-transfer + overcurrent synchro-check); wiring/cabling to existing bus; updated protection coordination study; FAT and commissioning test procedure.
MOD-02	Second MV incomer and MV bus-section breaker	Type C — single utility incomer / single MV bus	MV-SWGR , MV-MET	Provide a second independent utility feeder terminating on a second MV incomer panel with a normally-open bus-section breaker between the two MV bus sections. Dual independent MV metering. Auto-close bus-section on loss of either incomer (with protection coordination — bus-section relay must grade with both feeders). Second NER (MV-NER) for the second section.	New MV switchgear extension (panels, busbars, cabling to new utility termination); updated MV IDMT relay settings; NER sizing; protection coordination study update; utility connection agreement amendment.
MOD-03	Split duty/standby MCCs across Bus A and Bus B	Type D — shared MCC / shared cable for duty and standby loads	MCC-1 , MCC-2 , LV-MSB	Reassign the standby motor starter(s) from the same MCC panel as the duty motor to a separate MCC panel fed from the alternate LV bus. Route the standby feeder cable on a physically separate path (different cable tray / conduit / trench, minimum 1 m separation). This ensures a single bus, panel, or cable fault cannot remove both duty and standby simultaneously.	New MCC panel (or panel extension) on the second bus; new outgoing feed from LV-MSB Bus B; cable routing survey and trench/tray design; updated LV protection coordination; updated SLD.
MOD-04	Diesel generator set and ATS for essential loads	Type A/C — no backup for utility loss	DG , ATS , EDB	Install a diesel generator set sized for the confirmed essential load list, with automatic transfer switch and essential distribution board. Automatic start on utility loss detection (UV relay), ATS transfer within the specified dead	DG sizing (kVA, voltage frequency); ATS type and rating; EDB panel and distribution layout; essential load schedule and priority sequence; fuel system; acoustic enclosure; load test procedure; updated SLI and protection study.

Mod ID	Modification	SPOF type addressed (see)	Affected tag(s)	Concept description	Typical scope to spec
				time, load build-up sequence, anti-parallel interlock, and weekly / monthly test capability. Fuel tank for minimum 24 h autonomy at full essential load.	
MOD-05	Upgrade manual ATS to automatic transfer with PMS interlock	Type A/C — ATS in manual mode negates backup	ATS , DG	Replace existing manual or semi-automatic ATS with a fully automatic ATS unit interfaced to the PMS. PMS controls the transfer sequence (start DG, confirm voltage and frequency, transfer load, shed non-essential, confirm essential load is stable) and provides alarm and event records. Include hardwired UV trigger as a backup to PMS-commanded transfer.	New ATS panel or modification; PMS interface wiring / GOOSE; updated sequence-of-operation document; revised load build-up schedule; commissioning and test certificate.
MOD-06	Add or resize detuned PFC capacitor bank	Type — power factor penalty / harmonic resonance (not a SPOF but a reliability and quality risk)	PFC , LV-MSB	Install or replace a 7 % detuned capacitor bank (filter reactor + capacitor steps) at the LV-MSB to correct power factor to ≥ 0.95 lagging at full load. The 7 % detuning ratio prevents resonance with the 5th harmonic and above. Automatic power factor controller (APFC) switches stages in response to reactive demand. Size based on load-flow reactive demand — indicatively 400–600 kVAr for a 2 MW plant, confirmed by a reactive power survey.	Reactive power survey and load-flow study; capacitor bank rating a harmonic audit; APFC relay; installation space MSB; updated protection and harmonic study.
MOD-07	Separate cable routing of duty/standby feeders	Type D — common cable route for duty and standby	MCC-1 , MCC-2 , field loads	Where duty and standby feeders currently share a cable tray, conduit or buried trench for any part of their route, physically separate them to different routes with minimum 1 m lateral separation (or structural fire barrier). This ensures a cable fire, mechanical damage, or flooded trench cannot take out both the duty and standby supply simultaneously. Survey and document all current dual-feed routes.	Cable routing survey drawings (current and proposed); new cable tray / conduit / trench works; cable termination at both ends; fire barrier if separation is < 300 mm; updated cable schedule.
MOD-08	Install UPS and DCDB for control system independence	Type — control SPOF: protection and DCS lose supply during transfer/blackout	UPS , DCDB	Install an online double-conversion UPS (min. 30 min battery autonomy at full DCS / protection load) feeding all relay panels, DCS I/O, PMS servers, and communications. Install a separate 110 V DC DCDB (min. 2 h autonomy) for MV and LV protection relay trip/close coils, breaker charging motors, and	UPS kVA sizing (DCS + communications + relay panels + emergency lighting); DCDB Ah sizing (protection load + breaker charging currents $\times$ hold-up time); sealed lead-acid or Li-ion battery selection; charger and rectifier specification; essential bus feeder capacity; commissioning and autonomy test.

Mod ID	Modification	SPOF type addressed (see)	Affected tag(s)	Concept description	Typical scope to spec
				emergency lighting. Both systems fed from the essential bus downstream of the ATS.	
MOD-09	Arc-flash detection and zone-selective interlocking (ZSI)	Type — arc-flash severity / slow clearing of bus faults	LV-MSB , MCC-1..n , MV-SWGR	Fit arc-flash light + current detection sensors in LV-MSB busbar compartments and selected MCC panels. Link sensors to the incomer ACB trip coil via a dedicated arc-flash protection relay (response < 10 ms). Where IEDs support it, configure ZSI between the incomer relay and feeder relays to achieve instantaneous clearing of bus faults while maintaining discrimination for feeder faults. Re-run arc-flash incident energy calculations after implementation to verify reduction.	Arc-flash protection relay selection; sensor types and positions; wiring to trip coils; IED ZSI compatibility check; updated arc-flash study; arc-flash hazard label update; commissioning test.
MOD-10	Earthing remediation — NER resize and earth-grid extension	Type — earthing deficiency (MV unearthed or NER incorrect; LV earth grid resistance high)	MV-NER , LV-MSB , plant earth grid	If the MV-NER resistance value does not match the calculated earth-fault current requirement (see Part 2), replace with a correctly rated unit. If the LV earth grid resistance is above the target (typically < 1 Ω for a large plant), extend the electrode grid (driven rods, buried ring conductor, or copper mesh) to achieve the required resistance. Retest after installation. Update MV earth-fault relay settings to match the new NER current.	MV earth-fault current calculation; NER resistance and rating specification; LV earth electrode resistance test and target value; electrode design (rod lengths, grid spacing, resistivity); updated MV 51G relay settings; commissioning test certificate; updated earthing study.
MOD-11	Intelligent load-shedding PMS upgrade	Type — slow or absent load shedding on N-1 transformer or DG supply	LV-MSB , MCC-1..n , DG	Replace or supplement existing under-frequency relay load shedding with a PMS-based fast load-shedding scheme. The PMS monitors real-time MW at each feeder (via IED metering), detects a generation deficit event (utility loss or transformer trip), and issues GOOSE trip commands to non-essential feeder breakers within 50–100 ms. Priority groups are configurable; scheme can shed the minimum necessary load rather than a fixed block.	PMS software load-shedding module specification; IED GOOSE configuration for all shed feeders; priority load schedule and interlock list; PMS-to-DCS interface for shedding status alarms; factory acceptance test; commissioning test record; updated sequence of operation.
MOD-12	VFD ride-through and motor re-acceleration scheme	Type — motor stall / re-acceleration failure after supply restoration	VFD , MCC-1..n	Enable kinetic buffering and flying-restart in all VFD parameter sets. For direct-on-line (DOL) motors: install time-delay auto-restart relay modules in starters (essential motors only, per risk assessment). Prepare and commission a	VFD parameter review and update for each drive; auto-restart relay module procurement for selective starters; re-acceleration sequence document; PMS sequence configuration; updated motor control philosophy; commissioning test (simulate supply dip and

Mod ID	Modification	SPOF type addressed (see	)	Affected tag(s)	Concept description	Typical scope to spec
					motor re-acceleration sequence (priority groups and inter-step time delays) and load it into the PMS. Set VFD undervoltage ride-through window to minimum 500 ms (or to the expected transfer time, whichever is greater).	confirm successful re-acceleration).

✓ **Tip — filling the Recommendations Register:** Copy the Mod ID, Modification title, SPOF type and Affected tags directly into the Recommendations Register. The "Typical scope to specify" column becomes the basis for the concept modification package description. Priority (H/M/L) in the register should be driven by the FMECA criticality number, not by cost alone — a low-cost modification addressing a high-criticality SPOF should always be prioritised over an expensive modification addressing a low-criticality finding.

★ **Key takeaway:** the twelve modifications above cover the full set of SPOF types from Part 3. For each plant, scan the catalogue, identify which Mod IDs apply, and create the corresponding rows in the Recommendations Register — linked back to the FMECA finding that justifies them. All modifications are concept level only; each requires a detailed design phase before procurement or construction.

Part 4 — Self-Check

Where you are in the workflow: test your understanding of philosophies and modifications before moving to Part 5 field tasks.

Answer each question from memory, then expand the answer to check.

▼ **Q1 — Bus transfer speed.** A plant has a fast bus transfer (FBT) relay set to close the bus-tie within 80 ms. A reviewer proposes replacing it with a residual-voltage (RV) transfer that waits 5 s. What is the process consequence of the change, and when would the RV approach ever be preferred?

Answer. With the FBT relay, the bus transfer completes before the residual motor voltages drift significantly out of phase — motors keep spinning and the process continues with minimal dip. With the RV transfer, all motors on the dead bus will coast for 5 s (or until their residual voltage decays to the threshold). This typically means a full process stop: pump pressures drop, reactors lose agitation, and a restart sequence is needed. For a continuous process, this is a production loss event that the FBT was designed to avoid.

RV transfer is preferred when: (a) the relay set does not support synchro-check or the motor inertia is very low (residual decays fast anyway); (b) there are large synchronous motors whose residual voltage takes a long time to decay and out-of-phase closing would cause damaging mechanical torque; or (c) the process can tolerate a 5 s stop (e.g. a utility building, not the core process). The key trade-off is process continuity vs mechanical torque on motors during closing.

▼ **Q2 — Load shedding trigger.** A plant loses one of its two transformers. The surviving transformer is rated 1600 kVA. Before the fault, the total plant load was 2600 kVA (split equally, 1300 kVA per transformer). What happens if there is no load-shedding scheme? What should the scheme shed first?

Answer. Without a load-shedding scheme, the surviving transformer is immediately loaded to 2600 kVA — 162 % of its rated capacity. The transformer will trip on thermal overload protection within seconds (if it has a Buchholz / winding temperature relay) or will sustain damage and trip within minutes. This produces a total plant blackout — a worse outcome than the original single transformer loss.

The load-shedding scheme must shed enough non-essential load to bring the surviving transformer below its rated capacity, with a margin. If the essential load is ~1100 kVA and the transformer is rated 1600 kVA, the scheme needs to shed at least 1000 kVA ($2600 - 1600 = 1000$ kVA minimum, preferably more for stability margin). It should shed Priority 3 loads first (site lighting, workshops, offices), then Priority 2 (HVAC, secondary utilities), keeping the core process alive. The essential loads (DCS, fire pump, critical process drives) must never be in the shed list.

▼ **Q3 — DCDB importance.** A plant has a UPS for the DCS but no separate DCDB. The LV-MSB bus goes dead during a transformer fault. Can the bus-tie relay operate to close the bus-tie?

Answer. Only if the bus-tie relay trip coil is energised. If the relay coil is fed from the LV AC bus (e.g. via a 230 V AC adapter), it loses power when the bus goes dead — exactly when the auto-close command is needed. The relay cannot operate and the bus-tie stays open, leaving the dead bus dark even though the surviving transformer is healthy.

A DCDB supplies the relay trip and close coils from a battery-backed 110 V DC system that is independent of the AC bus state. With a DCDB, the bus-tie relay remains powered during the blackout and can execute the auto-close sequence. This is why the DCDB is a non-negotiable element of the control philosophy — the UPS protects the DCS, but the DCDB protects the protection system itself.

▼ **Q4 — IEC 61850 GOOSE vs Modbus.** A PMS designer proposes using Modbus TCP polling (250 ms cycle time) for the fast load-shedding trip signal to MCC feeder breakers. What is the risk, and what should be specified instead?

Answer. The risk is that on a generation shortfall event, the load-shedding command takes up to 250 ms to reach the feeder breakers plus the breaker trip time. On a DG set, frequency may decay at 2–5 Hz/s on a severe overload: in 250 ms the frequency could drop 0.5–1.25 Hz, potentially causing the DG governor to fail to recover or triggering spurious under-frequency trips on other loads before the shed command arrives. A 250 ms Modbus latency is too slow for a fast load-shedding scheme where the target response time is < 100 ms.

The specification should require IEC 61850 GOOSE messaging from the PMS (or directly IED-to-IED) to the feeder breaker trip coils. GOOSE latency on a local Ethernet network is < 4 ms, and the signal is self-supervised (heartbeat — the receiver knows if the publisher has failed). This achieves the required response time. Alternatively, hardwired trip outputs from a dedicated fast load-shedding relay provide a fully deterministic path if GOOSE is not available on the existing IEDs.

▼ **Q5 — Modifications catalogue.** During a site visit you find: (a) duty and standby cooling water pumps are both fed from MCC-1, which is on Bus A; (b) the bus-tie is present but auto-transfer is disabled; (c) the UPS is in static bypass. Write three rows for the Recommendations Register, citing the correct Mod ID for each.

Answer (indicative — adapt to actual plant data).

Rec. Ref	Finding	Mod ID	Recommended action	Priority
REC-01	Duty and standby cooling pumps both fed from MCC-1 (Bus A) — common MCC and bus SPOF defeats standby redundancy	MOD-03	Relocate standby pump starter to a new or extended MCC panel on Bus B with separately routed feeder cable. Update SLD and protection study.	High
REC-02	Bus-tie auto-transfer disabled — manual operation only; loss of one transformer will not result in automatic bus restoration	MOD-01	Re-enable auto-transfer function in bus-tie relay / PMS; confirm UV trigger settings; add DCDB-backed GOOSE or hardwired trip path; issue test certificate.	High
REC-03	UPS in static bypass — inverter not running; next AC supply disturbance will directly interrupt DCS and protection relay supplies	MOD-08	Restore UPS to online double-conversion mode immediately (check reason for bypass — typically a failed battery or faulty inverter). If UPS is unserviceable, procure replacement. Confirm battery age and conduct discharge test.	High

▼ **Q6 — Grid instability.** A plant engineer tells you the process trips unexpectedly two or three times a month, always on night shifts. The events appear to coincide with voltage "brownouts" visible on the DCS trend. No faults are found in the plant each time. What is the most likely cause, and what three actions would you recommend?

Answer. The pattern (recurring, night-shift timing, no internal fault, voltage sag visible) is characteristic of **external grid voltage disturbances** — likely voltage dips from the transmission or distribution network, possibly associated with renewable generation ramping or fault-clearance switching on the utility network. Night-shift timing often correlates with lower system demand (less inertia on the grid) and wind generation peaks.

Three recommended actions:

1. **Install or enable continuous power quality monitoring** at the MV-MET metering point (IEC 61000-4-30 Class A meter). Capture dip depth, duration, and frequency. This provides the evidence needed to characterise the disturbances and correlate them precisely with the process trips.
2. **Review and increase undervoltage ride-through time delays** on contactor coils and VFD undervoltage trip settings. If contactors are releasing at 70 % Vn after 20 ms, even a mild grid dip will cause mass dropout. Set the UV trip delay to ride through at least 500 ms of dips conforming to the applicable grid-code LVRT curve.
3. **Enable VFD kinetic buffering and flying-restart** on all process-critical drives. Check motor control philosophy to confirm whether auto-restart is permitted for each motor (safety review required). This allows the plant to self-recover from a short dip without a manual restart sequence.

If the disturbances are confirmed to be severe (e.g. dips below 60 % Vn lasting > 1 s, several times per month), escalate to a recommendation for a STATCOM or BESS for dynamic voltage support.

- ★ **You have completed Part 4. The eight philosophy templates give you a consistent framework for judging any MV/LV plant design. The modifications catalogue gives you the standard vocabulary for the Recommendations Register. Part 5 — Field Tasks now takes you through the complete per-plant workflow step by step: desk study, site visit, drawing verification, analysis, and final report assembly — starting with a worked walk-through of the first assignment.**

Next: [Part 5 — Field Tasks](#).

Desk Study Procedure

Where you are in the workflow: *Part 5 is your field task guide. Work through it in order for every plant. You are here — beginning the pre-site desk study, before you travel to the plant.*

The desk study is the preparatory phase that takes place in the office, typically 3–7 days before the site visit. Its purpose is to understand the plant's electrical architecture before you arrive, identify gaps in the document package, and frame the questions you need to answer on site. A thorough desk study shortens the site visit and substantially improves the quality of your FMECA.

Documents to request

Issue a formal document request to the plant owner / O&M team as soon as the plant is assigned. The table below lists the categories to cover and the specific detail to ask for.

Category	Documents to request	Why you need it
Single-line diagrams	Overall SLD (all voltage levels); MV switchgear SLD; LV main switchboard SLD; MCC & distribution SLD; standby (DG/ATS) schematic. Request <i>as-built</i> revision with date stamp.	Foundation of every analysis: SPOF identification, load-flow model, protection review, SLD verification on site.
Equipment schedules & load list	Electrical load schedule (kW, pf, demand factor per feeder); equipment schedule (make, model, year, rating for transformers, switchgear, generators, UPS, batteries, VFDs); cable schedule.	Input for transformer sizing, load-flow, short-circuit, and genset sizing calculations (see Part 2 — Calculations).
Protection coordination study	Most recent protection coordination / relay grading study with relay setting records (incomer, feeders, bus-section, transformer differential). Request the live controlled revision — not the original EPC study if the plant has been modified since.	Grading selectivity is a common SPOF class. An outdated study (predating expansions or modifications) is itself a risk finding.
Earthing design & test records	Earthing system design study (grid resistance, touch/step potential); soil resistivity report; as-installed grid resistance test results; bonding schedule. Date of last test.	Earthing is a latent safety SPOF: a degraded or untested earth grid impairs earth-fault protection and creates touch/step hazards.
Arc-flash study	IEEE 1584 / IEC 61482-1 arc-flash / incident energy study; PPE category labels in use; safe-working-distance schedule. Check that the study post-dates any switchgear replacement or protection re-grading.	Obsolete arc-flash studies are a QHSE SPOF. An out-of-date study gives incorrect PPE boundaries — a direct safety risk.
Transformer condition records	Factory acceptance test (FAT) and site acceptance test (SAT) reports; dissolved gas analysis (DGA) / oil-sample trend reports; temperature monitoring logs; thermography records.	Transformer internal faults are high-severity events. DGA trend is the primary condition indicator — missing DGA data raises the Detection (D) rating in the FMECA.
Standby generator records	Genset O&M manual; auto-start / load test log (date, duration, outcome); maintenance history; fuel-delivery records.	Genset fails-to-start is a hidden (latent) failure mode — high Detection rating if test logs are absent or intermittent.
DC / battery charger records	Station battery and charger O&M manual; battery capacity-discharge test records; DC-bus voltage trend; alarm-annunciator history.	DC supply is the trip power source for MV protection. Charger ageing with no capacity test on record is a top-priority latent SPOF.
Switchgear maintenance records	Circuit breaker service records (racking operations count, contact-wear status, timing tests, primary injection dates); switchgear inspection reports.	Ageing switchgear with no primary injection records has uncertain protection reliability. Identifies candidates for the FMECA switchgear rows.
Asset register	Plant electrical asset register (ISO 14224 format if available); tag list.	Maps tags to equipment for FMECA rows and document traceability.

How to review the received documents

Follow this sequence when documents arrive:

1. **Open the Document Inventory form (SF-01) immediately.** Log every document as it arrives: title, revision, date, format, quality (Good / Legible-only / Poor), and status (Received / Partial / Missing / Obsolete). Do not wait until after the visit.
2. **Study the master SLD first.** Trace the power flow from the utility point of supply through MV-MET → MV-SWGR → TX-1/TX-2 → LV-MSB → MCC and loads. Mark the bus-tie status (normally open or closed) and note any parallel-feed or alternative supply arrangements. Sketch a mental resilience map: what fails if component X is lost?
3. **Cross-check revision dates.** Compare the SLD revision date to the protection study revision date. If the protection study predates plant modifications (equipment additions, transformer replacements, feeder changes), flag it as a gap in SF-01 and a potential FMECA finding.
4. **Extract equipment ratings for the calculation model.** From the load schedule and equipment schedule, build the shell of your load-flow and short-circuit models. Fill in transformer ratings, impedances, and utility fault-level data where available; mark unknowns to be resolved on site.
5. **List the open questions.** For each missing or poor-quality document, write a specific question to ask the O&M supervisor on site. Annotate these directly on a working (print) copy of the SLD.

✓ **Tip — start SF-01 at the document-request stage, not on arrival.** When you send the document request, log the requested items in the Document Inventory with status "Requested". As documents arrive, change the status to "Received" or "Partial". By the time you travel to site, the inventory already reflects the pre-visit gap

picture — which becomes your data-call agenda for the visit.

Starting SF-01 before the visit

Every row in SF-01 where status is Missing, Partial, or Obsolete at the end of the desk study becomes an action item in the SF-04 Site Visit Report. Prioritise these:

- **Missing DGA / oil-trend data** → raises transformer FMECA Detection (D) rating.
- **Missing battery capacity-discharge test** → latent DC SPOF.
- **Obsolete protection coordination study** → grading selectivity SPOF.
- **Outdated or missing arc-flash study** → QHSE SPOF (immediate priority).
- **No confirmed relay setting records** → cannot validate relay grading.

★ **Key takeaway: The desk study is not a bureaucratic step — it is where you build the mental model of the plant and identify the biggest risks before you travel. Arriving with an annotated working SLD, a part-completed SF-01, and a list of targeted questions will make your site time twice as productive.**

Site-Visit Preparation

Where you are in the workflow: the desk study is complete; you are preparing to travel to the plant for the non-intrusive site survey.

QHSE and permit-to-work

Before travelling to any plant site, confirm the host organisation's QHSE requirements and the permit-to-work (PTW) procedure. You are a visitor operating under the host's safety management system. Key points:

- **Site induction:** most plants require a formal site-safety induction on first entry. Allow time for this in Day 0 of your schedule. Request the induction materials in advance if possible.
- **Permit to work (PTW):** entry to any switchroom or electrical plant room requires a valid PTW issued by the host's authorised person (AP). Arrange this in advance — do not assume you can enter on arrival.
- **Accompanying escort:** the host O&M electrical supervisor must accompany you throughout the electrical areas. Confirm their availability and schedule before travel.
- **Emergency procedures:** know the site's muster points, emergency contact numbers, and first-aid arrangements before entering the plant.

 **Warning — NON-INTRUSIVE surveys only.** This campaign operates under a strictly non-intrusive rule. You must not open any energised electrical panel, cubicle, or enclosure; operate any switch, circuit breaker, or isolator; perform any live electrical test; or take any measurement that requires direct contact with live conductors. Your tools for the survey are eyes, a camera, a torch, a notepad, and a tape measure for clearances only.

Visual inspection, nameplate reading, photograph capture, and accessible (non-contact) meter readings are permitted. Where a panel is already open and de-energised under a live PTW issued for O&M work, you may observe but must not touch equipment or interfere with the O&M activity.

Be aware of LOTO (lock-out / tag-out) points. Never remove or interfere with a lock or tag, even if you believe it is incorrectly placed. If you find LOTO absent where it should be present, record it as an observation and raise it with the escort — do not act on it yourself.

Pre-entry checklist

Complete this checklist before entering any electrical plant area:

Progress: 0 / 8

- ☐ Site induction completed and card / sticker received.
- ☐ PTW reviewed, understood, signed, and held on person.
- ☐ Escort (plant electrical supervisor or authorised person) confirmed and present.
- ☐ Personal protective equipment donned: arc-rated coveralls, hard hat, safety glasses, safety shoes, gloves (HV-rated if required by PTW).
- ☐ Camera / phone cleared for site use by host security — confirm no confidential-data or photography restrictions apply.
- ☐ Working copy of SLD and annotated question list in hand (printed).
- ☐ Blank or pre-headed Site Visit Report (SF-04) and Document Inventory (SF-01) in hand.
- ☐ Emergency muster-point location and site emergency contact number confirmed.

 **Tip — print the SLD at A3 or larger.** A reduced A4 SLD is almost unreadable for red-lining in a switchroom. Print at A3 and laminate it if possible. Carry two copies: one for red-lining discrepancies, one to keep clean as a reference.

 **Key takeaway: Safety first, always. The PTW and escort must be in place before you enter any electrical room. The non-intrusive rule is absolute — do not open any live panel regardless of circumstances. If in doubt, stop and consult the escort.**

Visual Inspection Checklist

Where you are in the workflow: you are on site, PTW in hand, escort present. Work through this checklist zone by zone.

This is the primary field inspection tool. Work through each zone in the order below, checking off items as you observe them. Record observations, nameplate data and discrepancies in SF-04 as you go. Where a finding seeds an FMECA row, note the preliminary FM-ID in the SF-04 observation table.

 **Warning — non-intrusive throughout.** Every check in this list is visual, nameplate-based, or photographic only. Do not open any energised panel, cubicle, or door. Do not operate any device. Do not remove any label, lock, or tag. If a cubicle or panel is already open and de-energised under an O&M PTW, you may observe but must not touch.

Zone 1 — Switchroom / electrical building

Progress: 0 / 8

- ☐ Building access: entry under valid PTW; door interlock or warning notice present and legible.
- ☐ Room condition: adequate lighting (emergency lighting present); no water ingress; no condensation on walls or ceiling; ventilation / HVAC operating.
- ☐ Housekeeping: no stored materials, rubbish, or non-electrical equipment inside the switchroom.
- ☐ Fire protection: fire extinguisher(s) present, in-date, correct type (CO₂ or clean agent — never water in live electrical rooms); smoke/heat detector present.
- ☐ Floor-plan organisation matches SLD layout (left-to-right / top-to-bottom incomer → bus → feeders); note any discrepancies.
- ☐ Cable entries: sealed with fire-stop / intumescent material; no open penetrations.
- ☐ Safety signage: arc-flash hazard warning labels and PPE requirement signs visible at room entry.
- ☐ Earthing: main earth bar accessible and visible; connections not corroded or loose.

Zone 2 — MV switchgear (MV-SWGR, MV-MET, MV-NER)

Progress: 0 / 11

- ☐ **MV-MET:** utility metering cubicle present; seals intact; meter readable through window; confirm utility voltage level on nameplate.
- ☐ **MV-SWGR:** switchgear make, type, voltage class, and year legible on main nameplate; photograph.
- ☐ Switchgear type: RMU (ring-main unit), metal-clad drawout, or fixed — record and compare to SLD assumption.
- ☐ Number of panel sections and their function (incomer, bus-section/tie, feeder) matches SLD; record any discrepancy.
- ☐ Breaker/switch positions visible through window or on mimic diagram: incomer CB — CLOSED or OPEN; bus-tie CB — CLOSED or OPEN (mark on working SLD).
- ☐ Relay front plates visible: manufacturer, type, and functional label (IDMT O/C, E/F) legible; compare relay type to protection study assumptions.
- ☐ Arc-flash hazard label on each cubicle: present, legible, and carrying an incident-energy value or PPE category; record the study revision date if stated. If absent or clearly out of date, flag as an FMECA finding.
- ☐ Indication lamps functioning (visible from outside — no cubicle entry); trip indication flags reset or showing.
- ☐ **MV-NER:** neutral earthing resistor present in its housing; nameplate rating (ohms, kA continuous) legible; no visible damage or corrosion on resistor elements or bushing terminals.
- ☐ Cable sealing ends visible through inspection windows: no apparent tracking, surface discoloration, or visible damage.
- ☐ Switchgear age: note year of manufacture; if ≥ 20 years, flag for FMECA.

Zone 3 — Transformers (TX-1, TX-2)

Progress: 0 / 10

- ☐ **TX-1 and TX-2** both present and identified; nameplate data photographed: kVA, primary/secondary voltage, vector group (expect Dyn11 or Dyn1), impedance (%), year of manufacture.
- ☐ Cooling type confirmed from nameplate: ONAN (oil natural air natural) or ONAF (with fans); fans operating if ONAF.
- ☐ Oil level visible in sight-glass: within normal (green) range; no oil staining on tank or bund.
- ☐ Oil temperature indicator: temperature reading noted; no over-temperature alarm showing.
- ☐ Buchholz relay (if ONAN oil-cooled): present; no gas-alarm flag showing.

- ☐ Pressure-relief device: present; no evidence of prior operation (staining, displaced disc).
- ☐ HV and LV cable / busbar connections: external connections clean, no surface tracking or discoloration visible (do not touch).
- ☐ Bund / drip-tray: present; clean; drain valve closed; oil-absorbing material present if required.
- ☐ Transformer bay segregation: both transformer bays accessible independently for maintenance (not shared bund or blocking arrangement).
- ☐ Earthing: transformer tank earth connection clearly bonded to earth grid; connection visible and accessible for future testing.

Zone 4 — LV main switchboard (LV-MSB, bus-tie BT)

Progress: 0 / 9

- ☐ **LV-MSB** nameplate: voltage (400/415 V), form type (Form 4 or similar), busbar rating (A), year of manufacture.
- ☐ Split-bus arrangement confirmed: Bus A and Bus B sections physically identifiable; incomer A and incomer B breakers present.
- ☐ **Bus-tie (BT)**: bus-tie breaker clearly labelled; position confirmed: OPEN (N/O) or CLOSED — record on SLD. If closed under normal conditions, flag for FMECA (negates split-bus SPOF mitigation).
- ☐ Incomer A and incomer B breakers: rated current, type, and manufacturer legible on front label or nameplate.
- ☐ Auto-transfer / bus-coupling scheme: any automatic changeover relay visible on the bus-tie panel front? If so, note make and type. If absent, flag the manual-only bus-tie as a potential SPOF.
- ☐ Metering and instrumentation: ammeter, voltmeter, power-factor meter, or energy meter present and readable (note readings if practical).
- ☐ Feeder breakers: count and compare to SLD. Note any spare (vacant) ways. Record any breakers in tripped condition and confirm with escort whether this is normal.
- ☐ Visible condition: no burn marks, scorch marks, or discoloration on busbar chambers or breaker faces (observe through front panels / viewing windows where present).
- ☐ Cable entries: armoured cables properly glanded; segregation between phases maintained; no overheating discoloration on cable-entry areas.

Zone 5 — MCCs and VFDs

Progress: 0 / 8

- ☐ Each MCC panel (MCC-1, MCC-2, etc.) identified by tag; verify count against SLD.
- ☐ MCC nameplate: voltage, busbar rating, form type, year of manufacture — photograph each.
- ☐ Starter modules: starters present and labelled by motor load; DOL, star-delta, or soft-starter type visible from module face label.
- ☐ Overload relay setting: record on-site if legible through the module window; compare to nameplate FLC of motor where known.
- ☐ **VFD**: VFD enclosure present; make, model, kW/kVA rating, and AC supply voltage on nameplate; photograph.
- ☐ VFD ventilation / cooling: air filter grille accessible; check for visible dust build-up or blocked vents (do not open enclosure). Note any over-temperature indicator on the VFD front panel.
- ☐ VFD control mode: local or remote; note any trip-history indicators showing on front panel HMI.
- ☐ MCCs fed from Bus A or Bus B: confirm from the LV-MSB SLD which bus feeds each MCC; verify any cross-bus feeding arrangement (if any, flag for SPOF review).

Zone 6 — PFC and distribution (PFC, DB, EDB)

Progress: 0 / 6

- ☐ **PFC**: power-factor correction panel present; nameplate: kVA rating, number of steps, detuning reactor type (for harmonic detuning); check that capacitor bank is connected to the correct bus section.
- ☐ PFC controller: power factor target setting visible on controller front panel (typically 0.95 or unity); note actual power factor on LV-MSB meter if readable.
- ☐ **DB / EDB**: each distribution board labelled; compare to SLD. Note any boards not on SLD — add to red-line discrepancies.
- ☐ EDB (essential distribution board): fed from ATS or UPS output (not directly from normal supply only); verify source with escort if unclear.
- ☐ Spare way capacity: note boards with no available spare ways (capacity constraint).
- ☐ Cable management: trays and conduit runs in good order; no unsupported sagging; no exposed conductors in non-hazardous areas.

Zone 7 — Standby plant (DG, ATS)

Progress: 0 / 8

- ☐ **DG**: diesel generator set present; nameplate photographed: kVA rating, voltage (400/415 V), frequency (50 Hz), make/model/year, power factor.

- ☐ DG fuel level: fuel-level gauge readable; adequate fuel for a minimum 8-hour run (indicative; confirm the site minimum with the escort).
- ☐ DG condition: coolant level visible; no oil or fuel pooling beneath unit; air-intake filters accessible and not visibly clogged.
- ☐ DG battery (starting battery): battery connections clean, no corrosion; battery charger indicator lamp showing charged. Note last documented start-test date from logbook if accessible.
- ☐ DG start-test log: request from O&M supervisor — is auto-start / load-test performed monthly? Note any gaps in the log.
- ☐ **ATS:** automatic transfer switch panel identified; nameplate rating and source labels (Normal / Standby) legible; confirm OPEN / CLOSED position of each source contactor or breaker with escort.
- ☐ ATS transfer time setting: note if visible on controller (typical: 10–15 s for non-critical loads, 3–5 s for essential loads — indicative).
- ☐ Load scope of ATS: confirm with escort which loads are on the essential (ATS-backed) circuit and which are non-essential — record on working SLD.

Zone 8 — Control and DC supplies (UPS, DCDB)

Progress: 0 / 8

- ☐ **UPS:** UPS present; nameplate: kVA, battery autonomy (hours at full load), AC input / output voltage; bypass switch position.
- ☐ UPS battery condition: battery cabinet(s) present; no swollen or leaking cells visible through cabinet vents or viewing points (do not open cabinet).
- ☐ UPS front-panel status: mains-OK, battery-OK, fault-alarm indicators; note any active alarms.
- ☐ **DCDB:** DC battery and charger panel identified; make, model, DC voltage (typically 110 V DC for MV switchgear trip circuits, or 24 V DC for control), charger rating.
- ☐ DC charger condition: note age; any audible fan noise, vibration, or elevated surface temperature? Check for corrosion on terminals. Note last capacity-discharge test date from records if accessible.
- ☐ DC battery: note battery type (VRLA / flooded); year of manufacture; any evidence of electrolyte leakage or cell swelling (visible without opening enclosure).
- ☐ DC-fail alarm: confirm with escort that a DC-supply loss alarm is active and routed to SCADA or local annunciator.
- ☐ Control panels: DCS / PLC / SCADA marshalling cabinets identified; physical condition (no open doors in normal service); UPS-backed supply confirmed.

Zone 9 — Visible earthing

Progress: 0 / 6

- ☐ Main earth bar: accessible, connections clean, no corrosion on the bonding strap or clamp; conductor cross-section approximately as designed (note mm² if labelled).
- ☐ Earth test point / pit: earth test disconnect link accessible; test tag (date of last test) present and legible. Note last test date. If no test tag, or last test > 3 years, flag as FMECA finding.
- ☐ Transformer tank bonds: each transformer tank earthed independently to the earth bar; bond visible and accessible.
- ☐ Cable-armouring bonds: MV cable armouring bonded at both ends (check at gland plate — visual only).
- ☐ Exposed metalwork bonded: switchgear frame, transformer bund, cable tray continuity bonds visible; no isolated metal sections.
- ☐ MV-NER earth connection: NER body earthed to earth bar with a clearly identifiable conductor (this is separate from the neutral point conductor — note if both are visible).

Photolog — mandatory photographs

The following 12 photographs are mandatory for every site visit. Capture them in order; check each off as taken. Each photo must include a scale reference (hand, A4 sheet, or tape measure) or a legible nameplate in frame. Label images sequentially (e.g. P01-IMG-01) and index them in SF-04 Section 5.

Progress: 0 / 12

- ☐ **PH-01 — Overall switchroom general view:** taken from the room entrance looking in; shows the full panel line-up and housekeeping condition.
- ☐ **PH-02 — MV switchgear line-up and arc-flash labels:** shows the incomer, bus-section, and feeder cubicles; arc-flash PPE-category labels visible (or their absence).
- ☐ **PH-03 — Bus-tie (BT / MV bus-section) breaker and position indicator:** close-up of the bus-tie cubicle front showing the position (O or I / OPEN or CLOSED) indicator.
- ☐ **PH-04 — TX-1 nameplate:** transformer 1 rating plate showing kVA, voltage ratio, vector group, impedance, year.
- ☐ **PH-05 — TX-2 nameplate:** transformer 2 rating plate (same fields as PH-04).
- ☐ **PH-06 — LV-MSB incomer and bus-tie section front:** shows incomer A, incomer B, and bus-tie breaker with their position indicators and labels.
- ☐ **PH-07 — DG nameplate and starting-battery condition:** generator rating plate and battery / charger in same frame if possible.

- ☐ **PH-08 — ATS panel front:** shows source labels (Normal/Standby) and transfer-switch position indicator.
- ☐ **PH-09 — DCDB / battery charger:** charger nameplate, battery type label, and any front-panel fault/alarm indicators visible.
- ☐ **PH-10 — UPS front panel:** nameplate and status indicators showing mains-OK, battery-OK, and any active alarms.
- ☐ **PH-11 — Main earth test point / pit:** test disconnect link and test tag (showing last test date); earth bar bonding connections.
- ☐ **PH-12 — Any notable defect or discrepancy found:** photograph the specific item (cable termination tracking, missing label, corrosion, apparent overheating discoloration, absent arc-flash label, etc.) that seeds an FMECA finding. Take one photograph per distinct finding.

 **Warning — do not photograph classified or sensitive information.** Some plants have restrictions on photographing SCADA screens, relay setting displays, production data, or utility metering. Confirm with the escort before photographing anything other than physical equipment nameplates and general views.

 **Key takeaway:** Work through this checklist zone by zone, checking off items as observed. The 12 mandatory photographs are non-negotiable — they provide the evidentiary basis for the site-visit report and support the FMECA Detection ratings. Everything is non-intrusive.

Drawing Availability & SLD Verification

Where you are in the workflow: alongside the visual inspection, you are verifying the single-line diagrams against the as-found plant and recording all discrepancies.

The SLD is the primary engineering document for the plant's electrical system. Verifying it against the as-found installation is a core deliverable of every site visit. Discrepancies between the SLD and the real plant are often the most direct route to identifying SPOFs: an unlabelled bus-tie that is closed in normal operation, a second transformer that has been added without updating the SLD, or a protection relay that has been replaced with a different type but the coordination study was never updated.

Drawing availability — what you need

Drawing	Minimum acceptable	Risk if missing
Overall master SLD (all voltage levels)	Latest controlled revision, readable at A3	Cannot verify plant architecture; FMECA incomplete
MV switchgear SLD / internal arrangement	Cubicle-by-cubicle layout, relay type per panel	Cannot verify relay types against protection study; miss selectivity gaps
LV main switchboard SLD	Bus A/B split, incomer and feeder labelling, BT position	Cannot confirm bus-tie scheme; miss LV SPOF
Standby (DG/ATS) schematic	ATS source selection, backed loads listed	Cannot confirm essential-load scope; genset sizing FMECA row uncertain
Earthing system plan / layout	Grid layout, test-point locations, bonding schedule	Cannot identify earthing SPOF; cannot locate test points on site
Cable schedule (MV feeders and LV main cables)	Cable ref, route, cross-section, length (MV feeders as minimum)	Cannot run voltage-drop or cable-sizing calculations

The red-line technique

Bring a printed working copy of the master SLD to site (A3 or larger, at a scale at which individual labels are legible). As you walk down the plant:

- Check every item on the SLD against the physical installation.** For each component — transformer, breaker, bus-section, relay, feeder — confirm it is present and correctly labelled. If confirmed, tick the item on your working SLD.
- Mark any discrepancy in red.** Cross out items not present; annotate added items not on the SLD; circle mislabelled items; note changed relay types or equipment ratings.
- Mark the as-found switch/breaker positions.** Use "C" (closed) or "O" (open) pencilled next to each breaker symbol. Pay particular attention to the bus-tie breaker and any normally-open-by-design devices.
- Transfer all red-line marks to SF-04** (observations and findings table) at the end of the visit. Each discrepancy becomes an observation with its tag and a note of what action is needed (e.g., SLD revision, FMECA row).

Common SLD discrepancies to look for

Progress: 0 / 8

☐

Unlabelled or wrong-status bus-tie: bus-tie BT shown as normally open on SLD but confirmed closed in normal operation by escort — negates N+1 redundancy; major SPOF finding.

☐

Missing bus-section breaker: SLD shows a split bus, but physically a solid busbar link (bolted connection) has replaced the bus-tie CB; no fault-isolation possible between halves.

☐

Obsolete single-transformer SLD: SLD shows only TX-1; plant actually has TX-1 and TX-2 (second transformer added in a later expansion but SLD not updated).

☐

Nominal vs as-found protection settings: relay type on SLD (e.g. IDMT overcurrent relay) does not match the relay on the actual panel (a different make/model with different characteristic curves installed).

☐

Added loads not on SLD: additional MCCs or distribution boards installed since SLD revision; not reflected in the load schedule used for calculations.

☐

Changed transformer taps: transformer on-load or off-load tap-changer position differs from SLD assumption; affects voltage-drop and protection-relay voltage setting calculations.

☐

Earthing system changes: additional earth electrodes driven or NER earthing arrangement changed without updating the earthing design document.

☐

ATS source labelling error: on site, Normal and Standby supply labels on the ATS are reversed from the SLD, or the ATS has been replaced with a different panel type.

When no SLD is available

If no SLD exists or the available SLD is so outdated as to be unreliable (e.g., a hand-sketch with no revision date), proceed as follows:

- Record the gap in SF-01** as Missing or Obsolete, with a data-call action to the plant owner for a redraw or as-built survey.

2. **Construct a sketch SLD during the visit.** Using a blank A3 pad, sketch the one-line from the incomer downwards as you walk the plant. Note every component you physically observe: tag, make, rating, and switch position. This becomes the red-line baseline.
3. **Note the absence of a controlled SLD as an FMECA finding** in its own right: operating a plant without an up-to-date SLD raises the likelihood of switching errors and impairs protection-setting maintenance (Detection rating raised for affected FMECA rows).
4. **Flag in SF-04** that calculations are based on site-observed data only and that a formal SLD should be drawn before the study is finalised.

✓ **Tip — take a photo of every nameplate to supplement the SLD.** Even when a good SLD exists, the nameplate photograph provides data that may not appear on the SLD: exact impedance %, serial number for warranty / age verification, precise kVA rating, and tap position. This photographic record is also the source for completing the calculation model back in the office.

★ **Key takeaway: The red-line SLD verification is a simultaneous deliverable alongside the visual inspection — not a separate post-visit task. Discrepancies found on site are direct inputs to the FMECA and to the SLD update recommendation. A missing or obsolete SLD is itself an FMECA finding.**

Post-Site Analysis

Where you are in the workflow: the site visit is complete; you are back in the office, turning your site data into the four deliverable forms.

Post-site analysis converts the raw data captured during the desk study and site visit into the completed FMECA, Recommendations Register, and finalised Document Inventory. The table below maps each form to its data sources and the relevant manual sections.

Form	Primary inputs	Methods / manual sections	Key actions
SF-01 Document Inventory	Desk-study document log; documents collected on site; data-call responses received post-visit.	Desk Study (this Part) ; SF-01 template in Appendix — Blank Forms .	Update status column (Received / Partial / Missing / Obsolete) for every row; close data-call actions as responses arrive; ensure gap summary reflects final state before submission.
SF-02 FMECA	Red-lined SLD; SF-04 observations; SF-01 gaps (each Missing or Obsolete document raises the Detection D rating of related rows); calculation results (see below).	Part 3 — SPOF Checklist ; Part 3 — FMECA Method ; calculations from Part 2 .	One row per failure mode; assign S, O, D on the 1–5 scales; compute $RPN = S \times O \times D$ and $Crit = S \times O$; flag $SPOF=Y$ where a single-component loss causes the end effect with no redundancy; link each row to a recommendation (REC-ID).
SF-03 Recommendations	Completed FMECA (SF-02) — one recommendation row per FM-ID; concept modification catalogue.	Part 4 — Modifications Catalogue ; Part 3 — Recommendations .	Priority derived mechanically from FMECA $RPN/SPOF$ ($P1 = SPOF \geq 4$ or $RPN \geq 60$; $P2 = RPN 36-59$ or $SPOF S=3$; $P3 = RPN 16-35$; $P4 < 16$). Assign cost band (L/M/H — no SAR figures). Include reliability/availability gain basis. Monitor-only findings carry no modification; note "monitor — no modification".
SF-04 Site Visit Report	Site-visit notes; SF-04 header completed on site; observations table from visual inspection; photolog.	Report Writing (this Part) ; photolog from Visual Inspection Checklist .	Finalise within 24–48 hours of the visit while recall is fresh; index all photos; include agreed action list from the site meeting; note open data-calls.

Calculations to run post-site

For each plant, complete the following calculation studies after the site visit using the as-found equipment data captured. The relevant manual sections are cross-linked.

- **Transformer sizing and loading check** — see [Part 2 — Transformer Calculations](#): confirm that the installed transformer kVA rating is adequate for the as-found load (with N+1 contingency). A transformer running at >80 % of rated kVA on normal load with no N+1 available is a SPOF candidate.
- **Full-load current (FLC) check** — see [Part 2 — FLC Calculations](#): verify that MCC starter and overload relay settings match the motor FLC. Significantly wrong settings (e.g., overload set above locked-rotor current) are a protection SPOF.
- **Short-circuit level** — see [Part 2 — Short-Circuit](#): calculate maximum prospective short-circuit current at the MV and LV busbars; confirm that switchgear rated fault withstand (kA) exceeds the calculated level. If not, this is a P1 safety SPOF.
- **Power factor correction sizing check** — see [Part 2 — PFC](#): verify that the installed PFC kVAr rating matches the current load's reactive demand. Oversized PFC on a lightly loaded bus can lead to over-voltage; under-sized PFC misses the utility penalty boundary.
- **Genset sizing check** — see [Part 2 — Genset](#): verify the standby generator kVA rating against the essential-load list. If the genset is sized to essential loads only (not full production), document this explicitly in the FMECA — it is a design limitation, not a fault, but it must be communicated.
- **Load flow (indicative)** — see [Part 2 — Load Flow](#): run a steady-state load flow to verify bus voltages and identify any heavily loaded feeders. Confirm voltage at the end of the longest cable run is within ± 5 % of nominal (IEC 60364 — consistent with the acceptance limit stated in [Part 2 — Load Flow](#)).

✓ **Tip — update the FMECA Detection (D) rating based on SF-01 gaps.** Every SF-01 item with status Missing, Partial, or Obsolete directly affects the detectability of a related failure mode. For example: no DGA oil-trend record → transformer internal-fault D = 4 or 5 (hidden); no battery capacity-discharge test → DC charger failure D = 5 (hidden). Mechanically update these D ratings as you finalise SF-01 — do not leave them as conservative estimates.

★ **Key takeaway: Post-site analysis is where the data turns into decisions. The FMECA rows are built from three sources: the red-lined SLD, the SF-04 observations, and the SF-01 document gaps. Every gap in SF-01 has a direct mechanical effect on the Detection rating of at least one FMECA row — follow the chain.**

Writing the Site Visit Report

Where you are in the workflow: you are finalising SF-04 within 48 hours of the site visit.

The Site Visit Report (SF-04) is the contemporaneous factual record of the visit. It is not an analysis document — analysis lives in the FMECA and Recommendations Register. SF-04's function is to record what you saw, where, when, with whom, and what data-calls remain open. It is the evidence base that supports the FMECA findings and allows the PM to quality-check that your FMECA rows are grounded in real observations.

Report structure

Section	Mandatory content	Notes
1 • Visit metadata	Date and time (start-finish); plant name and reference; weather / ambient conditions; escort name and title; access constraints (e.g. PTW number, energised panels not opened).	Complete on the day of the visit. Do not reconstruct from memory later.
2 • Minutes of meeting (MOM)	Attendees (name, organisation, role); agenda items discussed; key facts confirmed by the escort (e.g. bus-tie operating philosophy, genset essential-loads-only scope, date of last arc-flash study); decisions; action list (owner + due date).	The MOM is the primary source for confirming verbal statements made on site. Include any discrepancies the escort explained versus the SLD.
3 • Survey scope walked	List of all areas inspected and the equipment tags covered in each area.	If an area was not inspected (access denied, PTW not obtained), record that explicitly.
4 • Observations and findings	One row per observation: observation number; area; asset tag; what was observed; the FM-ID of the FMECA row it seeds. Each observation must be factual — describe what you saw, not an assessment of its risk (risk belongs in the FMECA).	Transfer directly from your site notes. Every FMECA row should trace back to at least one SF-04 observation or a SF-01 document gap.
5 • Photo log	Index of all photographs: photo ID, caption, asset tag, what it shows, finding reference. A minimum of the 12 mandatory photos from the Photolog checklist .	Caption every photo at the time of indexing — do not rely on memory 48 hours later. If photos were not permitted, state why.
6 • Safety / access notes	PTW conditions observed; any QHSE concerns identified on site (e.g. absent arc-flash labels, missing fire extinguisher, inadequate lighting); the non-intrusive rule adherence statement; heat-stress or other environmental controls applied.	The arc-flash label finding (if raised) is both a QHSE entry here and an FMECA row — cross-reference the FM-ID.
7 • Sign-off	Prepared-by (engineer name, organisation, date); acknowledgement by the host escort (name, signature or email confirmation); PM review date (leave blank until PM signs).	Obtain escort acknowledgement on the day, even if only by email confirmation.

Photo selection for the report

Include in SF-04 all 12 mandatory photos plus any additional photos that directly support an observation in the findings table. The following photos carry the most evidential weight and should always be included if captured:

- Bus-tie breaker position indicator — confirms normal operating mode (open or closed).
- Arc-flash label (or its absence) — supports the PPE / incident-energy FMECA finding.
- DC / battery charger condition — supports the latent DC-supply SPOF finding.
- Earth test point with date tag — confirms when the earthing system was last tested.
- Any visible defect: tracking on cable terminations, corrosion on connections, dust-clogged VFD filters, missing fire-protection equipment — each defect photo must link to an observation row.

Tone and accuracy

- **Factual, not inferential.** SF-04 records what you observed; the inference ("this is a SPOF because...") belongs in SF-02 FMECA. Example: *Good SF-04 text:* "Bus-tie CB confirmed by escort as manually operated only; no automatic transfer relay fitted." *Not for SF-04:* "The manual bus-tie is a critical single point of failure."
- **Attribute verbal information to the escort.** If the plant supervisor tells you something that is not visible on a nameplate or SLD (e.g. "we last tested the battery three years ago"), attribute it: "Confirmed by [Escort Name] on [date]".
- **Report absences explicitly.** "No arc-flash label observed on MV-SWGR incomer cubicle" is a valid finding. Do not omit negative observations.

★ **Key takeaway:** SF-04 is a factual record, not a risk assessment. Write it within 48 hours of the visit. Every FMECA row must trace to a SF-04 observation or a SF-01 gap. The photo log is mandatory — index all photos the same day they are taken.

Case Study — Jubail SWRO Desalination Plant

Where you are in the workflow: the methods are taught; now see all four forms filled in for one complete plant.

Introduction

The embedded document below is a **simulated, end-to-end example** showing all four assessment forms completed for a single imagined plant: the Jubail Seawater Reverse Osmosis (SWRO) desalination plant (reference P01). It has been deliberately constructed to exercise every form section and to illustrate the full range of SPOF types, priorities, and recommendation categories that an engineer is likely to encounter across the 21-plant campaign.

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Scope disclosure — why the Jubail plant is larger than the manual's reference basis. The Jubail SWRO study is a deliberately *larger, simulated plant* designed to show that the assessment *method scales* beyond the manual's indicative ~2 MW / 11 kV / 400 V reference. The Jubail plant uses:

- 132 kV grid incomers (not 11 kV)
- 2 × 40 MVA 132/13.8 kV transformers (not 2 × 1600 kVA 11/0.4 kV)
- 13.8 kV main switchboard (not 400 V LV-MSB)

The method — SPOF tracing, FMECA scoring on a 1–5 scale, priority banding — is identical. The scale difference simply exercises the forms on a larger system.

Because the Jubail plant is a distinct, larger facility, the document uses a **site-specific extended asset-tag taxonomy** (P01- prefix) rather than the manual's approved core tag scheme. The table below maps the principal Jubail site tags to their approved-scheme equivalents for cross-reference:

Jubail site tag	Approved core tag equivalent	Equipment
P01-SWGR-01	MV-SWGR	Main HV/MV switchgear
P01-BUSTIE-01	BT	Bus-tie circuit breaker
P01-GEN-01	DG	Standby / emergency generator
P01-DC-01	DCDB	DC distribution board / battery
P01-TX-01 / P01-TX-02	TX-1 / TX-2	Main power transformers
P01-UPS-01	UPS	Uninterruptible power supply
P01-ATS-01	ATS	Automatic transfer switch
P01-MCC-01 / P01-MCC-02	MCC-1 / MCC-2	Motor control centres

The manual's approved core tag scheme (MV-MET, MV-SWGR, MV-PROT, MV-NER, TX-1/2, LV-MSB, BT, MCC-1...n, VFD, PFC, DG, ATS, UPS, DCDB, DB, EDB) continues to govern all standard deliverables produced by this campaign.

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Note — simulated data. Every value, observation, equipment tag, document reference, personnel name, and cost band in this study is fictitious. Data items are tagged **SIM-...** throughout to confirm their illustrative-only status. Nothing in this document represents real Saudi Water Authority asset data, real equipment condition, or a real cost estimate. The study has been built to demonstrate the *method* — not to describe a real plant. All costs shown are indicative bands only (L / M / H) and must never be cited as a quote or an estimate.

Reading guide

Use the table below to navigate between the four embedded forms and the manual sections that taught the underlying method. Read each form, then return to the referenced section to confirm you understand why the form looks the way it does.

Form	What it demonstrates	Taught in
SF-01 — Document Inventory	15 documents logged; 7 Received, 6 Partial, 2 Missing, 1 Obsolete. Shows how document gaps (Missing DGA, Missing battery test, Obsolete arc-flash study) directly feed FMECA Detection ratings. The gap summary is the data-call agenda.	Desk Study Procedure (this Part)
SF-02 — FMECA	16 failure-mode rows covering the full spectrum: SPOF bus-tie, protection selectivity gap, DC supply latent failure (top RPN = 60), arc-flash PPE boundary, single service transformer SPOF, earthing untested, genset hidden failure. Three SPOF=Y rows; priority derivation from RPN and SPOF flags demonstrated.	Part 3 — FMECA Method; Part 3 — SPOF Checklist
SF-03 — Recommendations	16 recommendations closing the loop from every FMECA row. Shows the mechanical priority rule, the cost-band method, and the distinction between modification recommendations and monitor-only findings. Dependency / sequencing column demonstrates how recommendations can create critical-path constraints.	Part 4 — Modifications Catalogue
SF-04 — Site Visit Report	Full MOM, 12 observation rows each linked to an FM-ID, 10 photo-log entries, QHSE notes. Demonstrates the factual (non-inferential) tone and the traceability chain: observation → FM-ID → REC-ID.	Writing the Site Visit Report (this Part)

Top FMECA rows and matching recommendations — at a glance

The table below surfaces the three highest-priority FMECA rows from the Jubail study (all three are either SPOF=Y or RPN = P1/P2) and their matched recommendations. Row IDs use the Jubail site-specific `P01-` prefix (see tag mapping above). These rows are the ones most referenced in the peer-review commentary below — reading this table first lets you follow the commentary without opening the full embedded document. *All values are simulated for training only.*

FM-ID	Tag (Jubail)	Approx. core tag	Failure mode	S	O	D	RPN	SPOF?	Finding (end effect)
P01-FM-006	P01-DC-01	DCDB	Charger ageing → DC supply degraded / undetected loss	4	3	5	60	N (latent)	Latent loss of tripping → protection fails on demand (highest RPN; D=5 because no battery capacity test on record — hidden failure)
P01-FM-001	P01-BUSTIE-01	BT	No automatic transfer; manual close only	4	3	2	24	Y	Loss of ~half RO production for minutes-hours on transformer/incomer loss; manual restoration only (SPOF Type B pattern)
P01-FM-012	P01-SWGR-01	MV-SWGR	Busbar fault / flashover (whole common 13.8 kV board)	4	1	2	8	Y	Total loss of RO production — single common MV board is a whole-plant SPOF (SPOF Type C pattern); low probability but catastrophic consequence → P1 by SPOF+S≥4 rule

REC-ID	Linked FM-ID	Modification	Risk before	Risk after
P01-REC-06	P01-FM-006	Replace ageing battery charger; reinstate battery capacity-discharge testing; add on-line DC bus voltage monitoring alarm	High (RPN 60 — latent)	Low (latent DC-supply loss removed; protection reliability restored)
P01-REC-01	P01-FM-001	Add automatic bus-tie transfer scheme (auto-changeover relay + interlocks + protection re-grading)	High (SPOF=Y, S=4)	Low (half-bus restoration time: minutes → seconds; availability ~0.985 → ~0.999 indicative)
P01-REC-12	P01-FM-012	Switchgear integrity programme: bus-zone protection verification, primary-injection tests, arc-flash detection retrofit	High (SPOF=Y, S=4, whole-board)	Medium (reduces likelihood and consequence of whole-board bus fault)

✓

Traceability chain illustrated: SF-04 observation (missing battery test) → SF-01 gap (DOC-12 Missing) → FM-006 Detection raised to D=5 → RPN=60 → P1 priority → REC-06 (replace charger + reinstate testing). This is the canonical checklist → FMECA → register → modification chain, demonstrated end-to-end.

Jubail SWRO study — all four forms

P01 · Jubail Seawater Desalination Plant (SWRO)

Saudi Water Authority · Etimad 260539008749 · four-document study set

SIMULATED — ILLUSTRATIVE ONLY. Imagined plant conditions to demonstrate the standard forms; not real Saudi Water Authority asset data. All data is tagged SIM-... ; costs are bands only; not a quote.

SF-01
Document Inventory

SF-02
FMECA

SF-03
Recommendations

SF-04
Site Visit Report

SF-01 Document Inventory

Form: Document Inventory Form ID: SF-01 Rev: 0
 Plant: P01 Jubail Seawater Desalination Plant (SWRO) Sector: Central & Eastern Region: Eastern
 Prepared by: Electrical Engineer Date: 2026-06-02 Reviewed (SCE-stamped): [SCE name – pending stamp]
 Satisfies: REQ-DEL-01, REQ-SPEC-01 Standards: ISO 14224
 Status: [SIMULATED – ILLUSTRATIVE ONLY]

Register of every document collected or reviewed for the plant, with as-found status and

gaps. Missing/Partial items become data-call actions in the Site Visit Report (SF-04) and

lower the Detection confidence of related FMECA (SF-02) rows.

DOC-ID	Document title	Category	Source / holder	Format	Rev & date	Status	Quality	Gap / action	Used in	Source
P01-DOC-01	132 kV / 13.8 kV main single-line diagram (overall)	SLD	SWA O&M	DWG	Rev C, 2019-04	Received	Good	As-built; verify bus-tie status flag on drawing vs field	FMECA / SLD update	SIM-01
P01-DOC-02	13.8 kV switchgear SLD & internal arrangement	SLD	SWA O&M / OEM	PDF	Rev B, 2018-11	Received	Legible-only	Scanned copy; metering & interlock detail not legible	FMECA / Load-flow	SIM-02
P01-DOC-03	Protection coordination / relay setting study	Protection	Contractor (orig. EPC)	PDF	Rev A, 2014-09	Partial	Poor	Pre-dates 2019 RO train addition; feeder → incomer grading not revalidated — data-call	FMECA / RAM	SIM-03
P01-DOC-04	Relay setting records (incomer & feeder relays)	Protection	SWA O&M	paper	undated	Partial	Unverified	Marked-up by hand; no controlled rev; reconcile to live settings	FMECA	SIM-04
P01-DOC-05	Arc-flash / incident-energy study (IEEE 1584)	Test reports	Contractor	PDF	Rev A, 2015-06	Obsolete	Legible-only	Predates switchgear upgrade; 13.8 kV incident energy unverified — restudy	FMECA / RAM	SIM-05
P01-DOC-06	132/13.8 kV transformer test & commissioning reports (TX-01/02)	Test reports	OEM	PDF	2013-02	Received	Good	FAT/SAT only; no recent DGA/oil trend — request CMMS history	FMECA	SIM-06
P01-DOC-07	Transformer oil DGA / condition-monitoring trend	Maintenance records	SWA O&M	none	—	Missing	Unverified	No DGA trend held on site — data-call to central lab	FMECA / RAM	SIM-07
P01-DOC-08	13.8 kV cable schedule & termination register	Cable schedule	SWA O&M	DWG	Rev B, 2016-08	Partial	Legible-only	Covers main feeders; service-transformer spur cables omitted	FMECA / Load-flow	SIM-08
P01-DOC-09	Earthing / grounding study & test record	Earthing study	Contractor	PDF	2013-05	Partial	Poor	Soil resistivity only; as-installed grid resistance not re-tested	FMECA	SIM-09
P01-DOC-10	Load schedule / electrical load list	Load schedule	SWA O&M	PDF	Rev A, 2019-03	Received	Good	Confirms genset sized to essential loads only (no RO production)	Load-flow / RAM	SIM-10
P01-DOC-11	Standby diesel genset O&M & start-reliability log (GEN-01/02)	Genset records	SWA O&M	paper	partial 2022–24	Partial	Legible-only	Start-test log intermittent; auto-start reliability unconfirmed	FMECA / RAM	SIM-11
P01-DOC-12	Station DC / battery charger O&M & battery test record (DC-01)	Maintenance records	SWA O&M	none	—	Missing	Unverified	No capacity-discharge test on record — data-call (latent risk)	FMECA / RAM	SIM-12
P01-DOC-13	MV VFD O&M manual & trip-history log (VFD-01)	O&M	OEM	PDF	2017	Received	Good	Manual complete; on-site trip-history extract pending	FMECA	SIM-13
P01-DOC-14	Plant electrical asset register (ISO 14224 tagged)	Asset register	SWA O&M	PDF	Rev A, 2020-01	Partial	Legible-only	Tag list incomplete for 0.4 kV service area; reconcile in field	FMECA / SLD update	SIM-14

P01-DOC-15	Switchgear maintenance / CB service records (SWGR-01)	Maintenance records	SWA O&M	paper	2021–23	Partial	Legible-only	Routine only; primary injection / timing tests not evidenced	FMECA	SIM-15
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Gap summary: 7 Received · 6 Partial · 2 Missing · 1 Obsolete (of 15 total). Assessment-confidence note: The two Missing items (DGA trend P01-DOC-07, battery test P01-DOC-12) and the Obsolete arc-flash study (P01-DOC-05) directly raise the Detection (D) ratings for the related FMECA rows (FM-006 DC charger, FM-004 arc-flash, FM-007 transformer). The obsolete protection study (P01-DOC-03) underpins the selectivity finding (FM-002). Confidence is otherwise moderate-to-good: this is a relatively robust dual-incomer site and the core SLD/load data are held.

SF-02 FMECA

Form: FMECA (Failure Mode, Effects & Criticality Analysis) Form ID: SF-02 Rev: 0
Plant: P01 Jubail Seawater Desalination Plant (SWRO) Sector: Central & Eastern Region: Eastern
Prepared by: Electrical Engineer Date: 2026-06-02 Reviewed (SCE-stamped): [SCE name – pending stamp]
Satisfies: REQ-DEL-03, REQ-DEL-04, REQ-SPEC-01, REQ-SPEC-04
Standards: IEC 60812 (method) · ISO 14224 + OREDA (taxonomy/data) · IEEE 493 (electrical reliability)
Status: [SIMULATED – ILLUSTRATIVE ONLY]

One row per asset-function / failure-mode. S/O/D per the 1–5 scales in TEMPLATE_00.

RPN = S×O×D ; Crit = S×O . Flag **SPOF=Y** where a single component loss causes the end

effect with no redundancy. Carry the forward link to REC-ID (SF-03 Recommendations).

FM-ID	Asset tag	Asset / function	Class	Failure mode	Failure cause	Local effect	End effect	S	O	D	RPN	Crit	SPOF?	Existing safeguard
P01-FM-001	P01-BUSTIE-01	13.8 kV bus-tie — restore supply to half-bus on TX loss	BUSTIE	No automatic transfer; manual close only	No auto-transfer scheme; tie operated locally under permit	Half of 13.8 kV bus de-energised until operator closes tie	Loss of ~half RO production (half-capacity derate) for minutes–hours	4	3	2	24	12	Y	N+1 transformer trained operators; SCADA alert of bus dea
P01-FM-002	P01-RELAY-02	Feeder protection — clear faults selectively	RELAY	Mis-coordination / selectivity miss with incomer relay	Grading study (P01-DOC-03) predates 2019 RO addition; settings not revalidated	Feeder fault cleared by upstream incomer	Wider trip than necessary → larger section of bus lost	3	3	4	36	9	N	Two-stage OC/EF protection; SCADA event log
P01-FM-003	P01-GEN-01	Standby genset — back essential loads	GEN	Insufficient capacity for RO production (design limitation)	Gensets sized to essential loads only	RO trains not supported on total grid loss	Extended loss of RO production during grid outage	4	2	2	16	8	N	Dual 132 kV grid incomer (rare total loss); essential-load backup int
P01-FM-004	P01-SWGR-01	13.8 kV switchgear — safe operation / maintenance	SWGR	Incident energy / PPE boundary unverified (IEEE 1584)	Arc-flash study (P01-DOC-05) obsolete; predates switchgear upgrade	Arc-flash hazard category unknown at panel	Potential arc-flash injury to personnel during switching	5	2	4	40	10	N	PTW regime arc-rated PPE worn; restricted l access
P01-FM-005	P01-TX-03	13.8/0.4 kV service transformer — feed dosing/pre-treatment sub-area	TX	Loss of single transformer, no standby LV feed	Single service transformer; no alternate feed	LV sub-area de-energised	Loss of chemical dosing/pre-treatment sub-area (RO trains stay up)	3	2	3	18	6	Y	Manual portable-generator backup possible; sub-area not on critical path
P01-FM-006	P01-DC-01	Station battery charger / DC — protection & trip supply	DC	Charger ageing → DC supply degraded / undetected loss	Ageing charger; no capacity-discharge test (P01-DOC-12)	Battery support to trip/protection circuits weakened	Latent loss of tripping → protection fails on demand (escalated fault)	4	3	5	60	12	N	Single stat battery present; charger DC fail alarm (functional)
P01-FM-007	P01-TX-01	132/13.8 kV main transformer — step-down supply	TX	Internal winding/insulation fault	Insulation ageing; weak DGA evidence (P01-DOC-07)	One transformer tripped	N+1 redundancy holds; half-capacity if tie not yet closed (see FM-001)	4	2	2	16	8	N	N+1 (TX-01/TX-02); differential Buchholz protection
P01-FM-008	P01-VFD-01	MV VFD — drive HP RO feed pump train	VFD	VFD trip / power-electronics failure	Cooling-filter fouling; component ageing	One RO train drive offline	Loss of one RO train (partial derate)	3	3	2	18	9	N	Multiple trains; VFD fault alarm spare modules
P01-FM-009	P01-PMP-01	HP RO feed pump — high-pressure	PMP	Motor / bearing failure	Mechanical wear; thermal stress	One pump/train	Partial production derate for that train	3	3	2	18	9	N	Train-level partial

009		membrane feed				derated								redundanc vibration monitoring
P01- FM- 010	P01- PMP-02	Intake pump — raw seawater intake	PMP	Pump trip / blockage	Debris; mechanical wear	One intake pump offline	Minor intake-margin reduction	2	3	2	12	6	N	Installed spare intake pump; level/flow alarms
P01- FM- 011	P01- FEED- 01	132 kV grid incomer — primary supply	FEED	Loss of one grid incomer	Grid disturbance; incomer trip	One incomer lost	Other incomer carries load (N+1); no production loss	3	2	1	6	6	N	Dual independe 132 kV incomers; continuous SCADA monitoring
P01- FM- 012	P01- SWGR- 01	13.8 kV switchgear — single common busbar	SWGR	Busbar fault / flashover (whole board)	Insulation failure; foreign object; switchgear ageing	Entire 13.8 kV board faulted	Total loss of RO production (single common board)	4	1	2	8	4	Y	Bus-zone protection; routine maintenan arc-resista cubicles (partial)
P01- FM- 013	P01- RELAY- 01	Incomer protection — clear incomer- side fault	RELAY	Relay fails to trip / breaker-fail not backed up	Relay/aux ageing; no breaker-fail scheme	Fault not cleared by primary device	Escalated/wider fault before backup clears	4	2	4	32	8	N	Backup OC downstrea grading (impaired - see FM-00
P01- FM- 014	P01- EARTH- 01	Earthing grid — fault-current return & touch safety	EARTH	Grid resistance degraded / unverified	Corrosion; no re- test since 2013 (P01-DOC-09)	Elevated earth-grid resistance	Touch/step hazard; impaired EF protection sensing	3	2	4	24	6	N	Original design margin; periodic visual inspection
P01- FM- 015	P01- CABLE- 01	13.8 kV feeder cable termination — feed RO switchboard	CABLE	Termination breakdown / tracking	Ageing terminations (~12 yr); contamination	Single feeder cable fault	Loss of one feeder/train (partial derate)	3	2	3	18	6	N	Feeder OC/EF protection; periodic thermogra
P01- FM- 016	P01- GEN-01	Standby genset — start on demand	GEN	Fails to start / auto-start failure (hidden)	Battery/fuel/control faults; intermittent start-test log (P01- DOC-11)	Essential- load backup unavailable	Loss of essential loads during grid outage (latent until called)	3	3	4	36	9	N	Two gense (one may s start); periodic (intermitter start tests

Scale anchors (see TEMPLATE_00): S = end-effect severity 1–5 · O = OREDA/IEEE-493 rate band 1–5 · D = detection 1–5 (5 = hidden). **Action threshold:** RPN ≥ 60 or any SPOF with S≥4 → P1.

FMECA summary: 16 failure modes · 3 SPOFs · 1 row RPN≥60 · max RPN 60 .

SF-03 Recommendations

Form: Recommendations Form ID: SF-03 Rev: 1
Plant: P01 Jubail Seawater Desalination Plant (SWRO) Sector: Central & Eastern Region: Eastern
Prepared by: Electrical Engineer Date: 2026-06-02 Reviewed (SCE-stamped): [SCE name – pending stamp]
Satisfies: REQ-DEL-04, REQ-DEL-10, REQ-DEL-11, REQ-SPEC-01 (within REQ-DEL-05 battery limit – recommend only)
Standards: IEEE 493 / ISO 20815
Status: [SIMULATED – ILLUSTRATIVE ONLY]

Prioritised register closing the loop: each finding (FM-ID) → recommended modification/action

→ priority/urgency. **Priority is derived mechanically from the FMECA RPN/SPOF** (see TEMPLATE_00),

so it cannot drift. SPOF rows feed the portfolio ranked register (REQ-DEL-11). The modification now

lives inside each recommendation row; **high-level / conceptual only** (within the battery limit —

assess + recommend, no detailed design or construction). Cost is a **band L/M/H only**, never a SAR figure.

REC- ID	Finding (FM-ID)	Asset tag	Finding summary	SPOF?	Recommended modification / action	Type	Reliability / availability gain (basis)	RPN before → after	Cost band	Priority	Urgency	Dependency sequencing
P01- REC- 01	P01- FM-001	P01- BUSTIE- 01	Manual 13.8 kV bus-tie → half- capacity loss on	Y	Add automatic bus- tie transfer scheme (auto changeover	Automation/ATS	Removes the half- capacity SPOF; restoration time	24 → 8	M	P1	≤6mo	Requires brief switchgear ou scheme study

			capacity loss on TX/incomer loss until manual restoration		(auto-changover relay + interlocks) so half-bus restores on TX/incomer loss without operator action — install automatic bus-tie transfer scheme with interlocks; revalidate grading in parallel		restoration time minutes → seconds; half-bus availability ↑ ~0.985 → ~0.999 (indicative RBD)					SCHEME STUDY coordinate with 002 grading
P01-REC-06	P01-FM-006	P01-DC-01	Ageing battery charger; latent loss of DC trip/protection supply (no capacity test on record)	N	Replace ageing battery charger + reinstate battery capacity-discharge testing; add on-line DGA monitor / restart oil-trend programme on main transformers — replace charger, reinstate battery capacity-discharge testing, alarm DC-fail	Monitoring / Replacement	Removes latent DC-supply loss (protection reliability ↑); early warning of transformer insulation degradation	60 → 12	M	P1	Immediate	DC charger is RPN item; exp DGA monitor / lead
P01-REC-12	P01-FM-012	P01-SWGR-01	Single common 13.8 kV board — busbar fault would cause total RO production loss	Y	Switchgear integrity programme: bus-zone protection verification, primary-injection/timing tests, arc-resistant retrofit assessment — switchgear integrity programme: bus-zone protection verify, primary-injection/timing tests, arc-resistant retrofit assessment	Protection / Replacement	Reduces likelihood and consequence of whole-board bus fault (the total-loss SPOF)	8 → 4	H	P1	≤24mo	SPOF S≥4 (consequence though occur low); outage + specialist test
P01-REC-02	P01-FM-002	P01-RELAY-02	Protection selectivity gap (study predates RO addition) → wider-than-needed trips	N	Protection coordination restudy + relay re-grading; add breaker-fail element on incomer — protection coordination restudy + relay re-grading	Protection	Restores selectivity (narrows trips) and adds breaker-fail backup; reduces spurious wide trips	36 → 12	L	P2	≤6mo	Needs live set files (P01-DO(data-call) before grading
P01-REC-04	P01-FM-004	P01-SWGR-01	Outdated arc-flash study; 13.8 kV incident energy & PPE boundaries unverified (IEEE 1584)	N	Commission updated IEEE 1584 arc-flash / incident-energy study; apply PPE-category labels & boundaries — commission updated IEEE 1584 study; apply PPE-category labels and boundaries	Protection	Re-establishes verified arc-flash safety basis; correct PPE category and safe approach boundaries	40 → 10	L	P2	≤6mo	Study only (assessment & feeds switchin procedures are labelling
P01-REC-05	P01-FM-005	P01-TX-03	Single service transformer to dosing/pre-treatment sub-area — no standby LV feed	Y	Provide alternate LV feed / standby service transformer (or permanent backup-genset tie) to dosing/pre-treatment sub-area — provide alternate LV feed / standby service transformer to the sub-area	Redundancy	Removes the service-transformer SPOF for the sub-area; sub-area availability ↑	18 → 6	M	P2	≤24mo	Sub-area not critical path; low urgency
P01-REC-16	P01-FM-016	P01-GEN-01	Genset auto-start reliability unconfirmed (hidden) — essential backup may not start on demand	N	Genset reliability uplift: formal monthly load/auto-start test regime + controls refurbishment; concept study for partial-RO standby capacity — formal monthly load/auto-start test regime +	Monitoring / Redundancy	Restores essential-load backup confidence; quantifies cost/benefit of partial RO backup (availability study)	36 → 16	M	P2	≤6mo	Partial-RO standby a concept study (capacity/fuel/feasibility)

					genset controls refurbishment							
P01-REC-03	P01-FM-003	P01-GEN-01	Gensets sized to essential loads only — no RO production backup on total grid loss	N	Genset reliability uplift: formal monthly load/auto-start test regime + controls refurbishment; concept study for partial-RO standby capacity — concept feasibility study for partial-RO standby capacity (availability cost/benefit)	Monitoring / Redundancy	Restores essential-load backup confidence; quantifies cost/benefit of partial RO backup (availability study)	16 → 16	M	P3	Planned	Partial-RO starts a concept study (capacity/fuel/feasibility)
P01-REC-07	P01-FM-007	P01-TX-01	Main transformer internal-fault risk; weak DGA/oil-condition evidence (N+1 holds)	N	Replace ageing battery charger + reinstate battery capacity-discharge testing; add on-line DGA monitor / restart oil-trend programme on main transformers — reinstate DGA/oil-trend programme; consider on-line DGA monitor	Monitoring / Replacement	Removes latent DC-supply loss (protection reliability ↑); early warning of transformer insulation degradation	60 → 12	M	P3	≤24mo	DC charger is RPN item; exp DGA monitor lead
P01-REC-08	P01-FM-008	P01-VFD-01	MV VFD fault trips one RO train (other trains unaffected)	N	monitor — no modification — clean/replace VFD cooling filters; maintain spare-module holding; monitor trip history			18	L	P3	≤24mo	—
P01-REC-09	P01-FM-009	P01-PMP-01	HP RO feed-pump motor failure derates one train	N	monitor — no modification — vibration-based condition monitoring; planned bearing/seal replacement programme			18	L	P3	Planned	—
P01-REC-13	P01-FM-013	P01-RELAY-01	Incomer relay fail-to-trip with no breaker-fail backup → escalated fault	N	Protection coordination restudy + relay re-grading; add breaker-fail element on incomer — add breaker-fail element (covered in MOD-02 re-grading)	Protection	Restores selectivity (narrows trips) and adds breaker-fail backup; reduces spurious wide trips	32 → 12	L	P3	≤6mo	Needs live set files (P01-DO data-call) before grading
P01-REC-14	P01-FM-014	P01-EARTH-01	Earthing grid resistance/touch-step potential unverified since 2013	N	Re-test earthing grid (resistance, touch/step potential) and remediate corroded bonds — re-test earthing grid; remediate corroded bonds	Earthing	Restores verified earth-fault return path & personnel touch-safety margin	24 → 8	L	P3	≤24mo	Requires partial outage window grid testing
P01-REC-15	P01-FM-015	P01-CABLE-01	Ageing 13.8 kV feeder terminations; single-feeder fault risk	N	monitor — no modification — thermography programme; planned re-termination of oldest feeders			18	L	P3	≤24mo	—
P01-REC-10	P01-FM-010	P01-PMP-02	Intake pump failure — minor intake-margin reduction (installed spare)	N	monitor — no modification — monitor; retain installed spare; routine maintenance			12	L	P4	Planned	—
P01-REC-11	P01-FM-011	P01-FEED-01	Loss of one 132 kV incomer carried by the other (N+1) — no production loss	N	monitor — no modification — monitor; no action beyond existing SCADA surveillance			6	L	P4	Planned	—

Priority rule: P1 = SPOF S≥4 or RPN≥60 · P2 = RPN 36–59 or SPOF S=3 · P3 = RPN 16–35 · P4 < 16. **Cost bands:** L < 0.5 M · M 0.5–

2 M · H > 2 M SAR (indicative, pending concept-design pricing). **Summary:** 16 recommendations · P1 3 · P2 4 · P3 7 · P4 2 · SPOFs 3 · of which 11 carry a modification · cost mix L/M/H = 9/6/1.

Rev 1 merges the former SF-05 Recommendations Summary into SF-03; the modification now lives inside each recommendation row (one row per FM-ID); monitor-only findings carry no modification. Superseded SF-03 SIM-4x source tags are folded into the SF-05 SIM-5x rows.

SF-04 Site Visit Report

Form: Site Visit Report Form ID: SF-04 Rev: 0
Plant: P01 Jubail Seawater Desalination Plant (SWRO) Sector: Central & Eastern Region: Eastern
Prepared by: Electrical Engineer Date: 2026-05-19 Reviewed (SCE-stamped): [SCE name – pending stamp]
Satisfies: REQ-DEL-02 (also supports REQ-COMP-03 QHSE)
Status: [SIMULATED – ILLUSTRATIVE ONLY]

1 · Visit metadata

- Date / time:** 2026-05-19, 08:00–15:30 AST · **Weather / conditions:** clear, ~41 °C, high humidity (coastal)
- Escort:** SWA O&M shift electrical supervisor · **Access constraints:** live-plant; 13.8 kV switchroom entry under permit-to-work only, no internal panel opening on energised equipment.

2 · Minutes of Meeting

Attendees

Name	Organisation	Role
Eng. A. Al-Otaibi	SWA O&M	Plant Electrical Supervisor
Eng. M. Al-Harbi	SWA O&M	Maintenance Planner
Eng. R. Said	RWC (consultant)	Electrical Engineer
Eng. K. Nasser	RWC (consultant)	Protection / RAM Engineer

Agenda / discussion: Confirm electrical configuration vs SLD (P01-DOC-01); walk down 132/13.8 kV substation, 13.8 kV switchgear and bus-tie arrangement; review standby genset scope; identify SPOFs and protection/arc-flash gaps; agree the document data-call to close SF-01 gaps. SWA confirmed the 13.8 kV bus-tie is operated **manually** (no auto-transfer scheme) and that the two standby gensets back **essential loads only** — RO production is not supported on total grid loss. SWA noted the arc-flash study has not been refreshed since the switchgear upgrade.

Decisions & actions

Action	Owner	Due
Provide transformer DGA/oil trend (closes P01-DOC-07)	SWA O&M	≤4 weeks
Provide station battery capacity-discharge test record (closes P01-DOC-12)	SWA O&M	≤4 weeks
Provide current relay setting files for incomer & feeder relays (closes P01-DOC-04)	SWA O&M	≤2 weeks
Confirm bus-tie interlock/operating philosophy in writing	SWA O&M	≤2 weeks
RWC to revalidate protection grading on receipt of settings	RWC	≤6 weeks
RWC to scope IEEE 1584 arc-flash restudy (assessment only)	RWC	≤6 weeks

Open questions / data-call (links to SF-01 gaps): DGA trend (P01-DOC-07, Missing); battery test record (P01-DOC-12, Missing); live relay settings (P01-DOC-04, Partial); arc-flash restudy basis (P01-DOC-05, Obsolete); service-area cable schedule completeness (P01-DOC-08, Partial).

3 · Survey scope walked

Areas inspected: 132 kV incomer yard (FEED-01/02) · 132/13.8 kV transformer bays (TX-01/02) · 13.8 kV switchroom incl. bus-tie (SWGR-01 / BUSTIE-01) · MV VFD room (VFD-01) · HP RO feed-pump & intake-pump halls (PMP-01/02) · 13.8/0.4 kV service substation (TX-03) · standby genset compound (GEN-01/02) · station DC / battery room (DC-01) · main earthing pit & test points (EARTH-01).

4 · Observations & findings

Numbered; each tagged with the asset tag and the FM-ID it seeds in the FMECA.

Obs #	Area	Asset tag	Observation	Seeds FM-ID
O-01	13.8 kV switchroom	P01-BUSTIE-01	Bus-tie breaker operated manually; no auto-transfer relay. On loss of one 40 MVA transformer, half the 13.8 kV bus (and its RO trains) stays dead until an operator closes the tie under permit — minutes-to-hours of half-capacity loss.	P01-FM-001
O-02	13.8 kV switchroom / yard	P01-RELAY-01, P01-RELAY-02	Protection study (P01-DOC-03) pre-dates the 2019 RO addition; feeder and incomer relay grading not revalidated. Selectivity miss risk → a downstream feeder fault could trip the incomer and a wider section of the bus.	P01-FM-002
O-03	Standby genset compound	P01-GEN-01, P01-GEN-02	Two diesel gensets sized for essential loads (control, lighting, instrument air, critical pumps) only; no capacity to carry RO production on total grid loss. Availability limitation rather than a fault.	P01-FM-003

O-04	13.8 kV switchroom	P01-SWGR-01	Last arc-flash / incident-energy study is 2015 (P01-DOC-05) and predates the switchgear upgrade; 13.8 kV incident-energy levels and PPE category boundaries unverified against IEEE 1584.	P01-FM-004
O-05	13.8/0.4 kV service substation	P01-TX-03	A single 13.8/0.4 kV service transformer feeds the chemical-dosing / pretreatment sub-area with no standby or alternate LV feed. Single-component loss de-energises that sub-area (does not stop the main RO trains).	P01-FM-005
O-06	Station DC / battery room	P01-DC-01	Battery charger shows signs of ageing (audible fan noise, elevated cabinet temperature); no capacity-discharge test on record (P01-DOC-12). DC supply feeds protection/trip circuits — a latent failure could disable tripping.	P01-FM-006
O-07	Transformer bays	P01-TX-01, P01-TX-02	Two 40 MVA transformers in N+1; no on-site DGA/oil trend held (P01-DOC-07). Individual transformer internal fault is covered by redundancy but condition-monitoring evidence is weak.	P01-FM-007
O-08	MV VFD room	P01-VFD-01	MV VFD driving HP RO feed pumps; cooling-filter dust loading visible. VFD fault trips the associated RO train; other trains unaffected.	P01-FM-008
O-09	RO feed-pump hall	P01-PMP-01	HP RO feed-pump motor — multiple trains, partial redundancy. Single motor failure derates production for that train only.	P01-FM-009
O-10	Intake pump hall	P01-PMP-02	Intake pumps run with installed spare; single pump failure has minor effect on intake margin.	P01-FM-010
O-11	Main earthing pit	P01-EARTH-01	Earthing study is 2013 soil-resistivity only (P01-DOC-09); as-installed grid resistance not re-tested. Touch/step-potential and protection earth-return unverified.	P01-FM-014
O-12	13.8 kV feeders	P01-CABLE-01	Several 13.8 kV feeder cable terminations are original (≈12 yr); minor surface tracking dust observed at one termination box. Ageing termination could fault a single feeder.	P01-FM-015

5 • Photo log

Samples use captioned placeholders — no real images.

Photo-ID	Caption	Asset tag	What it shows	Finding ref
P01-IMG-01	[PHOTO P01-IMG-01: 13.8 kV bus-tie breaker — manual local control]	P01-BUSTIE-01	Bus-tie cubicle with local manual control, no auto-transfer relay fitted	P01-FM-001
P01-IMG-02	[PHOTO P01-IMG-02: incomer & feeder protection relay fronts]	P01-RELAY-01	Relay HMIs; setting labels not matching controlled study rev	P01-FM-002
P01-IMG-03	[PHOTO P01-IMG-03: standby genset rating plate]	P01-GEN-01	Genset nameplate confirming essential-load sizing	P01-FM-003
P01-IMG-04	[PHOTO P01-IMG-04: 13.8 kV switchgear line-up & arc-flash label]	P01-SWGR-01	Switchgear front with outdated/absent incident-energy label	P01-FM-004
P01-IMG-05	[PHOTO P01-IMG-05: single service transformer feeding dosing sub-area]	P01-TX-03	Lone 13.8/0.4 kV service transformer, no standby	P01-FM-005
P01-IMG-06	[PHOTO P01-IMG-06: station battery charger cabinet]	P01-DC-01	Ageing charger; cabinet temperature/condition concern	P01-FM-006
P01-IMG-07	[PHOTO P01-IMG-07: 132/13.8 kV transformer bay TX-01]	P01-TX-01	Transformer bay, no on-site DGA sampling rig	P01-FM-007
P01-IMG-08	[PHOTO P01-IMG-08: MV VFD cooling-air filters dust loading]	P01-VFD-01	VFD enclosure filters with dust accumulation	P01-FM-008
P01-IMG-09	[PHOTO P01-IMG-09: 13.8 kV feeder cable termination box]	P01-CABLE-01	Ageing termination with minor surface tracking dust	P01-FM-015
P01-IMG-10	[PHOTO P01-IMG-10: main earthing test point / pit]	P01-EARTH-01	Earthing pit and disconnect link, no recent test tag	P01-FM-014

6 • Safety / access notes (QHSE)

All switchroom entries under SWA permit-to-work; no work on or opening of energised 13.8 kV equipment. Site PPE enforced (arc-rated coveralls, helmet, glasses, safety shoes). The unverified arc-flash study (O-04) is itself a QHSE concern — current PPE category boundaries for the 13.8 kV switchgear cannot be demonstrated against IEEE 1584. Heat-stress controls in force (hydration, shaded breaks) given ~41 °C.

7 • Sign-off

Prepared by: R. Said (RWC Electrical Engineer) · SWA acknowledgement: A. Al-Otaibi · Date: 2026-05-19

P01 Jubail Seawater Desalination Plant (SWRO) — combined study (SF-01 Inventory · SF-02 FMECA · SF-03 Recommendations · SF-04 Site Visit). Simulated/illustrative; generated from the sample Markdown by build_sample_html.py.

Peer-review commentary — what an experienced engineer notices

WHAT AN EXPERIENCED ENGINEER NOTICES WHEN REVIEWING THIS STUDY

The following observations are the kind of quality-check commentary that a senior reviewer or the PM would raise when reviewing the Jubail study. Use them as a self-check when you produce your own forms.

- **The highest RPN row (FM-006, DC charger, RPN = 60) is rated P1 immediately — not because the occurrence is high, but because the Detection rating is 5 (hidden).** The DC supply feeds the protection and trip circuits; a failure that is undetected until a fault occurs is the most dangerous class of latent SPOF. The study correctly assigns "Immediate" urgency and cross-links to the missing battery capacity-discharge test in SF-01. On your own plants, always check whether the DC charger / battery-test record is in SF-01 before assigning a D rating — if it is missing, D = 5, and the row will likely be P1.
- **The protection-selectivity finding (FM-002, RPN = 36) exploits the document gap in SF-01 (P01-DOC-03 Partial, pre-dates expansion) to justify a high Detection = 4.** This is the correct method: a protection study that predates a plant modification is not evidence that grading is correct — it is evidence that grading

is *unknown*. The study appropriately flags this as Partial in SF-01 and elevates D accordingly. When you see a protection study dated before a major expansion on any of your plants, treat grading as unverified until proved otherwise.

- **Three SPOF=Y rows are identified: FM-001 (manual bus-tie), FM-005 (single service transformer), and FM-012 (common MV bus).** Notice the difference in severity: FM-001 and FM-012 carry S = 4 (significant production impact), while FM-005 carries S = 3 (sub-area only, not on RO critical path). Priority is assigned accordingly (FM-001 and FM-012 → P1; FM-005 → P2). When you identify a SPOF on your plants, always ask: does loss of this single component cause a plant-wide outage (S = 4 or 5) or only a partial / sub-system outage (S = 2 or 3)? The answer sets the priority class.
- **The recommendation for the manual bus-tie (REC-01) specifies "add automatic bus-tie transfer scheme with interlocks" — not just "close the bus-tie".** Simply closing the bus-tie permanently would eliminate the N+1 architectural benefit. The correct fix is an automatic transfer scheme that closes the tie on a monitored event (transformer loss or incomer undervoltage) and re-opens it once the fault is cleared. The study correctly notes that grading must be revalidated in parallel, because closing the bus-tie changes the fault-current contribution seen by each relay.
- **The earthing finding (FM-014) carries a moderate RPN (24) but a note that the last test was in 2013 — 13 years prior to the study date.** The study correctly escalates the Detection rating on the basis that no re-test has been performed since the original commissioning. Corrosion in a coastal (saline) environment like Jubail is significantly faster than in an inland site, making the time-since-test argument even stronger. On coastal plants, any earthing test older than five years should be treated as Partial or Obsolete in SF-01.
- **Seven of the 16 recommendation rows are "monitor — no modification".** This is disciplined use of the recommendations register: not every FMECA finding requires a capital modification. Items with low RPN and adequate existing safeguards are closed with a monitoring action and a cost band of L. A recommendations register that lists modifications for every single failure mode is a sign of under-calibrated FMECA scoring, not thoroughness.
- **The VFD cooling-filter dust loading (O-08 in SF-04) seeds FM-008, but the recommendation is to clean and replace filters rather than a capital modification.** The study correctly distinguishes between an O&M action (P3, cost band L) and a capital modification. On your own plants, always separate the "do now — O&M" actions from the "invest in — capital" recommendations, and be explicit about which is which in SF-03.
- **SF-04 Observation O-04 (arc-flash study predates switchgear upgrade) appears in Section 6 (Safety notes) as well as in the findings table.** This is correct practice: an unverified arc-flash boundary is both a FMECA finding (FM-004) and a QHSE concern that must be documented separately in the safety section of SF-04. On your own site visit reports, always copy any arc-flash label absence or outdated study finding into the QHSE section as well as the findings table.

- **The study's cost-band summary (L/M/H = 9/6/1) is stated explicitly in the SF-03 footer.** This is useful for the client's budget planning. On your own plants, include this summary line in the SF-03 footer — it helps the PM aggregate across all 21 plants for the campaign summary report.

- ★ **Key takeaway: Read the Jubail study as a worked example of the complete method — not as a template to copy blindly. The peer-review commentary above identifies the reasoning behind the scoring decisions. Apply that reasoning, not just the scores, on your own plants.**

Part 5 — Self-Check

Where you are in the workflow: test your understanding of the traceability chain and the non-intrusive rule before going to site for the first time.

Q1 — The traceability chain

Name the four links in the traceability chain that connects a site observation to a capital modification recommendation, and state which deliverable from each link lives in.

▼ Show answer

The four links are:

1. **Site observation (OBS-xx)** — recorded in SF-04 (Site-Visit Report), Section 4. Each observation is factual and tagged to an asset; it seeds a FMECA row via the FM-ID field.
2. **FMECA row (FM-xx)** — in SF-02 (FMECA Worksheet). The observation provides the evidence; the FMECA assigns S/O/D scores, computes RPN, flags SPOF=Y/N, and generates a recommendation reference (REC-ID).
3. **Recommendation row (REC-xx)** — in SF-03 (Recommendations Register). Each FMECA row that warrants action produces one recommendation row; the REC-ID cross-references the FM-ID so the chain is auditable.
4. **Modifications catalogue** — the concept modification packages (Part 4). Each REC row maps to one or more Mod IDs from the catalogue, which describe what to specify, verify and document.

A completed register in which every REC-xx traces back to a specific FM-xx, and every FM-xx traces back to an OBS-xx or a DOC-xx gap, is a fully traceable deliverable. This is what the PM quality-checks before approval.

Q2 — The non-intrusive rule

You arrive at the plant and notice that the MV switchgear front door is standing open. The O&M supervisor says: "Go ahead — it's been like that all morning, it's fine." What do you do, and why?

▼ Show answer

You **do not enter or touch** the open panel. The non-intrusive rule is absolute: you must not open energised panels, operate any device, or interfere with any equipment — regardless of whether someone else has already left it open, and regardless of verbal permission from local staff. Only the host's authorised person (AP) under a valid permit-to-work can authorise access to live electrical equipment.

The correct actions are:

- Record the open panel in SF-04 Section 4 as an observation: "MV-SWGR front door found open, no PTW in evidence" — with a photograph if permitted.
- Raise it in SF-04 Section 6 (Safety / QHSE notes) as a concern: an open live MV panel with no PTW is a QHSE finding in its own right.
- Inform the escort and request either a formal PTW to use the access for observation, or request the door be closed.
- Do not cite the open door as an opportunity to make visual observations inside the live panel without PTW authorisation.

Q3 — Document gap → FMECA Detection rating

The SF-01 Document Inventory shows that the battery capacity-discharge test record (DOC-15 UPS/DCDB battery maintenance records) is listed as **Missing**. How does this gap affect the FMECA row for the DCDB charger failure mode, and what priority band will the row most likely reach?

▼ Show answer

A missing battery capacity-discharge test means the true battery hold-up time is **unknown**. Without evidence of a recent discharge test, there is no way to know whether the battery will deliver its rated autonomy — it may have degraded to a fraction of nameplate capacity. This is the definition of a **latent (hidden) failure**: the failure is undetectable until the moment the battery is called upon.

Mechanical FMECA update: Set Detection D = 5 (effectively undetectable — hidden until a fault event). For DCDB charger failure with S = 4 (whole-bus protection loss, safety-critical) and a typical O = 3 (ageing charger, once per few years), $RPN = 4 \times 3 \times 5 = 60$. This meets the P1 threshold ($RPN \geq 60$). The row becomes Immediate priority — exactly as illustrated in the Jubail FM-006 row.

★ **Key takeaway: The traceability chain — SF-04 observation → SF-02 FMECA → SF-03 recommendation → concept modification — is the backbone of every deliverable. The non-intrusive rule is absolute: no observation is worth a safety violation. Document gaps in SF-01 mechanically raise Detection ratings in the FMECA — follow the chain, do not estimate.**

First-Assignment Walk-Through

Where you are in the workflow: you have received your first plant assignment. Follow this day-by-day sequence from first notification to submission.

This walk-through narrates a typical plant cycle from Day -5 (desk study begins) to Day 4 (quality assurance and submission). The timeline is indicative — a simple plant may compress to five days total; a complex plant with many document gaps may expand to eight or ten. Adjust as needed, but keep the sequence.

Day -5 — Desk study begins

Morning — receive the plant package and issue the document request

1. Open the four blank assessment forms from [Appendix — Blank Forms](#). Fill in the header of each form: plant name, plant reference number, engineer name, date, and status ("In preparation"). Save as working copies with the plant reference in the filename.
2. Issue the document request to the plant O&M contact using the standard request template. Request all ten document categories listed in the [Desk Study](#) section. Log each requested item in SF-01 with status "Requested".
3. Obtain the logistics information: site address, visitor access procedure, site safety induction requirements, escort contact details, and whether a PTW must be pre-arranged. Block the site visit date in your calendar.

Afternoon — first document review

1. As documents arrive, update SF-01 status (Received / Partial). If documents are already available in the campaign document store, retrieve them immediately.
2. Study the master SLD first. Trace the power path: utility → MV-MET → MV-SWGR → TX-1/TX-2 → LV-MSB (Bus A/B with bus-tie BT) → MCC-1 ... MCC-n, VFD, DG/ATS, UPS/DCDB. Annotate: how many transformers? Is the bus-tie N/O? Is there an auto-transfer scheme? Any single-feed sub-areas?
3. On a printed copy of the SLD, mark with a red pencil every component whose failure you believe might cause a plant-wide or significant partial outage. These are your preliminary SPOF candidates — you will verify them on site and refine them in the FMECA.

Day -2 — Desk calculations and pre-visit preparation

Morning — desk calculations

With the equipment schedule and load list in hand, build the calculation model shell and run the following desk-based checks (see [Part 2](#) for all methods):

- **Transformer sizing check** ([Part 2](#)): total active load (kW) ÷ assumed power factor → apparent power (kVA); divide by number of transformers; compare to installed transformer kVA. If >80 % loaded in normal operation, flag as a SPOF candidate.
- **FLC check** ([Part 2](#)): for the largest motor on each MCC, calculate rated FLC; compare to MCC overload relay setting from the schedule. Note any that appear set ≥ 115 % FLC.
- **Indicative short-circuit level** ([Part 2](#)): using transformer impedance and utility fault level (if known), estimate the maximum fault current at the LV main busbar. Compare to switchgear rated fault-withstand current. If utility fault level is not yet available, note this as a data-call.
- **Genset sizing check** ([Part 2](#)): sum the essential-load list; compare to DG rated kVA at the assumed power factor. Confirm whether the DG covers full production or essential loads only. If the load list has gaps, note as a site question.

Afternoon — pre-visit preparation

1. Print two A3 copies of the master SLD. Label one "RED-LINE COPY" and one "REFERENCE COPY". On the red-line copy, mark all preliminary SPOF candidates with a pencilled circle.
2. Print the [Visual Inspection Checklist](#) and the [Drawing Availability and SLD Verification Checklist](#) from this manual. These are your primary field tools.
3. Print a blank SF-04 Site Visit Report header and MOM template. Fill in the plant name, date, and visit agenda.
4. Confirm with the escort: PTW for switchroom entry (confirm time), induction requirements, photography restrictions, and any areas that will not be accessible.
5. Check PPE: arc-rated coveralls, hard hat, safety glasses, safety shoes, and gloves (confirm PPE category required by the PTW if known in advance).

Day 0 — Arrival and QHSE induction

On arrival

1. Report to the plant control room or administration office. Sign in. Confirm the escort is present and the site induction is ready.
2. Complete the site QHSE induction: emergency muster points, emergency contact numbers, permit-to-work procedure, specific hazards (process plant hazards are in addition to electrical hazards — confirm chemical, pressure, and process risks with the escort).
3. Collect the PTW for switchroom entry. Read it in full. Confirm the scope, validity period, and conditions. Sign it.
4. Attend the kick-off meeting with the escort and any other O&M staff available. Confirm the visit agenda, areas to be walked, and documents not yet received. Record attendees in SF-04 Section 2 (MOM). Confirm in writing (meeting minutes or email) the key verbal facts: bus-tie operating philosophy, genset scope, last arc-flash study date.

Day 1 — Site visit

Morning — MV and transformer walk-down

1. Enter the MV switchroom under the PTW with the escort. Work through **Zones 1–3** of the [Visual Inspection Checklist](#): switchroom general, MV-SWGR / MV-MET / MV-NER, then TX-1 and TX-2.
2. Photograph PH-01 (switchroom general), PH-02 (MV switchgear line-up and arc-flash labels), PH-03 (bus-tie position indicator), PH-04 (TX-1 nameplate), PH-05 (TX-2 nameplate) from the mandatory photo list.
3. On the red-line SLD, confirm every item visible in the switchroom. Mark confirmed items with a tick; mark discrepancies in red. Note the as-found breaker positions.
4. Ask the escort specifically: When was the protection coordination study last updated? Has the plant been expanded since that date? When was the arc-flash study last performed? Note answers in SF-04 Section 2.

Afternoon — LV, standby, DC, and earthing walk-down

1. Work through **Zones 4–9** of the [Visual Inspection Checklist](#): LV-MSB and BT; MCCs and VFDs; PFC and DBs; DG and ATS; UPS and DCDB; visible earthing.
2. Capture remaining mandatory photographs: PH-06 (LV-MSB), PH-07 (DG), PH-08 (ATS), PH-09 (DCDB), PH-10 (UPS), PH-11 (earth test point), PH-12 (any defect or discrepancy).
3. Collect any documents not previously received: relay setting records, genset test log, battery test record, asset register pages. Log each new document in SF-01.
4. Before leaving site: review your checklist — have all mandatory photographs been taken? Have all zones been walked? Are there open questions for the escort that can still be answered today? Ask them now.
5. Agree the data-call action list with the escort (documents to be sent within agreed timescales). Record in SF-04 Section 2, decisions and actions table.

Days 2-3 — Analysis and form completion

Day 2 — SF-04, SF-01, FMECA build

1. **Finalise SF-04 first** — while the visit is fresh. Complete all sections: metadata, MOM, survey scope, observations (one row per finding with tag and FM-ID), photo log (index and caption all photos), safety notes, and sign-off. Target: completed within 24 hours of the site visit.
2. Update SF-01: change the status of any documents collected on site; add any new documents to the register; update the gap summary. Mark any remaining Missing or Partial items as data-call actions.
3. Build the FMECA (SF-02): open a row for each SF-04 observation and for each SF-01 gap that represents a failure mode. Assign S, O, D; compute RPN and Crit; flag SPOFs. Cross-check: every SF-04 observation with an FM-ID should have a corresponding FMECA row. See the [S/O/D scale anchor table in Part 3](#) for the exact score definitions (what S=1..5, O=1..5, D=1..5 each mean).
4. Update D ratings for all FMECA rows affected by SF-01 gaps (as described in [the Post-Site Analysis section](#)).

Day 3 — Calculations update and SF-03 recommendations

1. Update the calculation model with the as-found nameplate data from the site visit. Re-run the transformer sizing, FLC, short-circuit, genset, and load-flow checks. Note any calculations that cannot be completed without outstanding data-call responses — flag these clearly in the calculation package.
2. Build SF-03 Recommendations: one row per FM-ID from the FMECA. Assign priority mechanically from the RPN / SPOF rule. Write the recommended modification or action. Assign cost band (L/M/H). Note monitor-only rows explicitly. See [Part 4 — Modifications Catalogue](#) for standard modification types and their descriptions.
3. Cross-check the traceability chain: every SF-03 row should carry an FM-ID from SF-02; every SF-02 row should carry a reference to an SF-04 observation or SF-01 gap. If any row is untraceable, either remove it or add the missing link.

Day 4 — Quality assurance and submission

1. **Self-check all four forms** using this list: **Progress: 0 / 6**

- ☐ SF-01: every row has a Status; gap summary count is correct; all data-call actions have owners and due dates.
 - ☐ SF-02: every row has S, O, D, RPN, Crit, and SPOF flag; every SPOF=Y row has a P1 or P2 recommendation in SF-03; no RPN cell is blank.
 - ☐ SF-03: every FM-ID from SF-02 has exactly one recommendation row; priority is consistent with the RPN/SPOF rule; cost band is L/M/H (no SAR figures).
 - ☐ SF-04: all seven sections are complete; MOM action list has owners and due dates; all 12 mandatory photos are indexed; escort acknowledgement received.
 - ☐ Calculation package: all required studies completed or flagged as pending data; indicative label on all values.
 - ☐ Traceability check: any FMECA row that cannot be traced to either an SF-04 observation or an SF-01 gap has been reviewed and either removed or sourced.
2. Send the completed forms and calculation package to the PM for review. Attach the indexed photo library. Note any items that are still outstanding (pending data-call responses) and their expected resolution dates.
 3. File the red-lined SLD with the plant document package. Note any SLD revision that the plant owner needs to carry out, and reference this in SF-03 as an action (if applicable).

★ **You are now ready. You have completed one full plant cycle: desk study → site visit → analysis → recommendations → submission. Repeat this sequence for each of the 21 plants in the campaign, adjusting the timeline as plant complexity and document availability dictate. Blank forms for every new plant are in [Appendix — Blank Forms](#).**

i **Note:** These appendices are quick-reference and printable tools — concise distillations of the manual's technical content intended for use in the field and at the desk. All figures are *indicative*; confirm against plant-specific data and the governing standards before acting on them.

A — Glossary of Terms and Abbreviations

Alphabetical table of abbreviations and terms used across this manual. Expand from the course overview glossary and includes terms introduced in Parts 4 and 5. All voltage and frequency references follow the IEC basis of this manual (11 kV MV, 400/415 V LV, 50 Hz).

Term / Abbreviation	Meaning and context in this manual
ACB	Air Circuit Breaker. Large LV breaker used for switchboard incomers and the bus-tie on LV-MSB . Ratings typically 1600–6300 A; short-time withstand up to 65–100 kA (1 s).
AFB	Arc-Flash Boundary. The working distance at which incident energy falls to 1.2 cal/cm ² (the onset-of-a-second-degree-burn threshold). Determined per IEEE 1584-2018.
ANSI	American National Standards Institute. Device numbering scheme (e.g. 50/51, 87T) used on SLD relay/breaker labels as industry convention alongside IEC.
ATS	Automatic Transfer Switch. Monitors the utility supply and, on failure, automatically transfers the essential bus to the standby DG . A failed-to-transfer ATS is one of the highest-RPN items in a typical FMECA.
Availability (A)	$A = MTBF / (MTBF + MTTR)$. The fraction of calendar time a component or system is in its functioning state. Expressed as a decimal (e.g. 0.9992) or as annual downtime hours.
AVR	Automatic Voltage Regulator. Controls the excitation of the diesel generator alternator to maintain output voltage within limits during load changes.
BESS	Battery Energy Storage System. Grid-connected or plant-connected battery system providing ride-through energy, inertia emulation, fast frequency / voltage support, or islanding capability.
BT (or B/T)	Bus-Tie. The normally-open ACB (or contactor) connecting Bus A and Bus B of the LV-MSB split busbar. Closes automatically on loss of one incomer to allow the healthy transformer to back-feed the dead bus. A critical SPOF-mitigation element.
Cable ampacity	The continuous current-carrying capacity of a cable under defined installation conditions (temperature, grouping, burial). Determined from IEC 60364 / manufacturer tables after applying derating factors.
CT / VT	Current Transformer / Voltage Transformer. Instrument transformers that step down MV/LV currents and voltages to safe measurement levels (e.g. 5 A / 110 V secondary). Feed protection relays, metering and SCADA.
DB	Distribution Board. Final-circuit LV panel fed from the LV-MSB or an MCC; supplies lighting, small power and HVAC loads via MCBs/RCBOs.
DCDB	DC Distribution Board (battery and charger system). Provides 110 V DC (typical) for MV switchgear trip/close coils and protection relay auxiliary supplies. A DCDB failure silences protection — very high severity in the FMECA.
DCS	Distributed Control System. Plant-wide process control system with operator HMI. Receives electrical status (breaker position, voltages, currents) and issues high-level electrical commands.
DG	Diesel Generator set. Standby power source sized for the essential-load bus (~800–1250 kVA indicative). Includes AVR, governor, fuel day-tank, battery starting and acoustic enclosure.
DOL	Direct-On-Line. Simplest motor starting method — motor connected straight to supply. Starting current typically 5–7 × FLC; suitable for motors up to ~7.5 kW or where the bus can absorb the inrush.
DVR	Dynamic Voltage Restorer. Series power-electronics device injecting a compensating voltage to protect plant loads against upstream voltage sags and swells.
EDB	Essential Distribution Board. Fed via the ATS from the DG (or utility essential bus). Supplies safety-critical loads that must remain energised during a utility outage.
FMECA	Failure Mode, Effects and Criticality Analysis. Structured IEC 60812 method: list failure modes for each component, assess effects on the system, score Severity (S), Occurrence (O), Detectability (D), compute $RPN = S \times O \times D$, and rank for action. The primary SPOF quantification tool in this campaign.
FLC	Full-Load Current. The rated continuous current at full load: $I = S / (\sqrt{3} \cdot V)$ for 3-phase. Basis for breaker sizing, cable selection and protection relay pickup settings.
GOOSE	Generic Object-Oriented Substation Event. IEC 61850 fast peer-to-peer message over the station bus used for inter-IED tripping, bus-transfer logic and load-shedding. Latency typically <4 ms.
IED	Intelligent Electronic Device. Numerical multifunction protection relay performing overcurrent (50/51), earth-fault (50N/51N), differential (87T), voltage (27/59) and other protection functions, plus measurement and control.
Isc	Short-Circuit Current (symmetrical). The prospective fault current at a bus under bolted three-phase conditions. Used to specify the breaking capacity (Icu) of switchgear. Indicative: ~38.5 kA per 1600 kVA, Z = 6 % transformer.
LPL	Lightning Protection Level. Classes I–IV per IEC 62305, determined by risk assessment. Sets air-termination geometry, down-conductor spacing and earth-electrode requirements.
LPS	Lightning Protection System. The complete system of air terminations, down conductors and earth electrodes protecting the plant structure and equipment against direct lightning strikes.
LV	Low Voltage. Utilisation voltage; in this manual 400/415 V, 3-phase, 50 Hz (IEC: ≤ 1 kV AC). Confirm per plant.
LVRT / UVRT	Low / Under Voltage Ride-Through. The capability or protection settings that allow plant equipment (particularly VFD-driven loads) to remain connected through a defined voltage dip rather than tripping and exacerbating the disturbance.
MCC	Motor Control Centre. LV assembly of motor starter bays (DOL, star-delta, soft-starter, VFD) feeding process motors. Tags MCC-1 ... MCC-n per the plant scheme.
MCCB	Moulded-Case Circuit Breaker. Compact LV protective device for outgoing feeders and motor circuits. Breaking capacity ratings from ~10 kA to 85 kA; motor-protection versions incorporate thermal/magnetic or electronic trip units.

Term / Abbreviation	Meaning and context in this manual
MMS	Manufacturing Message Specification. IEC 61850 protocol for client-server monitoring, control and data exchange between IEDs and SCADA/EMS. Complements GOOSE for non-time-critical data.
MTBF	Mean Time Between Failures. $MTBF = 1/\lambda$ (h). Average operating time between successive failures. Higher MTBF = lower failure rate = more reliable. See Appendix G for indicative values.
MTTR	Mean Time To Repair / Restore. Average time from failure occurrence to return to service (detection + logistics + repair + recommissioning). MTTR is the key lever that automation (auto bus-tie, ATS) acts on — reducing it from hours (manual) to seconds (auto-transfer) dramatically raises availability.
MV	Medium Voltage. Higher distribution voltage; in this manual 11 kV (IEC: $1\text{ kV} < MV \leq 36\text{ kV}$). Actual utility voltage must be confirmed per plant.
MV-MET	MV Utility Revenue Metering Cubicle. Contains utility-grade CTs/VTs and the tariff meter at the point of supply. Not a protection device — verify its presence and metering-point location during desk study.
MV-NER	Neutral Earthing Resistor. Connected between the MV transformer neutral and earth to limit earth-fault current (typically 300–400 A) while keeping faults detectable. $R \approx 21\ \Omega$ for 300 A at 11 kV (indicative).
MV-SWGR	MV Metal-Clad (or metal-enclosed) Switchgear. 11 kV bus with vacuum circuit breakers; incomer, bus-section and transformer-feeder panels. Rated for prospective fault level (indicative: 25 kA/1 s).
N+1 / N-1	N+1: one more capacity unit than the minimum required — so one can be lost without losing the function. N-1: the condition after one unit has failed; the system must still serve the essential load. The $2 \times 1600\text{ kVA} + \text{bus-tie}$ arrangement is N-1 secure for essential load (~55 % of total).
NER	Neutral Earthing Resistor — see MV-NER .
PCC	Power Control Centre. The main LV switchboard that distributes power to MCCs and outgoing feeders. In this manual, LV-MSB is the PCC. Distinct from an MCC (which only controls motors).
PFC	Power Factor Correction. Automatic capacitor bank (with 7 % detuned reactors to handle VFD harmonics) that raises plant power factor from ~0.85 to ≥ 0.95 , reducing reactive demand and transformer / cable loading.
PMS	Power Management System. Dedicated electrical controller responsible for: load shedding on generation failure, automatic bus transfer (bus-tie / ATS), generator synchronisation, and energy management. Interfaces with the DCS.
PPE	Personal Protective Equipment. Arc-rated PPE is required for work on or near energised electrical equipment; the required category is set by the arc-flash study (incident energy in cal/cm ²). Non-intrusive site surveys in this campaign do not require arc-rated PPE if live-panel access is not needed — but confirm with the host's QHSE procedure.
QHSE	Quality, Health, Safety and Environment. Management system framework (ISO 9001 / 14001 / 45001) applied to all site activities. All surveys in this campaign are non-intrusive; QHSE and permit-to-work rules of the host facility always take precedence.
RAM	Reliability, Availability, Maintainability. The quantitative study of system uptime: how often failures occur (R), what fraction of time the system works (A), and how quickly it is restored (M). See Part 3 — FMECA / RAM .
RBD	Reliability Block Diagram. Series / parallel model used to compute system availability from component data. Series elements degrade availability; parallel (redundant) paths multiply it up. See Part 3 .
ROCOF	Rate of Change of Frequency. Measure of frequency volatility on the grid (Hz/s). Increases as synchronous generation (with inertia) is displaced by inverter-based renewables. High ROCOF can trip plant protection (df/dt relays) prematurely.
RPN	Risk Priority Number. FMECA criticality score = Severity (S) × Occurrence (O) × Detection (D), each scored 1–5 (higher = worse outcome / more frequent / harder to detect). $RPN_{max} = 125$. Used to rank FMECA findings for action priority. Priority bands: P1 = SPOF with $S \geq 4$ or $RPN \geq 60$; P2 = $RPN\ 36\text{--}59$; P3 = $RPN\ 16\text{--}35$; P4 < 16. See Appendix C .
SCADA	Supervisory Control and Data Acquisition. Supervisory monitoring and control layer for the power system; aggregates IED data, trends energy, and provides alarm management to operators.
SLD	Single-Line Diagram. One-line schematic showing the power system's main components and their electrical connections, using IEC 60617 symbols. The primary document for SLD verification on site and for SPOF hunting.
SPOF	Single Point of Failure. A component or path whose failure alone disables a function or supply that must remain available. The central concept of this assessment campaign. See Part 3 and Appendix F .
STATCOM / SVC	Static Synchronous Compensator / Static VAR Compensator. Fast dynamic reactive-power devices used for voltage stabilisation and flicker / voltage-sag mitigation on weak or renewable-heavy grids.
TN-S	TN-S earthing system. IEC 60364 LV earthing arrangement where the neutral (N) and protective earth (PE) conductors are kept separate throughout. Used at 400 V LV in this plant.
TX-1 / TX-2	MV/LV power transformers 1 and 2. Each indicatively 1600 kVA, 11 kV / 0.4 kV, Dyn11, $Z \approx 6\%$, feeding the split LV-MSB busbar. Both together provide N-1 redundancy via the bus-tie.
UPS	Uninterruptible Power Supply. Battery-backed, no-break clean AC supply for the DCS/PLC/SCADA, field instrumentation and critical communications. Typical battery autonomy 15–30 min (to be specified per plant).
VFD / VSD	Variable Frequency Drive / Variable Speed Drive. Power-electronic converter providing continuously variable speed control of AC motors (pumps, fans, compressors). Reduces starting current to near FLC; generates harmonics requiring detuned PFC.
ZSI	Zone-Selective Interlocking. Communication scheme between LV ACBs and MCCBs that allows the faulted zone's device to trip instantaneously while upstream devices remain on their normal graded delay — achieving both selectivity and fast arc-flash clearance.
λ (lambda)	Failure rate (failures per hour). $\lambda = 1/MTBF$. The primary input to reliability calculations. See Appendix G for indicative values per equipment class.

★ Key takeaway: 46 terms defined. Use this table when reading the manual in the field or when completing FMECA and report forms. Cross-references to the

relevant manual sections are embedded in key definitions.

B — IEC / IEEE Standards Reference

The table below lists the principal standards cited across this manual and the reference training material. The "Where it applies" column states the specific use in an MV/LV process-plant SPOF assessment context. This manual is standalone — the descriptions below are sufficient context; you do not need to hold copies of these standards in the field, but you should be aware of their scope.

Standard	Title / Scope	Where it applies in this manual
IEC 60034 (series)	Rotating electrical machines — general requirements and ratings for AC motors.	Motor nameplate data verification; LV-vs-MV feed threshold (~150–200 kW); MCC starter selection.
IEC 60076 (series)	Power transformers — ratings, losses, impedance, tap changers, test methods.	Transformer rating and Z% verification (TX-1 / TX-2); N-1 loading check; fault-level calculation input.
IEC 60364 (series)	Low-voltage electrical installations — design, protection, cable selection, earthing (TN-S, TT, IT), derating rules.	Cable sizing and derating (§6, Module 2); LV TN-S earthing; distribution board design. Applied to LV-MSB , MCC , DB / EDB .
IEC 60812	Analysis techniques for system reliability — FMEA / FMECA procedure, including criticality analysis and RPN scoring.	The FMECA form (Appendix E, Form 2) follows IEC 60812 column structure. All FMECA worksheets in the campaign use this method. See Part 3 — FMECA .
IEC 60831	Shunt power capacitors for AC power systems — capacitor bank ratings, protection.	PFC bank specification and protection (PFC tag); detuned reactor sizing for VFD harmonic environment.
IEC 60909	Short-circuit currents in three-phase AC systems — calculation method for prospective fault currents.	Short-circuit study (§4, Module 2). Used with power-system software (ETAP / DigSILENT) to size breaking capacity of LV-MSB , MCC and all ACB/MCCB.
IEC 60947-2	Low-voltage switchgear and controlgear — circuit breakers (ACB and MCCB ratings, breaking capacity Icu/Ics, trip-unit types).	ACB / MCCB specification for LV-MSB incomers, bus-tie and outgoing feeders; breaking-capacity verification against fault level from IEC 60909 study.
IEC 61078	Reliability block diagrams — construction rules, series/parallel formulae, system availability computation.	RBD analysis in Module 5 and Part 3; quantifying availability benefit of N+1 transformer + bus-tie redundancy.
IEC 61439-1/-2	Low-voltage switchgear assemblies — type-tested assemblies: Form segregation, busbar bracing, temperature rise, short-circuit withstand.	Specification and verification of LV-MSB (Form 4b) and MCC assemblies. Check form segregation during site visit.
IEC 61800 (series)	Adjustable speed electrical power drive systems — ratings, EMC, protection, safety.	VFD specification; harmonic emission from VFD drives (input to PFC detuning requirement).
IEC 61850	Communication networks and systems for power utility automation — station bus, GOOSE (fast tripping/transfer), MMS (monitoring). Defines data models for IED interoperability.	PMS integration; GOOSE-based fast bus-transfer and load shedding (Module 4). Present on digital substations; check whether IEC 61850 is deployed during desk study.
IEC 62271-100	High-voltage AC circuit breakers — ratings, making/breaking capacity, tests.	MV circuit breaker specification (vacuum CBs in MV-SWGR).
IEC 62271-200	AC metal-enclosed switchgear and controlgear for rated voltages above 1 kV and up to and including 52 kV — construction, type testing, internal arc classification (IAC).	Specification and condition assessment of MV-SWGR during site visit; IAC rating relevant to arc-flash risk at MV level.
IEC 62305 (series)	Protection against lightning — risk assessment (Part 2), physical damage (Part 3), electrical and electronic systems (Part 4).	Lightning protection assessment (§12, Module 2); LPS class selection; bonding to earth grid. Applies to all plant buildings and outdoor structures.
IEEE 80	Guide for safety in AC substation grounding — touch voltage, step voltage, earth grid design.	Earth-grid design and touch/step voltage compliance (§12, Module 2); MV substation earthing verification.
IEEE 493	Recommended practice for the design of reliable industrial and commercial power systems ("Gold Book") — reliability data, design methods, redundancy principles, RBD analysis.	Primary source of indicative equipment failure-rate and MTBF data (see Appendix G). Referenced in Module 5 (FMECA & RAM) and the FMECA form.
IEEE 1584	Guide for performing arc-flash hazard calculations — empirical model for incident energy and arc-flash boundary at AC equipment.	Arc-flash / incident-energy study (§11, Module 2). Preferable (not required) in this campaign. Determines PPE category and arc-flash boundary at LV-MSB and MCC .
ISO 14224	Petroleum, petrochemical and natural gas industries — collection and exchange of reliability and maintenance data for equipment. Defines equipment taxonomy, failure modes and data formats.	Equipment taxonomy for populating the FMECA and Document Inventory (SF-01). Cross-referenced for failure-rate data alongside IEEE 493.
ISO 20815	Petroleum, petrochemical and natural gas industries — production assurance and reliability technology.	RAM / RBD availability analysis methodology (Module 5); preferable study deliverable in this campaign.
NFPA 70E	Standard for Electrical Safety in the Workplace (USA/ANSI). Arc-flash PPE categories, boundary definitions, energised work permits.	Arc-flash PPE selection and labelling (§11, Module 2); host QHSE requirements may reference NFPA 70E even in IEC-governed plants. Non-intrusive surveys should not trigger NFPA 70E energised-work requirements.

i **Note:** Standard numbers and editions are indicative. Always confirm the governing edition with the project specification and local regulatory authority. For Saudi Arabia plants, NEC/Saudi Electricity Company (SEC) technical standards may also apply and take precedence over the IEC defaults listed above.

★ **Key takeaway: 20 standards referenced. IEC 60812 (FMECA), IEC 60909 (fault level), IEC 62271-200 (MV switchgear) and IEEE 493 (reliability data) are the four you will use most in this campaign.**

C — Formula Reference Card

Consolidated printable reference of every key electrical calculation formula used in this manual. For worked examples and derivations see [Part 2 — Calculations](#). All quantities are SI; constants: $\sqrt{3} = 1.732$, 1 pu = 100 %. All values are *indicative*.

Quantity	Formula	Symbols	Indicative example (this plant)
Apparent power	$S = P / \text{pf}$	S: apparent power (kVA/MVA); P: active power (kW/MW); pf: power factor.	2.0 MW / 0.85 = 2.35 MVA
3-phase current (from S)	$I = S / (\sqrt{3} \times V)$	I: line current (A); S: apparent power (VA); V: line-to-line voltage (V).	1600 kVA @ 400 V → 2310 A (TX LV FLC)
Motor full-load current	$I_{\text{fl}} = P / (\sqrt{3} \times V \times \text{pf} \times \eta)$	P: shaft power (W); V: line voltage (V); pf: motor pf; η : efficiency.	110 kW, 400 V, pf 0.86, η 0.95 → 194 A
DOL starting current	$I_{\text{start}} \approx 6\text{--}7 \times I_{\text{fl}}$	Use 6.5 × as indicative; confirm with motor manufacturer's data.	6.5 × 194 A = 1260 A
Transformer loading (pu)	$\text{Loading} = S_{\text{load}} / S_{\text{rated}}$	Values > 1.0 pu indicate overload; N-1 loading must be ≤ 1.0 pu (continuous).	1.175 / 1.600 = 0.73 pu (normal); 1.29 / 1.60 = 0.81 pu (N-1)
Short-circuit current	$I_{\text{sc}} = I_{\text{fl}} / Z(\text{pu})$	I _{fl} : transformer LV FLC (A); Z(pu): transformer impedance as decimal (e.g. 0.06). Assumes infinite upstream source — conservative.	2310 / 0.06 = 38.5 kA per TX
Cable derating — required rating	$I_{\text{z(req)}} = I_{\text{B}} / (k_{\text{g}} \times k_{\text{t}})$	I _B : design (load) current (A); k _g : grouping factor; k _t : temperature factor. Select cable with tabulated I _z ≥ I _{z(req)} .	194 / (0.80 × 0.94) = 258 A → 95 mm ² Cu XLPE
Voltage drop (3-phase feeder)	$\Delta V = \sqrt{3} \times I \times (R \cdot \cos\phi + X \cdot \sin\phi) \times L$	I: current (A); R, X: conductor resistance / reactance (Ω/km); L: length (km); cosφ/sinφ from load pf.	194 A, 120 m, 95 mm ² Cu → 9.6 V = 2.4 % (limit ≤ 5 %)
PFC reactive power required	$Q_{\text{c}} = P \times (\tan\phi_1 - \tan\phi_2)$	P: active power (kW); ϕ_1 : present pf angle; ϕ_2 : target pf angle.	2000 × (0.620 – 0.329) = 582 kVAr → ~600 kVAr bank
Generator sizing (essential load)	$S_{\text{DG}} \geq (P_{\text{ess}} / \text{pf}) \times k_{\text{step}}$	P _{ess} : essential active power (kW); pf: generator operating pf (0.8); k _{step} : step-load factor (~1.25).	(0.65 MW / 0.8) × 1.25 = 1.01 MVA → ~1000 kVA DG
Transformer voltage regulation	$\Delta V\% \approx \text{load_pu} \times (R\% \cdot \cos\phi + X\% \cdot \sin\phi)$	load_pu: per-unit loading; R%, X%: transformer resistance and reactance in percent.	0.73 × (1.2×0.9 + 5.9×0.436) = 2.7 %
Protection grading margin	$t_{\text{up}} = t_{\text{down}} + \Delta t$	Δt: grading margin (typically 0.3 s) accounting for CB opening time, relay overshoot, CT error and safety margin.	LV incomer 0.20 s → MV feeder relay 0.50 s
Arc-flash boundary	$\text{AFB} \approx D \times \sqrt{(E / 1.2)}$	D: working distance (m); E: incident energy at D (cal/cm ²); 1.2 cal/cm ² = onset-of-burn threshold. Simplified inverse-square scaling (full IEEE 1584 model required for labelling).	0.6 m × $\sqrt{(10/1.2)}$ = 1.7 m
NER resistance	$R_{\text{NER}} = (V_{\text{LL}} / \sqrt{3}) / I_{\text{ef}}$	V _{LL} : MV line-to-line voltage (V); I _{ef} : target earth-fault current (A). Rate the NER for the fault duration (e.g. 10 s).	(11,000 / 1.732) / 300 ≈ 21 Ω
Earth conductor (adiabatic)	$A = I \times \sqrt{t} / k$	A: conductor cross-section (mm ²); I: fault current (A); t: clearing time (s); k: material constant (143 for Cu/PVC).	38,500 × $\sqrt{0.2}$ / 143 = 120 mm ² Cu
Availability	$A = \text{MTBF} / (\text{MTBF} + \text{MTTR})$	MTBF: mean time between failures (h); MTTR: mean time to repair (h). Annual downtime = (1 – A) × 8760 h.	TX: 200,000 / (200,000 + 168) = 0.99916 → ~7.4 h/yr
Series availability (RBD)	$A_{\text{sys}} = A_1 \times A_2 \times \dots \times A_n$	All elements must function; system availability is always lower than the worst individual element.	0.9992 × 0.9994 = 0.9986 (e.g. incomer + TX in series)
Parallel availability 1-of-2 (RBD)	$A_{\text{par}} = 1 - (1 - A_1)(1 - A_2)$	Function survives if at least one element works. Much higher than series for the same components.	1 – (1–0.99916) ² = 0.99999929 → ~22 s/yr down
FMECA Risk Priority Number	$\text{RPN} = S \times O \times D$	S: Severity (1–5); O: Occurrence (1–5); D: Detection difficulty (1–5; 5 = hardest to detect). RPN max = 125. Higher RPN = higher risk. Priority bands: P1 = SPOF with S ≥ 4 or RPN ≥ 60; P2 = RPN 36–59; P3 = RPN 16–35; P4 < 16. Rank and act top-down.	ATS: S=5, O=3, D=4 → RPN = 60 (P1 in example FMECA)
Reliability (survival function)	$R(t) = e^{(-\lambda t)}$	λ: failure rate (failures/h) = 1/MTBF; t: mission time (h). R(t) = probability no failure in time t.	λ = 5×10 ^{–6} /h, t = 8760 h → R = e ^{–0.044} = 0.957

✓ **Tip:** For arc-flash and short-circuit studies, always use recognised power-system software (ETAP, DigSILENT, SKM or equivalent) with the full IEC 60909 / IEEE 1584 models. The simplified formulas above are for quick manual checks and teaching purposes only.

★ **Key takeaway: 20 formulas on one page. The three you will use most in the field are the FLC formula, the short-circuit formula ($I_{sc} = I_{fl} / Z$), and the availability formula ($A = MTBF / (MTBF + MTTR)$). For full calculations, use the Part 2 worked examples.**

D — Tag Scheme and SLD Conventions

Full tag table and IEC 60617 symbol key used across this manual, the training modules, and the master single-line diagram. Use only these tags when completing assessment forms. Source: the reference design document and [Part 1 — Equipment Function Table](#).

Full Equipment Tag Table

Tag	Equipment	Voltage level	Function in this plant
MV-MET	Utility revenue metering cubicle	11 kV (MV)	Utility billing / revenue metering point. Contains utility-grade CTs and VTs and the tariff meter. Not a protection or switching device.
MV-SWGR	MV metal-clad switchgear	11 kV (MV)	The main 11 kV distribution assembly. Houses the utility incomer CB (52-I), MV bus-section CB, transformer-feeder CBs (52-T1, 52-T2) with protection IEDs (50/51, 50N/51N, 87T). Rated ~25 kA/1 s. Fault level basis for all downstream short-circuit ratings.
MV-PROT	MV protection and control IEDs	11 kV (MV)	Numerical IEDs providing overcurrent (50/51), earth-fault (50N/51N), transformer differential (87T), and under/over-voltage (27/59) protection. Relays are supplied from the DCDB so they remain operative even on AC failure. A latent FMECA row — loss of DC supply to MV-PROT disables protection.
MV-NER	Neutral Earthing Resistor	11 kV (MV)	Resistance-earthing of the MV system neutral. Limits earth-fault current to ~300–400 A (indicative), controlling arc damage while keeping faults detectable. $R \approx 21 \Omega$ (indicative at 11 kV, 300 A). Rated for fault duration (e.g. 10 s).
TX-1	MV/LV power transformer 1	11 kV / 400 V boundary	Steps down 11 kV to 400 V for LV distribution. Indicative: 1600 kVA, Dyn11, $Z \approx 6 \%$, ONAN or cast-resin. Feeds Bus A of LV-MSB. Normal loading ~73 % (indicative).
TX-2	MV/LV power transformer 2	11 kV / 400 V boundary	As TX-1. Feeds Bus B of LV-MSB. Together with TX-1, the two transformers provide N-1 redundancy via the bus-tie (BT).
LV-MSB	Main LV switchboard / PCC	400 V (LV)	The Power Control Centre. Split busbar (Bus A / Bus B) with ACB incomers from TX-1 and TX-2, a motorised normally-open bus-tie (BT), and MCCB outgoing feeders to MCCs, VFDs, PFC, UPS and DBs. Form 4b segregation (indicative). Bus rated ~2500–3200 A, 50–65 kA withstand.
BT	Bus-tie circuit breaker	400 V (LV)	The normally-open ACB (or contactor) connecting Bus A and Bus B of LV-MSB. Closes automatically (via PMS/ATS logic) on loss of one LV incomer to allow the surviving transformer to back-feed the dead bus. The bus-tie is a critical SPOF-mitigation element; verify its auto-transfer scheme is commissioned and periodically tested.
MCC-1 ... MCC-n	Motor Control Centres	400 V (LV)	LV assemblies of motor starter bays (DOL, star-delta, soft-starter, VFD). Compartmentalised to IEC 61439; withdrawable or fixed. Each MCC feeds a group of process motors. For SPOF assessment: check whether duty and standby motors share the same MCC (common element — see Appendix F).
VFD	Variable Frequency Drive(s)	400 V (LV)	Power-electronic speed controllers for pumps, fans and compressors. Injects harmonics into the supply — requiring detuned PFC capacitors. Also provides LVRT capability and kinetic buffering through voltage sags.
PFC	Power factor correction bank	400 V (LV)	Automatic capacitor bank with 7 % detuned reactors. Corrects plant pf from ~0.85 to ~0.95+. Sized ~580–600 kVar (indicative). Detuning prevents resonance with VFD harmonic currents. Loss of PFC causes reactive demand increase and potential utility penalty.
DG	Diesel generator set	400 V (LV)	Standby power source for the essential bus. Indicative rating ~800–1250 kVA; includes AVR, governor, day-tank, battery starting. Connected via the ATS. Verify weekly auto-test-on-load schedule, fuel level and battery condition during site visit.
ATS	Automatic Transfer Switch	400 V (LV)	Detects loss of utility supply and automatically starts and connects the DG to the essential bus. A P1-priority item (RPN 60) in the example FMECA. A failed-to-transfer ATS on a utility outage = total loss of essential load. Verify commissioning records and test schedule.
UPS	Uninterruptible Power Supply	400 V AC / battery	No-break, clean AC supply for DCS/PLC/SCADA, field instrumentation and critical communications. Battery autonomy typically 15–30 min (to be specified per plant). Loss of UPS during a supply disturbance can cause a DCS/PLC crash — verify battery condition and bypass availability.
DCDB	DC distribution board (battery and charger)	110 V DC	Provides DC supply for MV switchgear trip / close coils and protection relay auxiliary circuits. A DCDB failure can silence all MV protection — extremely high severity. Verify battery condition (float voltage, impedance test) and charger operation.
DB	Distribution board (normal)	400 V (LV)	Final-circuit panel for lighting, small power, HVAC, and sockets. Fed from LV-MSB or MCC outgoing feeders. MCB/RCBO protection.
EDB	Essential distribution board	400 V (LV)	As DB, but supplied via the ATS from the essential bus (DG-backed). Feeds safety-critical loads that must remain live during a utility outage (e.g. emergency lighting, fire-alarm panel, safety system auxiliaries).

IEC 60617 Symbol Key

The master SLD and all SPOF example diagrams use IEC 60617 symbols rendered by the `slddraw.py` library. The table below summarises the symbols you will encounter when reading or red-lining the drawings on site.

Symbol description	IEC 60617 representation (text description for field use)	Used in this plant for
Utility / grid source	Circle with AC sine-wave mark inside	11 kV utility feeder supply connection

Symbol description	IEC 60617 representation (text description for field use)	Used in this plant for
Circuit breaker (closed)	Solid filled square on conductor	All normally-closed breakers: MV CBs, LV ACBs (incomers), MCCB feeders
Circuit breaker (open / N.O.)	Hollow (outline) square on conductor	Bus-tie (BT) — normally open; auto-closes on N-1
Isolator / disconnecter	Open knife-contact symbol on conductor	Maintenance isolation points upstream of CBs
Fuse	Rectangle on conductor	MV HRC fuses (where used); LV MCB/fuse in final circuits
Two-winding transformer	Two interlinked circles (Dyn11 — delta primary, star secondary with neutral)	TX-1 and TX-2
Neutral earthing resistor	Resistor symbol in series to earth on the transformer neutral conductor	MV-NER (resistance earthing of the 11 kV neutral)
Earth / ground	Three horizontal bars of diminishing length	All earth connections; transformer neutral solidly earthed at 400 V (TN-S)
Capacitor bank (PFC)	Two parallel plates (capacitor symbol)	PFC detuned capacitor bank on LV-MSB Bus A or B
Motor	Circle with letter M	Process motors (pumps, fans, compressors) fed from MCCs
Generator	Circle with letter G	DG alternator symbol
Labelled equipment block	Rectangle with tag name and sub-label	MCC, VFD, UPS, ATS, DB, EDB, DCDB
Busbar	Bold horizontal line	MV bus (firebrick colour); LV Bus A / Bus B (navy colour) in the SLD
SPOF indicator (red-lined SLD)	Red dashed ring around the failure-point element with red caption	Used on red-line mark-ups to flag identified SPOFs. Not a standard IEC symbol.

✓ **Tip — reading the SLD on site:** When verifying the drawing against the as-found plant, check: (1) bus-tie switch — is it normally open and does it have an auto-transfer scheme? (2) transformer taps — what position are they set at? (3) ATS — is the transfer scheme commissioned? (4) DG — is the auto-start enabled? These four items account for most SPOF findings on routine site visits.

★ **Key takeaway:** 17 equipment tags (including MV-PROT), 14 IEC 60617 symbols. The tag scheme is fixed — do not invent new tags; update the reference document if a new item is found on a plant that does not fit any existing tag.

E — Blank Assessment Forms

Printable blank versions of the four per-plant deliverable forms. Open all four as soon as a plant is assigned; fill the header during desk study so the form is ready on site. See [Part 0 — The Four Deliverable Forms](#) for descriptions of what each form contains and when it is completed. The filled-in example in [Part 5](#) shows how a completed set looks.

SF-01 — Document Inventory

Record all electrical documents requested, received and reviewed. Add rows as required. Documents not received should be chased via the site contact before or during the site visit. *Form IDs SF-01...SF-04 match the worked example in [Part 5 — Jubail Case Study](#).*

Plant name	
Plant reference / ID	
Assessment reference	
Engineer (lead)	
Date desk study started	
Date last updated	

Ref.	Document title	Drawing / document number	Revision	Date	Source (client / site / engineer)	Status (received / requested)
DOC-01	Master Single-Line Diagram					
DOC-02	MV Switchgear Single-Line Diagram					
DOC-03	LV Switchboard (LV-MSB) Single-Line Diagram					
DOC-04	MCC-1 Single-Line / Schematic					
DOC-05	MCC-2 Single-Line / Schematic (if present)					
DOC-06	Equipment schedule / load list					
DOC-07	Transformer test certificates (TX-1, TX-2)					
DOC-08	Protection relay settings (MV-SWGR IEDs)					
DOC-09	Protection coordination study					
DOC-10	Short-circuit study					
DOC-11	Load-flow study					
DOC-12	Earthing / lightning protection study					
DOC-13	Arc-flash study (if available)					
DOC-14	DG / ATS commissioning records					
DOC-15	UPS / DCDB battery maintenance records					
DOC-16	Previous SPOF / reliability assessment (if any)					
(add rows as required)						

Documents outstanding at site-visit date	
Data calls issued (number)	
Form status	☐ Draft ☐ For review ☐ Final

SF-02 — FMECA Worksheet

Failure Mode, Effects and Criticality Analysis per IEC 60812. One row per item / failure mode. Score S, O, D on a **1–5 scale** (5 = worst); $RPN = S \times O \times D$ (max 125). Priority bands: P1 = SPOF with $S \geq 4$ or $RPN \geq 60$; P2 = $RPN\ 36\text{--}59$; P3 = $RPN\ 16\text{--}35$; P4 < 16. See [Part 3 — FMECA method](#) for scale definitions. Scores are indicative; calibrate scales using IEEE 493 / ISO 14224 data for real studies.

Plant name	
Assessment reference	
Engineer	
FMECA revision	
Date	

Item ref.	Tag	Function	Failure mode	Local effect	System effect (worst case)	Existing controls / detection	S (1-5)	O (1-5)	D (1-5)
FM-01									
FM-02									
FM-03									
FM-04									
FM-05									
FM-06									
FM-07									
FM-08									
FM-09									
FM-10									
(add rows as required — one row per item/failure-mode combination)									

Total items assessed	
P1 items (SPOF S ≥ 4 or RPN ≥ 60)	
P2 items (RPN 36–59)	
Form status	☐ Draft ☐ For review ☐ Final

SF-03 — Recommendations Register

Ranked list of recommended modifications and improvements derived from the FMECA. One row per recommendation. Link each row back to the FMECA reference (Rec. ref. = FM-xx). Risk ratings: H = High, M = Medium, L = Low. Cost: H / M / L indicative only.

Plant name	
Assessment reference	
Engineer	
Register revision	
Date	

Rec. ref.	Finding / SPOF description	Recommended modification	Affected tag(s)	FMECA ref.	Risk before (H/M/L)	Risk after
REC-01						
REC-02						
REC-03						
REC-04						
REC-05						
REC-06						
REC-07						
REC-08						

Rec. ref.	Finding / SPOF description	Recommended modification	Affected tag(s)	FMECA ref.	Risk before (H/M/L)	Risk after
(add rows as required)						

Total recommendations	
Priority 1 (High risk, act now)	
Priority 2 (Medium risk, plan)	
Priority 3 (Low risk, monitor)	
Form status	☐ Draft ☐ For review ☐ Final

SF-04 — Site-Visit Report

Factual record of the site visit. Non-intrusive only — do not open energised panels, operate switches or perform live tests. Complete on the day of the visit; review with PM before issue.

Section A — Site and Visit Details

Plant name / location	
Plant reference / ID	
Assessment reference	
Site visit date(s)	
Lead engineer (assessor)	
Accompanying engineer / technician	
Plant contact / host representative	
Host organisation	
Permit-to-work reference(s)	
Areas of plant accessed	
Areas NOT accessed (reason)	

Section B — Scope and Non-Intrusive Statement

Survey scope (tick all that apply)	☐ MV switchroom ☐ LV switchroom ☐ Transformer yard / enclosure ☐ MCC room ☐ Generator room / enclosure ☐ UPS / battery room ☐ Control room / DCS panels ☐ Earthing / earth pits ☐ Other: _____
Non-intrusive confirmation	☐ Confirmed — no live panels opened, no switching operations performed, no live electrical testing. Visual inspection, nameplate reading and photographic documentation only.
Thermographic survey performed	☐ Yes (host permit obtained) ☐ No

Section C — Key Findings

Overall as-found condition (brief)	
SLD discrepancies found	
Equipment nameplate data confirming SLD (Y / N / partial)	
Bus-tie (BT) status observed	☐ N.O. (correct) ☐ Closed ☐ Locked out ☐ Not verified
ATS auto-transfer status	☐ Auto enabled ☐ Manual only ☐ Not verified
DG auto-start status	☐ Auto enabled ☐ Manual only ☐ Not verified
UPS battery condition observation	
DCDB / battery charger observation	
Visible housekeeping / condition concerns	
Thermographic hot-spots noted (if surveyed)	

Section D — Photolog Index

The 12 mandatory photographs (PH-01 through PH-12) match the photolog defined in the [Visual Inspection Checklist, Part 5](#). Tick each row when the photo is captured and indexed.

☐	Photo ref.	File name	Equipment / location	Description / relevance to assessment	OBS-ref (if finding)
☐	PH-01		Switchroom / main entry	Overall switchroom general view — panel line-up and housekeeping	

☐	Photo ref.	File name	Equipment / location	Description / relevance to assessment	OBS-ref (if finding)
☐	PH-02		MV-SWGR line-up	MV switchgear line-up and arc-flash labels (or their absence)	
☐	PH-03		Bus-tie (BT / MV bus-section)	Bus-tie breaker cubicle front — position indicator (O/I) clearly visible	
☐	PH-04		TX-1 nameplate	TX-1 rating plate: kVA, voltage ratio, vector group, impedance, year	
☐	PH-05		TX-2 nameplate	TX-2 rating plate (same fields as PH-04)	
☐	PH-06		LV-MSB incomer and bus-tie section	LV-MSB front: incomer A, incomer B, bus-tie breaker with position indicators and labels	
☐	PH-07		DG nameplate and starting battery	Generator rating plate and battery / charger in same frame if possible	
☐	PH-08		ATS panel front	ATS source labels (Normal/Standby) and transfer-switch position indicator	
☐	PH-09		DCDB / battery charger	Charger nameplate, battery type label, front-panel fault/alarm indicators	
☐	PH-10		UPS front panel	UPS nameplate and status indicators: mains-OK, battery-OK, any active alarms	
☐	PH-11		Main earth test point / pit	Test disconnect link, test tag (last test date), earth bar bonding connections	
☐	PH-12		Notable defect or discrepancy	Specific item seeding an FMECA finding (one photo per distinct finding; add rows as needed)	
	(add)			(additional photos beyond mandatory 12)	

Section E — Actions and Data Calls

Action ref.	Action / data call description	Owner	Required by date	Status
A-01				
A-02				
A-03				
A-04				
	(add rows as required)			

Report prepared by	
Date	
Reviewed by (PM)	
Review date	
Form status	☐ Draft ☐ For review ☐ Final


Warning — NON-INTRUSIVE surveys only. Do not open energised panels, operate switches or breakers, or conduct any live electrical test. Visual inspection, nameplate reading and photography (where permitted) are the only permitted activities. Observe host permit-to-work, LOTO and QHSE rules at all times.


Form ID mapping — blank forms to worked example (Jubail, Part 5): SF-01 = Document Inventory (rows: DOC-xx); SF-02 = FMECA Worksheet (rows: FM-xx); SF-03 = Recommendations Register (rows: REC-xx); SF-04 = Site-Visit Report (observations: OBS-xx, photos: PH-xx, actions: ACT-xx). These IDs align with the Part 5 Jubail case study.


Key takeaway: Four blank forms — SF-01 Document Inventory, SF-02 FMECA Worksheet, SF-03 Recommendations Register, SF-04 Site-Visit Report. Open them at plant assignment, fill the header during desk study, and update iteratively through site visit, analysis and recommendations stages.

As-Found Verification Sheet

Use this sheet during post-site analysis to verify that nameplate data captured on site matches the calculated parameters and to record a Pass/Fail verdict for each key item. This directly supports the "assessment = verification" thesis: every FMECA row for a sizing or rating finding must trace to a completed row here. *All calculations are indicative; confirm with project-specific studies.*

Plant name	
Assessment reference	
Engineer	

Date completed				
Tag	Nameplate: kVA (or kW)	Nameplate: Z% (TX) or FLC (motor)	Nameplate: V (primary / secondary)	Nameplate: rated A
TX-1				
TX-2				
LV-MSB Bus A incomer ACB	—		400 V	
LV-MSB Bus B incomer ACB	—		400 V	
BT bus- tie ACB	—		400 V	
DG	—		400 V / 415 V	
MCC-1 incomer MCCB	—		400 V	
MCC-2 incomer MCCB	—		400 V	
UPS	—		230/400 V	
DCDB charger	—		110/48/24 V DC	
(add rows)				

Pass criteria (indicative): TX loading $\leq 80\%$ nameplate kVA in normal; $\leq 100\%$ in N-1. ACB / MCCB rated current $\geq$ calculated FLC $\times 1.25$ (service factor). Switchgear Icu $\geq$ calculated Isc. Cable ampacity $\geq$ FLC after derating. DG kVA $\geq$ essential load kVA $\times 1.1$. UPS autonomy ≥ 15 min. DCDB autonomy ≥ 2 h. All criteria indicative — confirm against governing specification.

F — SPOF-Hardening Principles (One-Page Summary)

Distilled from [Part 3 — SPOF Analysis](#) and Module 3. Use this page as a rapid scan-reference when reviewing an SLD or writing recommendations. Every principle has a trade-off in cost, complexity and footprint — apply redundancy where the consequence of failure justifies it (critical / essential loads first). All values and configurations are indicative.

Principle	What it means in practice	Typical implementation (tags)	SPOF it eliminates	Cost / complexity
N+1 redundancy	Provide at least one more capacity unit than the minimum. The surviving unit(s) must carry the essential load on loss of one unit.	2 × TX-1 / TX-2 (each capable of carrying essential load at ~81 % loading on N-1); N+1 MCC feeders for critical motors.	Single transformer / single source	H capital; H benefit for critical loads
Bus segregation	Split the LV busbar into independent sections (Bus A / Bus B) so a bus fault or maintenance on one section does not take the whole plant.	LV-MSB split bus; Form 4b compartmentalisation of MCC	Single un-split LV bus (Diagram A)	M capital; H benefit (contained fault zone)
Bus-tie with auto-transfer	The normally-open bus-tie closes automatically within seconds on loss of one LV incomer, connecting the healthy transformer to the dead bus. N-1 loading must be verified (≤ 1.0 pu continuous).	BT ACB with auto-transfer scheme via PMS or hard-wired interlock; loss-of-voltage / breaker-position logic. Periodic on-load test required.	Two buses with no tie (Diagram B)	M capital; very H benefit (MTTR reduced from hours to seconds)
Dual utility incomers	Two independent utility feeds to the MV bus from different utility substations or circuits, with a closeable MV bus-section between them.	Two incomer panels in MV-SWGR (52-I1, 52-I2); MV bus-section CB (52-BS).	Single utility feeder / single MV incomer (Diagram C)	H capital (utility connection); H benefit for essential-load continuity
Separate routing — duty vs standby	Duty and standby motors / loads must be fed from <i>different</i> sources, MCCs and cable routes. A single cable, MCC bus or MCC incomer serving both duty and standby is a common-element SPOF.	Duty pump on MCC-1 (Bus A), standby pump on MCC-2 (Bus B); cables in separate trays on opposite sides of plant.	Shared cable / single MCC for duty + standby (Diagram D)	M capital; H benefit (eliminates common-mode failure)
Standby generation (DG + ATS)	A standby diesel generator, automatically started and connected via the ATS on utility loss, backs up the essential bus for safety-critical and process-critical loads.	DG set (~1000 kVA indicative) + ATS + EDB. ATS transfer time typically < 10 s (depends on DG run-up time). Auto-test on load weekly; fuel and battery monitoring continuous.	Total loss of supply to essential loads on utility outage	H capital per kVA; H benefit for safety loads
UPS for control / protection	No-break, clean AC supply for the DCS/PLC/SCADA and instrumentation. Prevents a supply disturbance from crashing the control system and causing a process trip or safety incident.	UPS (battery autonomy 15-30 min indicative) + DCDB (110 V DC for MV switchgear trip / close coils).	Loss of control / protection on any supply disturbance	M capital; H benefit for control continuity and protection dependability
Condition monitoring and testing	Scheduled on-load testing (DG / ATS weekly; bus-tie transfer periodically) and continuous condition monitoring (transformer oil / temperature, battery impedance, UPS health) reduces Detection score (D) in the FMECA without changing failure rate.	Monitored via PMS / DCS; oil/temperature sensors on TX-1 / TX-2; battery monitoring on UPS and DCDB.	Latent failures in standby equipment (DG fails to start; bus-tie fails to close)	L-M capital; H benefit (lowers RPN through better detection)
Fault containment and selectivity	Graded protection (time-current coordination) ensures the device closest to a fault trips first, isolating the smallest possible zone. Zone-selective interlocking (ZSI) maintains both selectivity and fast arc-flash clearance.	Relay / trip-unit settings in MV-SWGR IEDs and LV-MSB ACBs; ZSI between ACB and outgoing MCCBs.	Upstream breaker trips on a downstream fault (wider zone lost than necessary)	L capital (settings only, unless extra relays); H benefit for zone containment
Earthing integrity	A continuous, low-impedance earth system (grid + bonding + NER) ensures earth faults are reliably detected, current limited, and touch/step voltages kept safe.	MV-NER (R ≈ 21 Ω indicative); LV TN-S; earth grid (≤ 1 Ω indicative); 120 mm² Cu earth conductors (indicative); bonding to LV-MSB, MCC, cable armour.	Undetected earth faults; personnel electrocution; equipment damage from high fault energy	M capital; safety-critical

SPOF-Hunt Checklist — Quick Scan of an SLD

Progress: 0 / 10

- ☐ Is there a single utility incomer / single MV bus? (Diagram C SPOF — check for dual incomers or DG/ATS coverage).
- ☐ Is there only one transformer? (Diagram A SPOF — check for N+1 or bus-tie capability).
- ☐ Is the LV bus split into sections A and B? If not, the whole LV plant is a single zone.
- ☐ Is there a bus-tie (BT) between Bus A and Bus B? Is it normally open with auto-transfer?

☐ Are duty and standby motors fed from different MCCs on different buses, via separately routed cables? (Diagram D SPOF).

☐ Is the DG connected via an ATS? Is the ATS auto-transfer commissioned and tested regularly?

☐ Is there a UPS for DCS/PLC/SCADA? Is there a DCDB for MV switchgear trip/close supplies?

☐ Are transformer differential protection (87T) and earth-fault protection (50N/51N) commissioned on both TX-1 and TX-2?

☐ Is the NER present and rated correctly? (Check neutral CT and earth-fault relay are connected).

☐ Is protection coordination current — does the study reflect the as-found breaker / relay configuration?

★ **Key takeaway: Ten hardening principles; ten SPOF-hunt checklist items. The five most common findings are: no bus-tie auto-transfer, duty and standby on the same MCC, single utility incomer with no DG coverage, ATS not commissioned, and outdated protection coordination study. Address these first.**

G — Indicative Failure-Rate and MTBF Data

Indicative λ (failure rate), MTBF and MTTR figures per equipment class, drawn from the reliability analysis in Module 5 and typical industrial power-system data. Use these values for preliminary FMECA scoring and RBD availability estimates only.

**Warning — indicative values only.** The figures in this table are *indicative and tentative*. They are representative of industrial power-system equipment in good condition under reasonable maintenance regimes. Actual failure rates vary significantly with age, maintenance history, environment (tropical, desert, coastal), load factor, manufacturer and installation quality. **For any real reliability or availability study, these figures must be replaced with values confirmed against IEEE 493 (Gold Book), ISO 14224, or plant-specific historical maintenance data.** Do not cite this table as a primary source in a formal study.

Equipment class	Tag(s)	Indicative λ (failures/yr)	Indicative MTBF (h)	Indicative MTTR (h)	Indicative Availability A	Notes
MV metal-clad switchgear (panel)	MV-SWGR	~0.006 / yr	~150,000 h	168 h	~0.99889	Per p (bus incom TX fe Inclu vacu and prot relay inclu spar logis
MV/LV power transformer	TX-1 , TX-2	~0.004 / yr	~200,000 h	168–720 h	~0.99916 (at 168 h MTTR)	Cast units lowe but s elect failu oil-fi MTT exte 720-rewi requ Oil/t mon redu later
LV switchboard / PCC (complete assembly)	LV-MSB	~0.010 / yr	~90,000 h	8–72 h	~0.99920 (at 72 h MTTR)	Busk rare seve mos are i indiv brea auxi Form segr limit zone redu effec MTT
Air Circuit Breaker (ACB)	LV-MSB incomer / BT	~0.015 / yr	~60,000 h	4–24 h	~0.99960 (at 24 h MTTR)	Inclu failu (mos mod failu close BT fa auto tran Regul testi impr Dete scor
Moulded-Case Circuit Breaker (MCCB)	Feeders in LV-MSB , MCC	~0.020 / yr	~45,000 h	2–8 h	~0.99982 (at 8 h MTTR)	High rate large popl mos affect the i feed Sele (coo prev upst trip.

Equipment class	Tag(s)	Indicative λ (failures/yr)	Indicative MTBF (h)	Indicative MTTR (h)	Indicative Availability A	Notes
Motor Control Centre (MCC, complete)	MCC-1 ... MCC-n	~0.030 / yr	~30,000 h	4–24 h	~0.99920 (at 24 h MTTR)	Aggr figur asse inclu cont over and A sir start affec one bus inco failu all m that
Variable Frequency Drive (VFD)	VFD	~0.050 / yr	~18,000 h	4–48 h	~0.99733 (at 48 h MTTR)	High rate simp cont pow elect are s to temp harm and MTTR long pow mod be o Cons bypa cont critic drive
Power factor correction bank (PFC)	PFC	~0.020 / yr	~45,000 h	8–48 h	~0.99893 (at 48 h MTTR)	Capa elerr failu dom mod detu reac gene more Failu incre reac dem does typic caus supp outa Med seve
Diesel generator set (complete)	DG	~0.060 / yr (start-to-run)	~15,000 h (run)	4–48 h	Conditional: P(start success) ~0.95 per demand	Disti betw failu start dem indic and while (< 1 Wee teste load P(fai start battu cooli conc the r later
Automatic Transfer Switch (ATS)	ATS	~0.030 / yr	~30,000 h	2–24 h	~0.99920 (at 24 h MTTR)	Failu tran: dem the c mod (high in ty FME Redu cont posit mon and

Equipment class	Tag(s)	Indicative λ (failures/yr)	Indicative MTBF (h)	Indicative MTTR (h)	Indicative Availability A	Notes
UPS (online, double-conversion)	UPS	~0.030 / yr	~30,000 h	2–24 h	~0.99920 (at 24 h MTTR)	on-line mode reduces occurrence and detection of faults. Battery aging dominates long-term failure. Invest in electrical components. Battery impedance testing (annual) identifies degraded cells. Failure bypass and reduction of UPS improvement system availability.
DC battery and charger (DCDB)	DCDB	~0.020 / yr	~45,000 h	4–48 h	~0.99893 (at 48 h MTTR)	Battery failure silent protection. Extreme high float monitoring and discharge testing. Keep battery mode site critical plan.
Neutral Earthing Resistor (NER)	MV-NER	~0.002 / yr	~500,000 h	8–24 h	~0.99998	Pass element failure. Open or short circuit both earth protection behavior. Verify resistance and CT in duration of assembly.
MV/LV feeder cable (per km)	Feeders between tagged equipment	~0.003 / yr / km (LV XLPE)	~300,000 h / km	8–72 h	~0.99976 (at 72 h MTTR)	Buried cable rate increase with installation. Damage proximate other service. Single shared duty station is a critical element. SPOF (Diagram) use separate routing cable.

RBD Availability Examples

Configuration	Formula	Indicative result	Annual downtime
Single transformer (series)	$A = 0.99916$	0.99916	~7.4 h/yr
Two transformers 1-of-2 (parallel)	$A = 1 - (1 - 0.99916)^2$	0.99999929	~22 s/yr
Transformer + ACB in series	$A = 0.99916 \times 0.99960$	0.99876	~10.9 h/yr
ATS (manual MTTR ~8 h vs auto ~0.003 h)	Manual: $A = 30000/(30000+8) = 0.99973$ Auto: $A = 30000/(30000+0.003) = 0.99999990$	Manual: 0.99973 Auto: ~1.000	Manual: ~2.3 h/yr Auto: ~90 s/yr
DG + ATS (both must work)	$A = A_{DG} \times A_{ATS} = 0.9980 \times 0.99973$ <i>A_{DG} derivation: DG failure-to-start probability ≈ 5% per demand (1 demand/yr); MTBF running ≈ 15,000 h; MTTR ≈ 24 h. Effective A_{DG} = 1 – P(fail-to-start) ≈ 0.9980 (indicative — dominated by start-reliability; full RAM model needed for real studies).</i>	~0.99773	~19.6 h/yr (manual ATS)

i Note — MTTR as the key lever: The ATS example illustrates why auto-transfer (MTTR reduced from 8 h to ~10 s) delivers a far larger availability improvement than any realistic reduction in failure rate. Auto-close bus-tie and auto-transfer ATS are the highest-value reliability upgrades in a typical MV/LV plant. This principle is central to the recommendations generated by this campaign.

★ Key takeaway: 14 equipment classes tabulated. Use these indicative figures for initial FMECA scoring only. Replace with IEEE 493 / ISO 14224 / plant maintenance records for formal studies. The single most impactful design choice visible in the MTBF data is the auto-transfer MTTR reduction: from ~8 h (manual) to seconds (auto), turning ~2 h/yr loss into < 2 min/yr.